



Can a cheap EEG headset test what people report hearing?



A buildable experiment design, the instrument and software it requires, and how it could run at scale on TI One Voice

A proposal seeking partners

Funding · an academic or clinical partner for ethical review · people who experience voice-to-skull · people who do not · volunteers in electronics, software and logistics.



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Version 4.2 — a buildable experiment design, the instrument and software it requires, how it could run at scale on TI One Voice, the full flow in screens, what the other branch would require, what a participant can and cannot be told, what is needed to start, the first external review incorporated — and what academic and medical contribution would cost

Every figure in this article is a schematic drawing of a predicted result. None of them show measured data. No study described here has been run.

Where this stands today. This is a proposal, not a programme. Nothing described here has been built and no data has been collected. There is a design, a platform that already carries most of the software, and a first stage costed at two thousand euros — and there is nothing else. What it needs, in order of urgency: funding to buy the first kits and pay for independent ethical review; people who experience these voices and are willing to be measured, knowing the result may not be the one they hope for; people who do not experience them, to serve as controls; and volunteers who can build, ship, refurbish and operate. The Background section that follows sets out the phenomenon for a reader coming to it cold; Parts 18 and 19 set out each of the asks in detail.

People who report hearing voices they believe are transmitted to them — often called V2K, short for "voice-to-skull" — are usually offered one of two responses. Either they are told the experience is a symptom and the conversation ends, or they are told it is definitely a transmission and the conversation also ends. Both responses share a defect: neither of them measures anything.

This article is about what happens if you actually measure. It sets out an experiment that could be run with hardware costing about the price of a used car engine rebuild, on a group of volunteers, in a spare room, and that would produce a result meaning something regardless of which way it came out. It is deliberately designed so that it cannot only confirm. If a design can only produce one answer, it is not an experiment; it is a ceremony.

I want to be precise about scope from the first paragraph, because the precision is the whole point.

What this design does test: whether something measurable happens in the hearing regions of the brain when a person reports a voice; whether the brain's own "this is me" tagging system is working normally; and whether a person who reports replying silently to the voices is genuinely producing an act of speech in their head, as opposed to simply thinking.

What this design does not test: whether anything is being transmitted. An EEG headset is a voltmeter taped to a scalp. It cannot receive radio. It cannot decode words. It cannot tell an individual person where their experience came from. Nothing in the following pages should be



read as offering that.

If you have limited patience, here is the short version. There is a decades-old, well-replicated measurement in neuroscience that distinguishes sounds your own brain caused from sounds it did not cause. That measurement is cheap, it works on eight electrodes, and nobody appears to have applied it to this population under proper conditions. It should be applied. The rest of this article explains how, and what each possible result would mean.

Where this came from

The starting point was not mine. Mykhailo Sokolov, a fellow signatory on the TI One Voice platform, sent me a white paper from the NeuroLaws Foundation proposing a very different approach to the same problem: mail a standardised EEG kit to hundreds of people and run centralised anomaly detection on what comes back. Thinking about why that design would and would not work is what produced the one in this document, and the two are complementary rather than competing. Part 13 sets out his proposal in full, what it gets right, where it would need strengthening, and how the two fit together. Part 14 describes how the protocol could actually be delivered through TI One Voice. My thanks to him for the prompt. Version 4 owes a parallel debt: an external academic reviewer, who has asked not to be named, read the experiment sections cold and returned the disconnected-generator objection that now shapes Parts 4, 16 and 16b. A design that claims to welcome adversarial review should show what it does when it receives some; this version is that.



What changed in version 4.2

Version 4.2 is an attribution release. It changes no design decision, no figure, no cost and no conclusion.

- **The external reviewer is no longer named.** The reviewer whose critique shaped version 4 has asked that their name and institutional affiliation not appear in this document, and that request has been honoured throughout. Every passage that carried the attribution now reads as "the external reviewer" or "the review".
- **Nothing else moves.** The disconnected-generator objection in Parts 16 and 16b, the laboratory tier, the cautions attached to the Outcome B fork and the costings of Part 18b all stand exactly as they did in version 4.1. Removing a name removes no argument, and the review is still credited as external and adversarial — which is the part a reader judging this design needs to know.

What changed in version 4.1

Version 4.1 is a costing release. It adds one part and changes no design decision, no figure and no conclusion.

- **New Part 18b — What academic and medical contribution costs.** Part 18 prices academic involvement at zero cash and says so. That is the plan, and it is still the plan, but a funder or a journal reads an uncoded dependency as an undisclosed one. Part 18b states where the academic and clinical hours actually go, what they are worth at loaded salary cost, what they would cost as honoraria or at consultancy rates, and what each scenario does to the stage totals in Part 18. It prices the laboratory tier of Part 16b separately, because that tier is partner-led and partner-funded.
- **The medical role is named.** A neurologist or psychiatrist appears in the walkthrough as Dr Okafor and in Part 18 as "clinicians", but neither Part 18 nor Part 19 states what a medically qualified person is being asked to do or what those hours cost. Part 18b does both: protocol and safety review, diagnostic confirmation of the clinical comparison group, medical escalation cover behind the participant-support role, and the debrief of the difficult outcome and of the individual feedback in Part 16c.
- **Part 18 gains two cross-references** — from the ethical-review paragraph and from the stage table — pointing to Part 18b, and a named contingency line for honoraria. The stage totals themselves are unchanged.
- **Version 4.1 does not** change the participant contribution models in Part 12. It notes, in Part 18b, that neither model currently recovers any medical or academic cost, and says what the per-session figure would be if Model 2 were asked to.



What changed in version 4

Version 4 incorporates the first external methodological review of this design. An external academic reviewer, who has asked not to be named, read the experiment sections and replied with the sharpest critique the document has yet received. Every change below follows from that review, received on 28 August 2026. None of its points weakens the case for running the study; several sharpen what it can honestly claim, which is worth more.

- **The disconnected-generator alternative is now named.** The reviewer's central objection — that a voice-generating brain region insufficiently wired to the rest of the system would reproduce every "non-self" signature this design measures — is stated in Part 16, added as a third caution to the fork in Part 16b, and the Outcome B reading is reworded accordingly.
- **New section: if Outcome B is proven, what could explain it.** The known mechanisms that produce an Outcome B signature in one person, the demonstration that no single recognised pathology accounts for it across a cohort of voice-to-skull reporters, and the three qualifications that keep that statement honest.
- **New section: what a partner's laboratory adds.** MEG source localisation as a route to neural detection of voice onsets — the timestamp this document names as its weakest link in four places — plus the magnetically shielded room, fMRI, and rTMS as a partner-led, ethically distinct extension.
- **Test Two is stated at the right level.** Part 4 now says which stage of the auditory pathway the probe-tone N100 indexes, cross-referenced to the pathway stages of Part 16b.
- **The suppression score is treated as a continuous measure.** Pre-registered regressions against symptom frequency, medication and demographics join the cluster analysis in Part 10.
- **A volume-sourcing note in Part 14:** what Chinese board assembly genuinely changes about the cost floor, what it does not, and the counterfeit-converter risk the per-unit validation already defends against.
- **The N1/N100 terminology is corrected** in Part 2.
- **The individual-feedback debrief wording is strengthened** in Part 16c: "consistent with B" must never be glossed as "consistent with transmission."

What changed in version 3

Version 3 adds three parts and a set of screenshots, and changes no conclusion. The additions come from the animated walkthrough of the module (the nine-minute film that accompanies this document), which follows a participant, a control, an administrator, a neurologist and a statistician through the whole flow and then walks the branch this document had not written out: what it would look like if the voices did not behave like inner speech, and what instrument that branch would need.

- Part 15b — The walkthrough, screen by screen. The full flow of the animation as still frames, participant side and administrator side. The administrator screens (cohort overview, the



controls-first sanity gate, the pre-registered contrasts, the confound step, the cross-subject time-locking test, the discipline checklist) were not in version 2 at all.

- Part 16b — The other branch: what Outcome B would require. The fork between "behaves like inner speech" and "behaves like an external sound", the order-of-arrival logic through the hearing pathway, envelope tracking against silent microphones, the otoacoustic-emission check, the environmental arm, and a plain statement of what the two-thousand-euro instrument cannot do.
- Part 16c — Can a participant be told A or B? A re-evaluation of individual feedback after one or several sessions, with the arithmetic, and a bounded individual-feedback protocol that the design can support without becoming a machine for false certainty. This changes the session plan for participants who opt in: three sessions rather than one.
- Part 15 now carries the animation frames for each of the nine screens described there.
- The repeated phrase in the Part 7 cost note has been corrected.

Decisions the author still has to make

The animation and version 2 disagree in five places. Version 3 flags them rather than silently choosing, because each is a protocol decision, not an editorial one.

1. Block 6. In the animation, Block 6 is an eight-minute passive oddball (mismatch) block — "just listen, there is nothing to do" — with a difference wave shown to the participant after the block. In version 2, Block 6 is the repeat of Blocks 1 and 2 for within-session reliability (Figure 9), and the "surprise test" of Part 6 is an expected/unexpected rating after voice episodes. If the oddball block is wanted, it should be added as a seventh block and the timeline in Part 8 revised; the reliability repeat should not be dropped. Showing any waveform to a participant also has to be reconciled with rule 14.10, "no individual interpretation anywhere in the module".
2. Baseline block. The animation has an explicit resting block — four minutes eyes open, four eyes closed — with the rationale "what the room and the headset look like when nothing at all is being asked". The text of Part 8 does not describe one. It is cheap, it is what the fleet protocols in Part 12 already record, and it should be written into Part 8.
3. Sampling rate. Part 7 says 250 samples per second, the frame in Part 14 says 1,000, the animation says 500 Hz. One figure should be fixed for the fleet unit and used everywhere.
4. Silent-reply gate. The animation's session report has a "Marginal" state for silent replies marked; the report table in Part 13 has only pass and fail. The third state is useful and should be added.
5. The persona cast. Sarah (hears voices), Marcus (control), Priya (administrator), Dr Okafor (neurologist), Dr Lindqvist (signal processing and statistics) appear in the screenshots. They are fictional and map onto the roles in Part 19; Part 15b says so.



Background — the phenomenon, for a reader encountering it cold

This section exists for one reason: if you are reading this as a prospective academic or clinical partner, you should not have to take anything on trust, and you should not have to go and find the literature yourself. What follows is an attempt to state what is actually known, what is contested, and what is simply unstudied, in enough detail that you can judge whether the study proposed in the rest of this document is worth your name.

It is written by someone who is not neutral. I am part of the community described below and I run a platform that serves it. I have tried to compensate for that by stating the mainstream clinical position at full strength rather than in a form that is easy to knock down, and by designing a study whose most likely outcome is not the one my community hopes for. You should read the rest sceptically anyway.

What people report

A large, internationally distributed group of people — who mostly refer to themselves as targeted individuals, abbreviated TI — report being subjected to sustained, deliberate, technologically mediated harassment. The reported experiences cluster into a few recognisable categories:

- **Auditory experiences without an external acoustic source.** Voices, sometimes conversational and responsive, sometimes abusive or commanding, frequently described as perceived inside the head rather than in the room. Within the community this is called V2K, short for voice-to-skull — a term borrowed from a declassified US Army glossary entry. This is the phenomenon the present study addresses.
- **Bodily sensations.** Localised heat, pressure, vibration, sudden pain, or forced muscle movement, often reported as coinciding with the voices.
- **Sleep disruption.** Waking at the same time nightly, being woken repeatedly, or being unable to reach sleep, described as externally imposed rather than as insomnia.
- **Cognitive intrusion.** Thoughts experienced as inserted, or the sense that one's thoughts are being read and responded to.
- **Organised social harassment.** Coordinated following, workplace interference, and repeated encounters interpreted as staged. This is often called gang stalking and is reported both with and without the perceptual phenomena above.

The attributions people give vary widely: intelligence agencies, military research programmes, organised crime, private contractors, or an unspecified network. The attribution is not what this study examines. What this study examines is whether the auditory experiences correspond to anything measurable, and if so, what kind of thing.

How many people, and who

There is no reliable prevalence figure, and anyone who gives you one is estimating. What can be said with more confidence:



- Online communities dedicated to the experience run to hundreds of thousands of accounts across platforms, with the largest single forums in the tens of thousands of members. Account counts are not people, and this is an upper bound of a very soft kind.
- The population is international, spans a wide age range, and includes a substantial proportion of people with higher education and professional histories — a point often raised within the community as evidence against a purely clinical account, and one that should be treated cautiously, since neither education nor prior competence is protective against the conditions in question.
- The best-known formal study of the population is Sheridan and James's 2015 analysis of self-reported cases, which examined narrative accounts and concluded that the reports were consistent with delusional belief. It is a real piece of work and it is the paper you will be pointed to. It is also, as its authors note, based on self-report rather than on any physiological measurement — which is precisely the gap this proposal is aimed at.
- Since then the population has grown substantially and the literature has not. There is no instrumented study of this group at all. That is a strange state of affairs for a phenomenon reported by this many people, and it is the strongest single argument for doing the work.

The mainstream clinical account, stated properly

The standard psychiatric reading is that these reports describe auditory verbal hallucinations occurring alongside a persecutory delusional system, most often within the schizophrenia spectrum or delusional disorder. The account is coherent and it explains a great deal:

- **Auditory verbal hallucinations are common** and well characterised. They occur in perhaps 60 to 80 per cent of people with schizophrenia, and also in around 5 to 15 per cent of the general population who never come to clinical attention at all.
- **The content matches.** Running commentary, second-person address, command phenomena and abusive content are all typical of clinical AVH and are all typical of what this population reports.
- **The technological attribution follows the era.** Persecutory delusions have historically taken the form of whatever technology is culturally salient — mesmerism, radio, television, satellites, implants, and now directed energy. This is a well-documented pattern and it is a genuinely strong argument.
- **There is a mechanism with evidence behind it.** The source-monitoring account holds that inner speech is misattributed to an external agent because the neural tagging that marks self-generated content is impaired. Part 2 of this document is built on the measurement that tests exactly this, and the finding it rests on has replicated for a quarter of a century.

A partner considering this study should assume that the clinical account is the most probable explanation for most of this population, because on the current evidence it is. Nothing in this proposal argues otherwise. What the proposal argues is that "most probable" and "tested" are different states, and that the second one is achievable for a few thousand euros.



Why the community rejects that account

Stated as its holders would state it, because a partner will encounter these arguments and should have heard them in a fair form:

- The experiences frequently begin abruptly in mid-life, without prodrome and without the functional decline typically seen in first-episode psychosis.
- Many report intact functioning in every other domain — employment, relationships, finances — which sits awkwardly with a florid psychotic illness.
- Antipsychotic medication is reported by many to have no effect on the voices, though this is self-report and adherence is not verified.
- The technology being described is not entirely fictional, which distinguishes it from a delusion about, say, thought-broadcasting via dental fillings. That point deserves its own paragraph.

The physics, neither dismissed nor overstated

The microwave auditory effect is real, replicated and uncontroversial. Pulsed radiofrequency energy absorbed in tissue causes rapid thermoelastic expansion, which launches an acoustic wave that reaches the cochlea and is perceived as a click or buzz. It was described by Allan Frey in 1961, has been reproduced many times since, and is standard material in radiofrequency bioeffects literature.

What is not established is the leap from that to sustained intelligible speech delivered covertly at a distance. The energy deposition required, the thermal limits of tissue, the beam control, and the modulation problem are all serious obstacles, and no peer-reviewed demonstration of covert intelligible speech transmission at range exists in the open literature. Demonstrations that do exist involve short phrases, close range, and laboratory conditions.

The Havana syndrome episode brought institutional attention to pulsed radiofrequency as a candidate mechanism for a cluster of reported injuries among diplomatic staff; assessments diverged substantially over subsequent years, and it is not a settled matter in either direction. It is mentioned here only to establish that the general class of question has been taken seriously by serious bodies, not as evidence for anything.

The honest position on mechanism. A partner does not need to hold any view about whether covert transmission is possible in order to participate in this study, and that is deliberate. The design measures brain state and self-monitoring function. It does not measure transmissions and cannot. If the mechanism question is what interests you, this is the wrong study. If the question of what is actually happening in these people's auditory systems interests you, this is the right one, and it is answerable.

Why this is worth an academic's time

Four reasons, in descending order of how much I would expect them to move a department head.



1. **There is a genuine empirical gap.** A population this large with zero instrumented studies is unusual. The corollary discharge paradigm has been applied to clinical AVH extensively and to this group never once. The study is cheap, the measurement is well-validated, and the result is publishable whichever way it falls.
2. **The methodological problem is interesting in its own right.** How do you study a contested experience in a population that distrusts researchers, without a design that can only confirm what you already believe? Solving that has application well beyond this topic — to contested illness generally, to vaccine injury claims, to any domain where lived report and clinical interpretation diverge. A group that publishes a credible answer to that has a contribution that outlives the specific finding.
3. **The population is under-served and the harm is real regardless of cause.** Whether the voices originate internally or externally, these are people in serious distress who are largely disengaged from clinical services because they experience those services as dismissive. A study that treats them as research participants rather than as patients to be corrected is, at minimum, a route back into contact with care.
4. **The design is unusually defensible.** Pre-registered, blinded, powered, with a clinical comparison group and a commitment to publishing the null. The reputational risk of association is lower than it first appears, because the protocol is built to survive exactly the scrutiny it will attract.

What a partner would actually be signing up for

We would be asking for	We would not be asking for
Ethical review through your institution's committee, and the standing that comes with it.	Funding. The programme is separately funded and the figures are in Part 18.
Methodological challenge to the protocol before pre-registration — the harder the better.	Endorsement of any claim about the origin of the phenomenon. No such claim is made anywhere in this document.
Help recruiting a clinical comparison group of people with an established explanation for their voices.	Clinical responsibility for participants. Referral pathways are separate and defined.
A named methodological co-author on the resulting papers.	Lab space, staff time in volume, or equipment. The instrument ships to participants.
Advice on the statistical plan and the analysis, ideally with a veto over anything that overreaches.	Agreement with the community's interpretation of their experiences.

The last row is the important one. A partner who thinks the most likely finding is reduced corollary discharge suppression — that is, the pattern associated with internally generated voices — is exactly the kind of partner this study needs, because that person will make the design harder to fool. The study is not looking for allies. It is looking for a method that both sides would accept before the data comes in.



Part 2 — What an EEG can actually see

EEG stands for electroencephalography, which is a long word for a simple thing: small metal discs on the scalp, wired to an amplifier, recording voltage changes. When large groups of brain cells fire in a loosely synchronised way, the electrical fields they generate add up and a very small fraction of that reaches the skin. We are talking about signals measured in microvolts — millionths of a volt. A single AA battery is 1.5 volts. A brain response is roughly a millionth of that.

Three consequences follow from that, and they shape everything else.

5. **EEG sees the surface of the brain best, and the depths badly.** The signal comes overwhelmingly from the outer layer, the cortex. Deep structures are largely invisible.
6. **EEG has superb timing and poor geography.** It can tell you what happened at 96 milliseconds versus 118 milliseconds. It cannot tell you reliably which cubic centimetre of tissue produced it — not with eight sensors, and not really with sixty-four either without a lot of extra machinery.
7. **A single brain response is buried in noise.** You cannot see it in one go. You have to do the same thing a hundred times and average the recordings together. The random noise cancels itself out; the consistent response survives. That averaged waveform is called an **event-related potential**, or ERP. Everything in this design is built on ERPs.

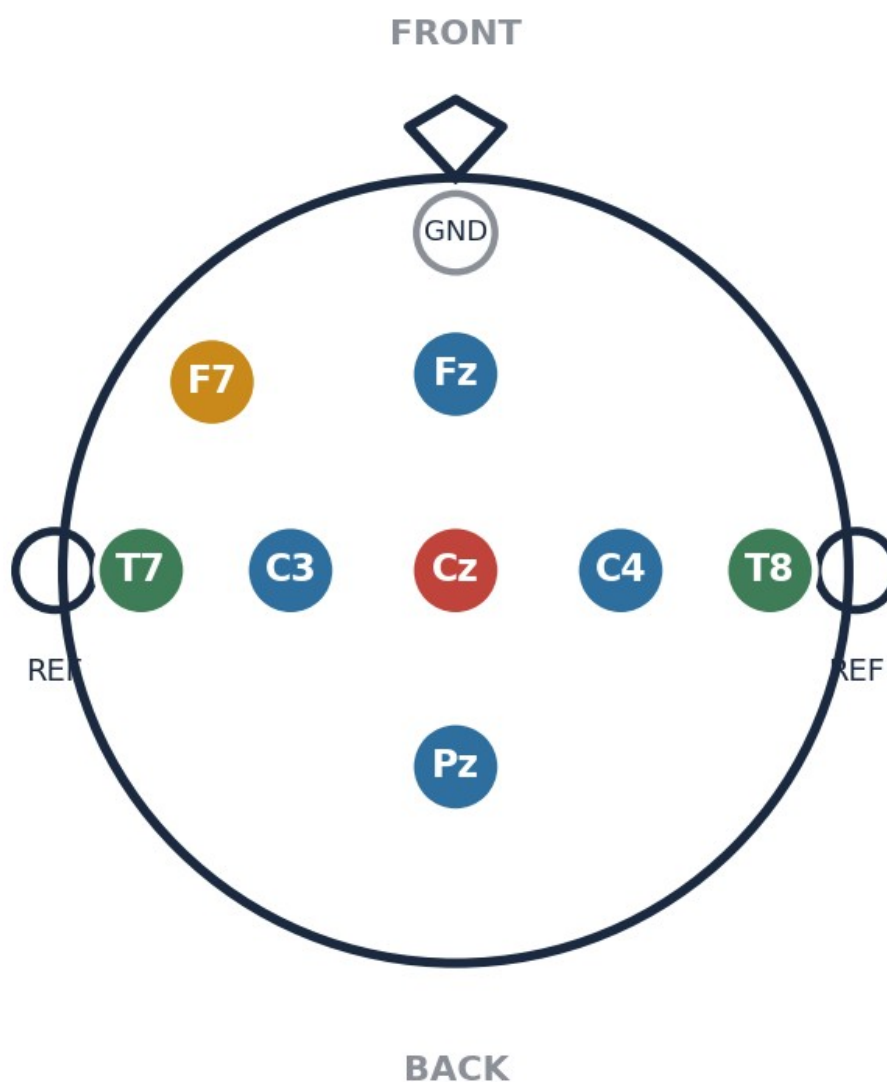
The third point is worth dwelling on, because it explains why every block in this experiment is repetitive and slightly boring. Boring is a design requirement. You need roughly sixty to a hundred usable repetitions of each condition before an average becomes trustworthy.

The one measurement everything hinges on: the N100

When any reasonably abrupt sound reaches your ears, the electrical trace at the top of your head does something characteristic: about a tenth of a second later, it dips. Because the dip is a negative-going voltage and it happens around 100 milliseconds, it is called the N100 — N for negative, 100 for the timing. In older papers you will see it written N1. Strictly the two are not synonyms — N1 names the first negative deflection whatever its latency, which shifts with the stimulus — but for the abrupt sounds used throughout this design the first negative deflection falls at about 100 milliseconds, so the labels coincide here.

The N100 is the workhorse of auditory brain research because it is large, reliable, appears in essentially everyone, and sits right on top of the head where a cheap electrode can reach it. It is not a "meaning" response. It does not care what the sound means. It is closer to a registration signal: the hearing system noting that something arrived.

Figure 1 — Where the eight sensors go



- Cz — main recording site for the N100 'surprise dip'
- Midline + central — spread of the same response
- T7 / T8 — sit over left and right hearing regions
- F7 — over the left-front speech-planning region
- REF (earlobes) = baseline; GND = electrical ground

Figure 1 — The eight-sensor layout. The names are from an international standard (the "10–20 system") that lets any lab reproduce any other lab's placement. Cz sits at the crown of the head and is where the N100 is largest.



A short glossary, because the jargon is unavoidable

Everything technical in this article is defined below. Skip it and come back if a term ambushes you.

Term	What it actually means
EEG	Electroencephalography. Metal discs on the scalp reading tiny voltage changes caused by brain activity.
Electrode / channel	One metal disc and the recording it produces. An "eight-channel" system records from eight places at once.
Wet vs dry electrode	Wet electrodes use conductive gel between metal and skin. Dry ones skip the gel and are far noisier. For this study, wet is mandatory.
Impedance	How much the skin resists the signal getting through. Low is good. You check it before recording; if it is high, you re-gel.
Reference / ground	EEG measures a difference between two points. The reference (usually the earlobes) is the "compared to what". The ground stabilises the whole circuit.
Microvolt (μV)	A millionth of a volt. Brain responses are a few to a few dozen microvolts.
Sampling rate	How many times per second the amplifier takes a measurement. 250 per second is ample here.
ERP	Event-related potential. The averaged brain response to a repeated event, extracted from noise by repetition.
N100 (N1)	The negative dip in that average about 100 ms after an abrupt sound starts. Strictly, N1 is the first negative peak at whatever latency the stimulus produces; for the stimuli used here the two coincide. The main measurement in this study.
Epoch	One short slice of recording cut around one event — here, from 200 ms before to 600 ms after.
Artifact	Non-brain electrical junk in the recording: blinks, jaw clenching, cable movement, mains hum.
EOG	Electrooculography. An extra electrode near the eye that records blinks so they can be subtracted out.
ICA	Independent component analysis. A mathematical technique that separates mixed signals into sources, used here to strip blinks out.
EMG	Electromyography. The same kind of electrode, but placed over muscle, reading muscle activity instead of brain activity.
Efference copy	A copy of a motor command sent to sensory areas as advance warning. The brain's internal spoiler alert. Also called corollary discharge.
Readiness potential	A slow voltage drift over the front of the head that begins one to two seconds before a self-initiated action.
AVH	Auditory verbal hallucination — the clinical term for hearing speech with no external source. Used here as a neutral label for a comparison group, not as a diagnosis of



Term	What it actually means
	anyone.
Effect size (Cohen's d)	How big a difference is, relative to how much people vary. 0.2 small, 0.5 medium, 0.8 large.
Statistical power	The chance your study detects a real effect if one exists. Below 80% you are mostly gambling.
Pre-registration	Publishing your full analysis plan before collecting data, so you cannot quietly change the target after seeing the result.
Blinding	Making sure the person analysing the data does not know which group each file came from.

Part 3 — Test One: the self-generated sound discount

This is the centrepiece, and it deserves the most careful explanation, because if you understand this one measurement you understand why the whole study is worth running.

The idea

Your brain does not passively receive the world. It predicts it. Every time you issue a motor command — move an arm, blink, speak — a copy of that command is routed to the sensory regions that are about to be affected. That copy is the efference copy. Its job is to say: whatever you are about to feel, I caused it, so do not get excited.

This is why you cannot tickle yourself. The touch is real, the nerve endings fire, but the sensory response is damped because the system was warned. The same mechanism operates in hearing. When you speak, your auditory cortex receives advance notice that a specific sound is coming, and its response to your own voice is reduced.

You can measure this directly, and the measurement is elegant in its simplicity. You record a person saying a syllable out loud. Then you play that exact recording back to them. Physically, the sound hitting the eardrum is nearly identical in both cases. But the N100 is markedly smaller when they are speaking than when they are listening to the playback. The difference between the two is the size of the discount their brain applies to its own output.

Figure 2 — The self-generated sound discount

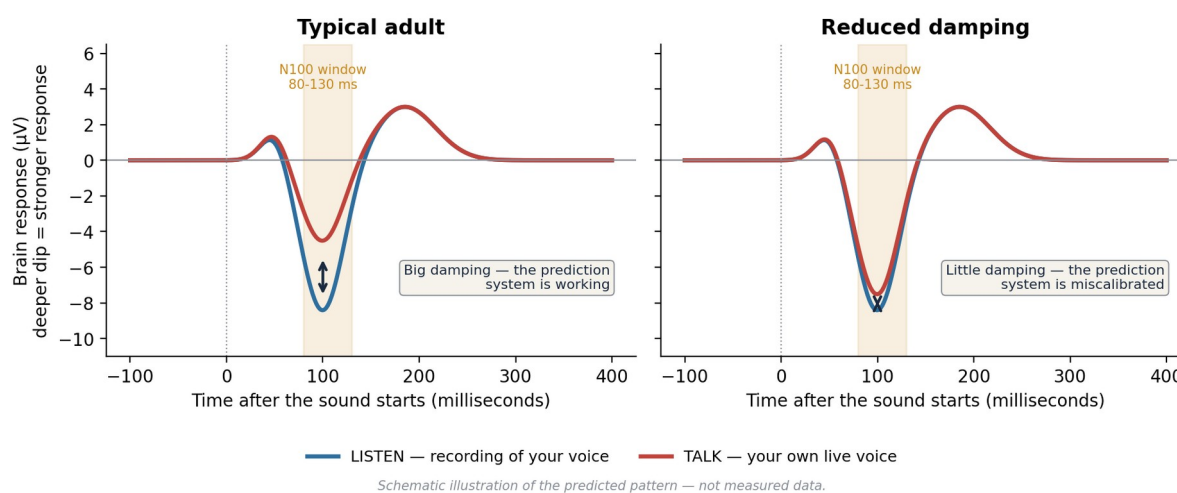


Figure 2 — Left: the normal pattern. The same sound produces a much smaller brain response when you generate it yourself, because the prediction system warned the hearing regions in advance. Right: what a miscalibrated prediction system looks like — the discount is much smaller.

Why this bears on voices

For twenty-five years, one of the strongest findings in hallucination research has been that in people who hear voices, this discount is reduced. Their brain fails to apply the full "this was me" tag to self-generated speech. The leading explanation is straightforward: if your own inner speech is not being

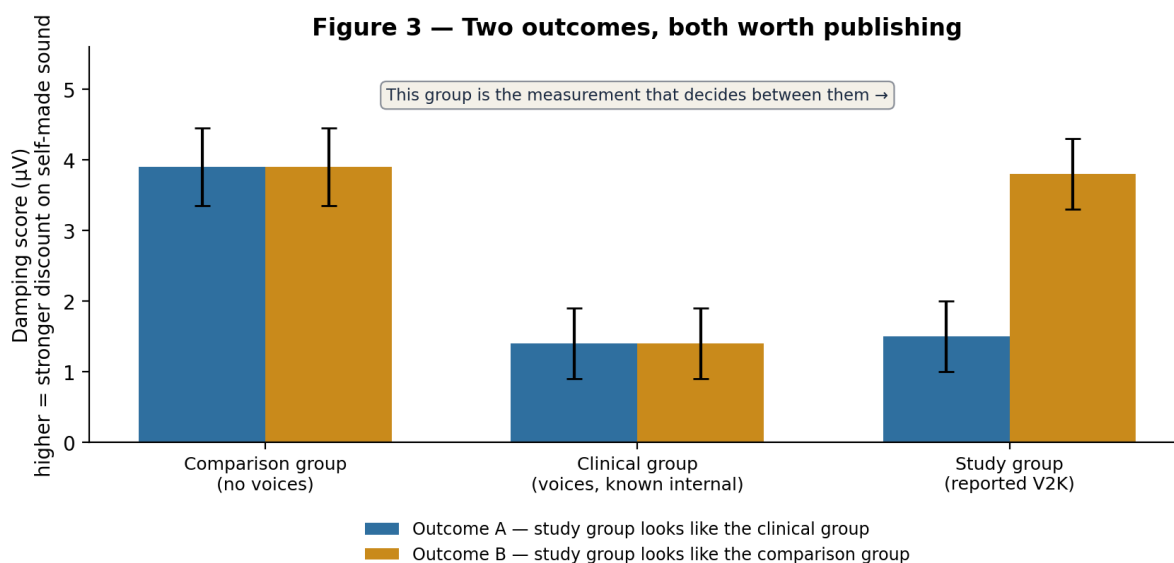


properly labelled as yours, it arrives in consciousness untagged, and an untagged voice feels like it came from somewhere else.

This gives us a genuine, discriminating measurement. Not a measurement of transmissions — a measurement of whether the brain’s internal authorship-tagging system is working.

What each result would mean

Here is why this is a real experiment and not a ceremony. Both outcomes are informative, and I do not know in advance which one you would get.



Illustrative values showing the two competing predictions — not measured data.

Figure 3 — The two competing predictions. Whichever way the study group falls, you learn something you did not know before.

Result	What it would mean	Why it matters
Study group shows a reduced discount, like the clinical comparison group	The authorship-tagging mechanism is miscalibrated in this group too.	Supports the internal-generation account. Not proof — but it makes the standard explanation harder to dismiss.
Study group shows a normal discount, like the no-voices comparison group	The tagging mechanism is intact. These people’s brains correctly label their own speech as their own.	This is the genuinely surprising outcome. It would mean the standard misattribution model does not fit this group, and would demand an explanation nobody currently has.
Study group sits between the two, or splits into subgroups	The population is not homogeneous.	Arguably the most likely and most useful outcome. It would say the label covers more than one phenomenon and they should be studied separately.

That third row deserves emphasis. Anyone who has looked at self-reported populations at scale knows they are rarely uniform. A result that cleanly separates a heterogeneous group into distinguishable subgroups on an objective physiological measure would be, in research terms, a



substantial contribution — and one that helps the people involved far more than any amount of argument about the population as a whole.

Version 4 note: following review, the discount score is treated as a continuous measure throughout, not only as a group mean. The pre-registered analysis includes regressions of the score against symptom frequency, medication status and demographics, alongside the cluster analysis of Part 10 — because a distribution of discount values is itself a finding, whichever way the group means fall.

How to run it

- **TALK block.** The volunteer says a short "ah" roughly a hundred times, at their own pace, about two seconds apart. A microphone records every one.
- **LISTEN block.** That recording is played back through the same headphones at the same level. Same sounds, same person, same ears — the only difference is who caused them.
- **Measurement.** Average both blocks, find the N100 at electrode Cz in the window 80 to 130 milliseconds, and subtract. That difference is the discount score.
- **Order.** Counterbalance across volunteers — half do TALK first, half do LISTEN first — so that fatigue and gel drying do not systematically favour one condition.

The single most common way to ruin this experiment: timing. You are measuring a dip that lasts fifty milliseconds. If your computer does not know exactly when each sound started, to within a few milliseconds, the averaging smears the response into nothing and you will conclude, wrongly, that there was no effect.

Wireless timestamps are not good enough. Bluetooth introduces variable delays of tens of milliseconds. The fix is to route the actual audio signal into a spare analogue input on the amplifier, so the sound and the brain response are recorded on the same clock. Before you test a single volunteer, verify the whole chain with an oscilloscope loop-back. If you skip this step, nothing else in the study matters.

Part 4 — Test Two: is the hearing system actually busy?

Test One asks about a mechanism. Test Two asks a blunter question: at the moments a person says they are hearing a voice, is their auditory cortex in a different state than at moments when they say they are not?

The logic borrows from a simple principle. If a system is already occupied, it responds less strongly to a new demand. So if the hearing regions are genuinely engaged during a reported voice episode — whatever is engaging them — an unrelated beep played during that episode should produce a smaller N100 than the same beep played during a quiet period.

The volunteer holds a button down whenever a voice is present and releases it when it stops. Meanwhile the computer plays a plain 1000 Hz tone at random intervals of three to eight seconds.

Afterwards you sort the tones into two piles — those that landed inside a reported voice period and those that landed outside — and compare the averages.

Figure 4 — Testing whether the hearing system is genuinely busy

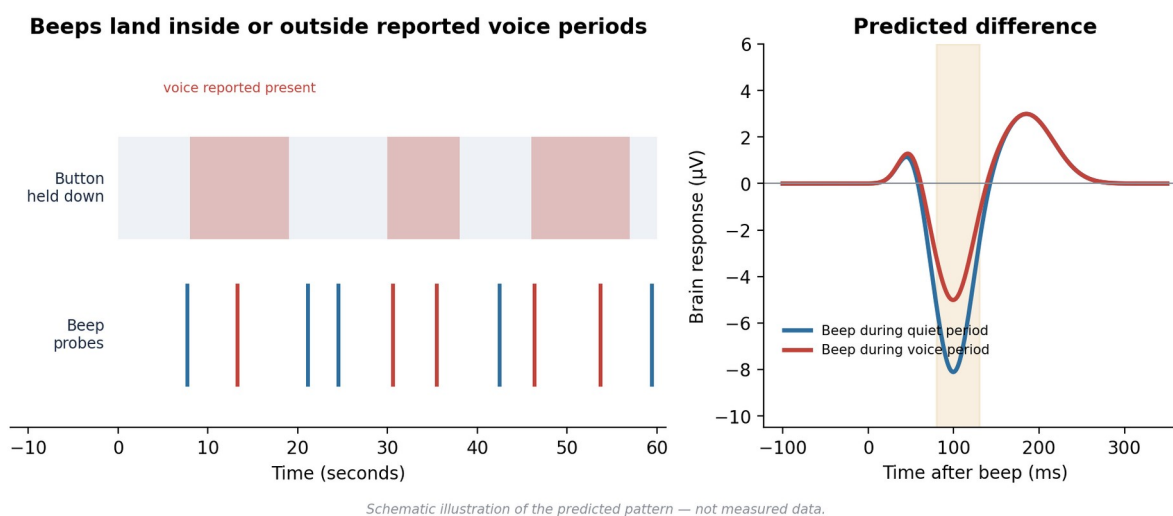


Figure 4 — Left: probe tones are scattered at random, so some fall inside reported voice periods and some outside. Right: if the hearing system is genuinely engaged during those periods, the response to the probe should be smaller.

Practical points that decide whether this works

- Only recruit volunteers who report voices frequently. If someone hears a voice twice a week, a twenty-minute block will yield nothing. You need enough episodes to accumulate sixty usable probes in each pile.
- Allow slack around the button presses. Nobody presses a button the instant an experience begins. Treat the reported onset as approximate and exclude probes falling in a window from three seconds before to one second after each press, so that ambiguous cases do not pollute either pile.
- Record a button-only control block with no probes, to check that the act of pressing does not itself produce artifacts that could be mistaken for a result.
- Keep the tone boring. A plain sine tone at a comfortable level. Anything startling introduces its own responses.

An important caveat: a positive result here does not identify a cause. Attention alone can reduce the N100. If someone is concentrating hard on an internal experience, their response to an external beep will drop. That is exactly what you would expect from an internally generated experience too. So Test Two establishes that the reports correspond to a real change in brain state — which is not nothing, and which has never been properly established for this population — but it does not, on its own, say where the change comes from.

One precision, prompted by review: "is the hearing system busy" is loose language, because if a person perceives a voice then some level of the auditory system is by definition active. What the probe-tone N100 indexes is the early cortical stage of the pathway laid out in Part 16b — the response of primary auditory cortex to a fresh external input — not the cochlea below it and not the



interpretive stages above it. The question Test Two actually asks is therefore whether that specific stage is already occupied during reported episodes. Which stage an experience engages is exactly what separates the outcomes in Part 16b, and it is the reason the pathway logic there, not this test alone, carries the interpretive weight.

When the voices say nothing: provoking a reply

There is a gap in the paradigm as described so far. It assumes episodes occur on their own during a twenty-minute block, and for many participants they will not. A recording session is an unusual, self-conscious hour, and several people have made the same observation: the voices may simply be quiet, but they can often be drawn out by addressing them aloud.

That observation is worth taking seriously as a design input rather than as an anecdote, because it converts Test Two from something observational into something experimental. Instead of waiting for an event and hoping, you elicit one at a moment you control. Everything that follows in this section depends on that shift.

It also introduces an obvious problem, which is that speaking aloud produces a great deal of electrical noise. The rest of this section is about whether that noise can be worked around on eight channels. The short answer is that it can, and not because the artifact is small, but because it is over before the interesting part starts.

What speaking actually does to the recording

Three artifacts, with different characters:

- **Muscle activity from jaw, lip and neck.** Very large — hundreds of microvolts against an N100 of five to ten. It is broadband and dominates above 20 Hz, which is one more reason the design already caps analysis at 30 Hz. Crucially, it stops within roughly 200 to 500 milliseconds of the mouth closing.
- **The tongue potential.** The underrated one. The tongue is a standing electrical dipole, and moving it drags a slow potential across the scalp. It sits below 4 Hz, is largest at frontal electrodes, and can persist for a second or more after speech ends.
- **Head and cable movement.** Low frequency, and largely handled by a properly seated headband and a secured cable.

The decisive point is that all three are time-locked to the speech, and the phenomenon under test is time-locked to what follows it. If replies arrive a second or more after the participant stops talking, the artifact has decayed before the analysis window opens. You do not need to clean the speech period. You need to step over it.

Figure 4a — Provoking a reply, and stepping around the artifact

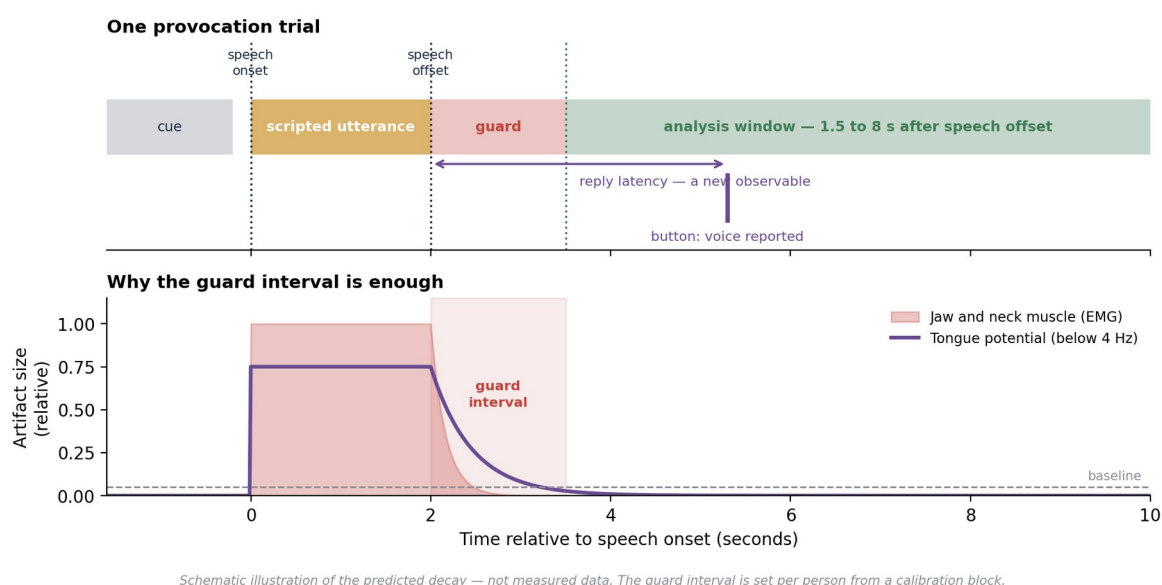


Figure 4a — The trial structure, and the artifact decay that makes it workable. The guard interval is the whole trick: nothing inside it is analysed, and by the time it ends the muscle and tongue artifacts are back at baseline.

Two further points make this more comfortable than it first sounds. The montage helps: the tongue potential is maximal frontally and falls away toward the crown, so Cz — where the N100 lives — is among the least affected sites on the head. F7 will be badly contaminated during these blocks and should be treated as an artifact monitor rather than a signal channel.

And the corollary discharge literature underlying Part 3 measures brain responses during active speech, and has done so reliably since the late 1990s. Measuring a window that begins after speech has stopped is strictly easier than what is already routine.

Setting the guard interval per person, not globally

A fixed guard would silently favour participants who hold less tension in their jaw. So the session opens with a two-minute calibration block: speak a line, then hold still and silent for five seconds, twenty times. Measure how long that individual's muscle activity and low-frequency drift take to return to baseline, and set their guard interval to the larger of 1,500 milliseconds or their own 95th-percentile recovery time. The number is recorded in the session metadata and reported in the paper.

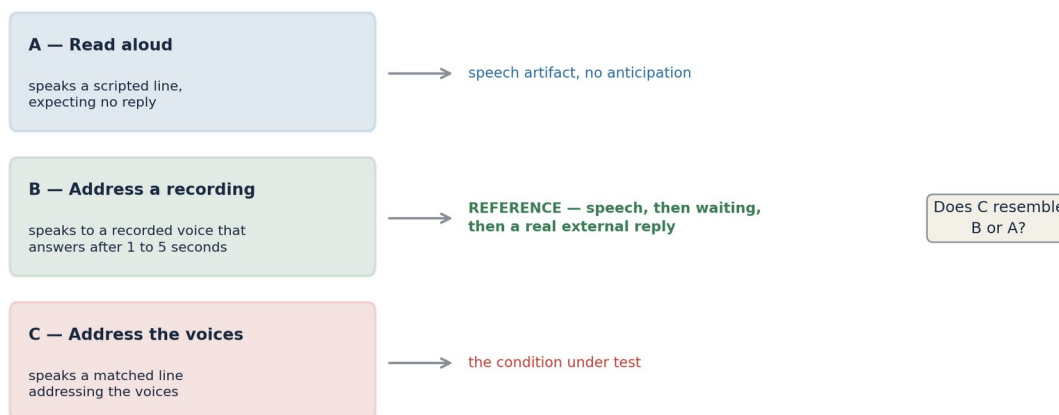
The confound that actually threatens this

It is not the artifact. It is expectancy.

When you address the voices and wait for an answer, your brain produces a slow anticipatory shift that appears whenever someone expects an imminent event. It looks like a real effect, it sits in exactly the window of interest, and it would appear identically whether or not anything answered. A design that compared “spoke and waited” against “sat quietly” would find a difference every time and it would mean nothing.

So the paradigm needs conditions matched on speech and on anticipation, differing only in what follows — the same structure that makes Test Three work.

Figure 4b — The three provocation conditions



All three interleaved at random, with identical scripted phrases matched for length and prosody.

Figure 4b — Three conditions, interleaved at random. Condition B manufactures ground truth: the participant's own brain, speaking and then waiting for a voice that certainly is external.

Condition	What the participant does	What it gives you
A — Read aloud	Speaks a scripted line on a cue, with no expectation of any reply.	The speech artifact profile and the motor activity, with anticipation removed.
B — Address a recording	Speaks the same line to a recorded interlocutor, which answers after a variable delay of one to five seconds.	The reference pattern: speech, then waiting, then a genuinely external reply arriving at a time the system knows exactly.
C — Address the voices	Speaks a matched line addressing the voices, then presses the button if a reply comes.	The condition under test.

Script the utterances. Fixed phrase sets matched across conditions for syllable count, duration and prosody, so that the motor and auditory activity is as close to identical as it can be and the only thing that differs is who is being addressed. Free improvisation would let the speech itself vary between conditions, and the two would never be untangled afterwards.

Block structure

- Trial: cue, scripted utterance of about two seconds, ten seconds of silence, button press if a voice occurs, then a jittered gap of three to six seconds.
- Roughly five trials per minute, so a twenty-minute block yields about a hundred trials and around sixty surviving artifact rejection.
- Conditions A, B and C interleaved trial by trial rather than run in blocks, so that arousal, fatigue and gel drying cannot align with condition.



- Catch trials in which the participant speaks but is instructed to address nobody, appearing unpredictably.
- The primary contrast is within Condition C: trials where a reply was reported against trials where none was. Same speech, same expectancy, differing only in outcome. It is the cleanest comparison in the extended design.

Where the microphone fits, and what the server can and cannot fix

This is worth stating precisely, because it is easy to assume that recording the voice and the EEG into the same place solves the problem. It solves most of it. It does not solve the timing.

Timing is not solved on the server. If the microphone is a separate stream — browser audio capture, a phone, anything with its own clock — then aligning it to the EEG afterwards means correlating two independently sampled sources, and that carries tens of milliseconds of uncertainty. Tolerable for a one-second guard interval; fatal for anything measured in an evoked-response window. The fix is the one already built into the instrument in Part 14: the microphone envelope is digitised on the same converter as the brain signal, so speech offset arrives as another column of the same sample row. Nothing has to be correlated because nothing was ever separate.

Everything else is solved on the server, in one place. Once that co-registered stream lands, every step of the compensation happens against a single file with a single timeline.

Step performed on the server	What it operates on
Detect speech onset and offset	The envelope channel, accurate to the sample.
Set the guard interval for this person	Envelope and muscle channels from the calibration block.
Remove residual speech artifact	The rectified muscle envelope, regressed out of the EEG channels — more reliable at eight channels than component-based separation.
Cut the post-speech analysis window	Sample index arithmetic, nothing more.
Build the reply latency distribution	Speech offset against the button channel, from the same rows.
Contrast reply against no-reply trials	All of the above, in one recording.

Two microphone paths, not one

They do different jobs and have different requirements, and conflating them is how the timing gets lost.

- **The envelope, into a converter channel.** Low bandwidth, sample-accurate, carries no intelligible content. This is the timing anchor and the only one the analysis depends on.
- **Full-bandwidth audio, streamed as a separate track.** It does not need millisecond precision, because it can be aligned to the envelope afterwards by cross-correlation. It is what lets you verify the participant actually spoke the scripted line, measure vocal amplitude as a covariate, and confirm they did not speak during the silent window.



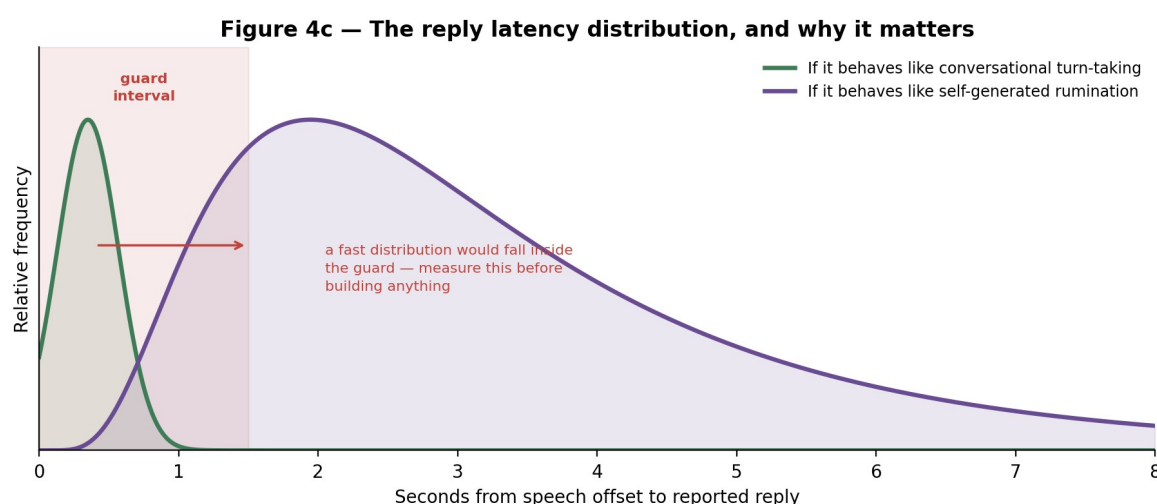
The second track also buys an acoustic negative control, which no amount of EEG can provide. If a real sound occurred in the room at the moment a voice was reported — a neighbour, a television, a phone — the microphone recorded it. Being able to state that reported episodes were not accompanied by any detectable acoustic event closes an obvious objection before a reviewer raises it.

The privacy cost of the second track is real and has to be designed down. Continuous room audio is the most re-identifying data anywhere in this study — more than the EEG, more than the metadata. It is someone's home for two hours, and it may capture family members who never consented to anything. So: retain full audio only for the scripted utterance windows and a couple of seconds either side. For the silent intervals, keep the envelope and a spectral summary sufficient to detect that an acoustic event occurred, and discard the waveform at ingest rather than at some later cleanup. That preserves the negative control while ensuring the recording which reaches storage cannot reconstruct a conversation. It also matters for consent: “we record what you say during the test phrases, and otherwise only whether a sound occurred” is a sentence a participant can evaluate, which “we record your room” is not.

What this unlocks, and what could sink it

Provocation is not merely a workaround for quiet sessions. It produces an observable the passive design cannot.

With a known provocation time you get a measurable interval from speech offset to reported reply, across roughly sixty trials per participant. That is a distribution, not an anecdote, and it is diagnostic in an interesting way. Natural conversational turn-taking clusters tightly around a fifth of a second with a narrow spread. Internally generated rumination is slower and far more dispersed. Where this distribution actually falls is an empirical question nobody has asked, and the answer would be informative regardless of which way it came out.



Two competing predictions, drawn schematically. Which one is right is an empirical question nobody has asked.

Figure 4c — The reply latency distribution. Note the practical consequence: if replies typically arrive in under half a second, they fall inside the guard interval and the paradigm as described would not work.



It also allows proper time-locked averaging. Instead of averaging on a lagged button press, you average on speech offset — a hardware-precise timestamp on the same clock as the brain signal. That alone recovers a great deal of the sensitivity that a delayed subjective report throws away.

Two honest risks

- **If replies typically arrive in under half a second, the paradigm breaks.** The guard interval would swallow the response. This is cheap to check before building anything: collect button-press latencies alone, with no EEG at all, and look at the distribution. If it is fast, a different artifact strategy is needed — probably accepting the speech period and doing full muscle regression across it. Measure this first.
- **Demand characteristics get worse, not better.** The participant is now being asked explicitly to produce the phenomenon. Blind the analysis to condition, keep the catch trials, and pre-register the primary contrast before any data is seen.

One line for the pre-registration. State in advance that a null result here — provocation produces no differentiable post-speech signature — is a publishable finding rather than a failed block. Without that commitment the temptation to keep adjusting the guard interval until something appears is considerable, and that is precisely how a good paradigm turns into a garden of forking paths.

Part 5 — Test Three: talking back without speaking

This is the part of the design I find most interesting, and it came out of a good question: if people report replying to the voices in their heads without moving their mouths, would that show up as something distinguishable from ordinary inner monologue?

The answer is yes, probably, and on more than one channel at once. Not because we can read the words — we cannot, not with this equipment — but because a deliberate act of silent speech directed at something has structural features that idle thinking does not.

There are four such features, and the design tests all four. Together they are considerably more informative than any one alone.

3a. The three conditions

The whole test rests on a comparison that gives you a known reference pattern. That reference is the crucial design element, and it is what makes this more than hand-waving.

Figure 5 — The three talking-back conditions

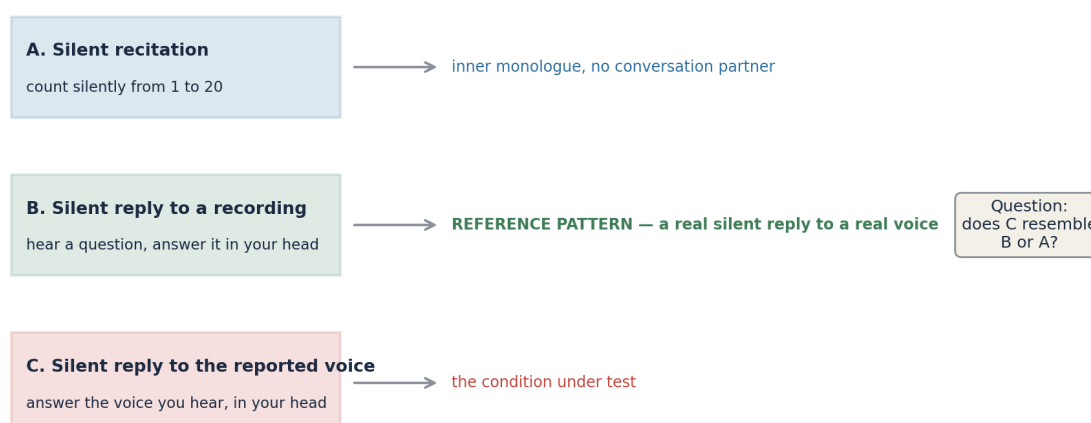


Figure 5 — Condition B is the key. It gives you a recording of what this person's brain looks like when they silently answer a voice that definitely exists outside their head. That becomes the yardstick against which Condition C is measured.

Condition	What the volunteer does	What it gives you
A — Silent recitation	Counts silently from one to twenty, or recites something memorised, on a cue.	The signature of pure inner monologue: language production with nobody on the other end.
B — Silent reply to a recording	Hears a short recorded question through headphones and answers it in their head, without moving their lips.	The reference pattern. A genuine, addressed, silent reply to a real external voice — with the timing of the question known exactly.
C — Silent reply to the reported voice	Replies in their head to the voice they report hearing, pressing a button at the moment they begin their reply.	The condition under test.



Condition B is not filler. It solves the hardest problem in this kind of work, which is that you have no ground truth for Condition C. By constructing an artificial version of exactly the thing under test — same person, same session, same electrodes, same act of silent replying, but with a stimulus you control and timestamp perfectly — you manufacture your own ground truth. Then the question becomes tractable: does C look more like B or more like A?

3b. The muscles that move even when you are silent

When most people speak silently in their heads, the muscles used for speech do not stay completely still. There is faint activation of the lips, tongue, jaw and larynx — far too small to produce sound or to be visible, but well within reach of an electrode. This is called subvocalisation, and it is the basis of the "silent speech interface" research that has been going on for years, in which people control devices by mouthing words internally.

You do not need extra equipment for this. Your amplifier has spare inputs. Two small electrodes on the cheek and under the chin, over the muscle rather than the skull, record EMG — muscle activity rather than brain activity.

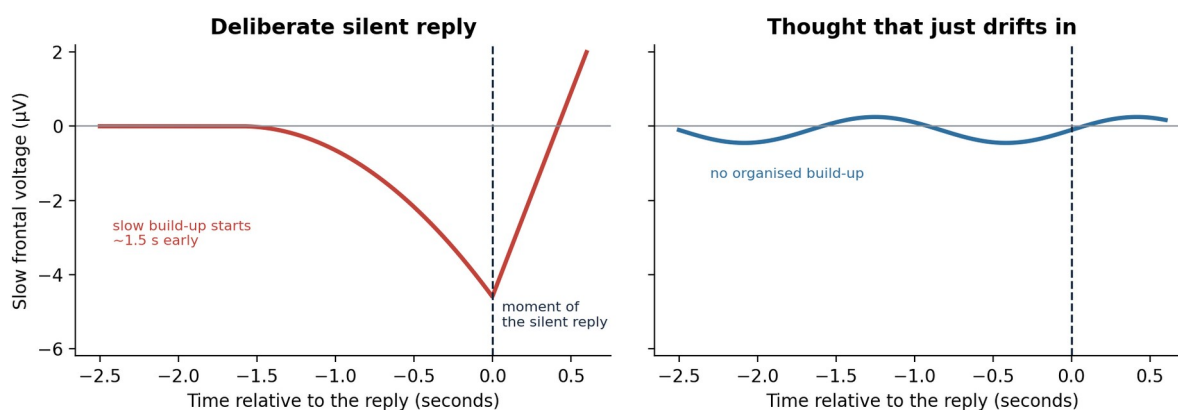
The prediction is that silent replying produces a burst of low-level speech-muscle activity that idle listening does not, and that the timing of those bursts is measurable to within a few tens of milliseconds. That gives you something you otherwise lack entirely: a precise, objective timestamp for when a silent reply happened, instead of relying on a button press the volunteer makes after the fact.

Note the double-edged nature of this. Not everyone subvocalises, and the extent varies enormously between people. An absence of muscle activity does not mean an absence of inner speech. Use this channel as a bonus timing source when it is present, never as a criterion for whether the person is "really" doing the task.

3c. The build-up before a willed act

In 1965 two German researchers discovered that voluntary movements are preceded by a slow negative voltage drift over the front of the head, starting one to two seconds before the movement itself. They called it the Bereitschaftspotential; in English it is the readiness potential. It is one of the more philosophically inconvenient findings in neuroscience, and for our purposes it is extremely useful, because it marks the difference between something you decide to do and something that simply happens.

Figure 6 — The 'getting ready' signal that precedes a willed act



Schematic illustration of the predicted pattern — not measured data.

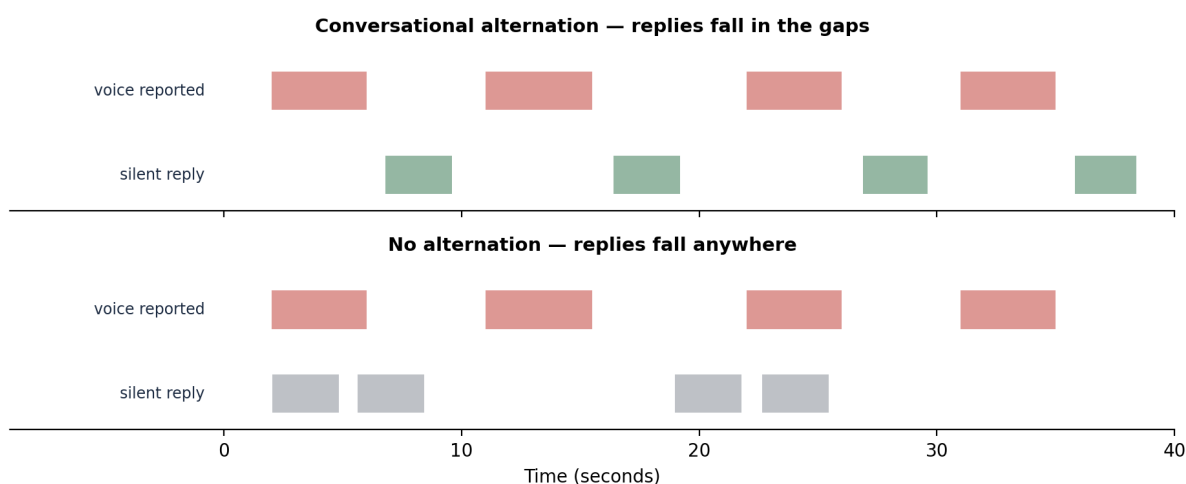
Figure 6 — A deliberate, self-initiated act is preceded by a slow build-up. A thought that arrives unbidden is not. This is the difference between "I chose to say something" and "something occurred to me".

Applied here: if a volunteer is genuinely composing and issuing a reply — an intentional act — you should see this build-up in the seconds before it. If their reported "replies" were something more like intrusive thoughts arriving on their own, you should not, or you should see a different shape. Condition A and Condition B give you that person's own readiness potential for cued and for conversational silent speech, so you have their personal baseline rather than a textbook one.

3d. Turn-taking

Conversation has a shape. Speakers alternate, and the gaps between turns cluster in a fairly narrow band. If someone is genuinely in a back-and-forth exchange, their replies should fall in the gaps between the other party's turns, not scattered at random across them.

Figure 7 — Turn-taking is a testable structure



Schematic illustration of the two competing patterns — not measured data.

Figure 7 — Alternation is a statistical property you can test for, using only the times of the reported voice periods and the reported replies. It requires no decoding of content whatsoever.



The analysis is a straightforward one: take the sequence of reported voice onsets and offsets, take the sequence of reply onsets, and test whether replies cluster after offsets more than chance would predict. You generate a null distribution by shuffling the reply times within the session thousands of times and asking how often the shuffled version looks as structured as the real one. If it looks structured less than five per cent of the time, you have a real alternation pattern.

This is the cheapest test in the entire design — it costs nothing beyond two button channels — and it is one of the more informative, because a person merely imagining a conversation under instruction would have difficulty producing consistent, naturalistic turn-taking timing across a twenty-minute block without knowing that was what you were measuring.

3e. Putting the four channels together

The reason to run all four is that a machine-learning classifier can use them jointly. The procedure is:

8. Train a classifier to distinguish Condition A (recitation) from Condition B (silent reply to a recording), using the EEG, the muscle channels, and the timing features. Confirm it performs well above chance on data it was not trained on.
9. Apply that trained classifier to Condition C, which it has never seen.
10. Ask which category it assigns Condition C to, and how confidently.

This is called cross-condition decoding, and it is a much stronger design than simply comparing averages, because the classifier is defined entirely by data collected before Condition C is examined. There is no room to adjust the criterion after seeing the answer.

If Condition C is classified as...	The reasonable reading
B — replying to an external voice	The person is performing a genuine, addressed act of silent speech, structured like a real reply to a real interlocutor. Something is being replied to, and it is being treated by their nervous system as an other.
A — inner monologue	What is happening resembles self-directed thinking more than conversation. That does not make the experience less real to the person, but it changes what it is.
Neither, with low confidence	The honest and quite likely outcome with eight channels. Report it as such. An underpowered null is not evidence of absence.

The limit you must state out loud. None of this identifies the interlocutor. A person conversing with an internally generated voice is genuinely conversing — the reply is a real reply, the turn-taking is real turn-taking, the readiness potential is real. Test Three establishes whether a dialogue is occurring. It says nothing about what is on the other end. Only in combination with Test One, which probes the authorship-tagging mechanism, does the picture start to constrain interpretations. Anyone who presents Test Three alone as evidence of an external source is overreaching, and I would rather say so here than have it said for me later.



Part 6 — Test Four: the surprise test

One further test is worth attempting, though it is the most technically fragile of the four and should be labelled exploratory from the outset.

The brain responds distinctively to content it did not expect. In language research this shows up as a response about four hundred milliseconds after an unexpected word — for instance, at the end of a sentence about spreading butter on warm socks. The size of that response tracks how badly the prediction failed.

Now consider the implication. If the content of a voice is generated inside the same brain that is hearing it, that brain has no reason to be surprised by it; it wrote the script. If the content genuinely originates elsewhere, it should be as unpredictable as another person's speech, and unpredictable content should produce surprise responses.

The design: after each reported voice episode, the volunteer rates on a simple scale whether the content was expected or unexpected. You then compare the brain activity preceding "unexpected" reports against "expected" ones.

Why this is fragile, stated plainly

- You have no precise timestamp for when the words arrived. A button press lags the experience by an unknown and variable amount, and averaging across variable lags smears out exactly the response you are hunting for.
- The rating is subjective and made after the fact, which invites reconstruction rather than recall.
- Any effect will be smaller than the ones in Tests One and Two, and eight channels are marginal for it.

So why include it? Because it is nearly free once the rest of the session is running, because a clear positive would be genuinely striking, and because a null result costs you nothing provided you declared in advance that it was exploratory. What you must not do is run it, get a marginal result, and then present it as a primary finding. Declare it as exploratory in the pre-registration and report it as exploratory in the write-up.

Part 7 — The hardware

The equipment list is genuinely modest. This is the part of the argument that surprises people: the reason a study like this has not been done is not that the instruments are expensive.

Item	Specification	Indicative cost
Amplifier	Eight channels, 250 samples per second or better, with at least one spare analogue input for the audio timing signal. Research-grade open hardware or a hybrid consumer unit both work.	€800 – 1,200
Electrodes	Wet silver / silver-chloride with conductive gel. Not dry. This is not a preference; dry-electrode noise will destroy the N100.	€100 – 200
Headphones	Closed-back, calibrated to a fixed output level so every volunteer receives the same intensity.	€100
Microphone	Headset or throat microphone on a boom, recorded into a spare analogue input so voice onsets are on the same clock as the EEG.	€60
Muscle electrodes	Two or three surface electrodes for cheek, chin and neck, using the same amplifier.	€40
Response buttons	Two hardware buttons wired to analogue inputs. Not keyboard keys — keyboards add unpredictable delay.	€50
Oscilloscope	For verifying end-to-end timing before you record anything. Borrowable. Non-negotiable.	€0 – 300
Software	Open-source acquisition and analysis toolchains exist for all of this and are the standard in the field.	€0

There is a much cheaper route, and it is the one this programme takes. The prices above are for buying an amplifier. Building one from the same converter chip the commercial units use brings the whole kit to between €230 and €300 in parts before the audio subsystem before the audio subsystem before the audio subsystem — see Part 14. That is not a compromise on quality, because it is literally the same front-end silicon; it is a compromise on convenience, support and the time of whoever assembles it. For a single-site pilot, buy. For a fleet, build.

The one thing you must not do

Do not analyse activity above 30 Hz. There is a substantial literature on high-frequency brain activity and hallucinations, and it is tempting territory. It is also exactly where jaw, neck and scalp muscle activity contaminates the recording most severely, and on consumer hardware in a room that is not electrically shielded, you will find effects that are muscle tension rather than brain activity. Filter the recording to below 30 Hz and stay there. If your result depends on gamma-band activity from an eight-channel headset in a spare room, your result is probably about clenched teeth.

Part 8 — A session, start to finish

Everything above fits into a single visit of a little over three hours, of which about a hundred minutes is actual recording. The rest is preparation, breaks and a proper debrief.

Figure 9 — One session, start to finish (about 3¼ hours)

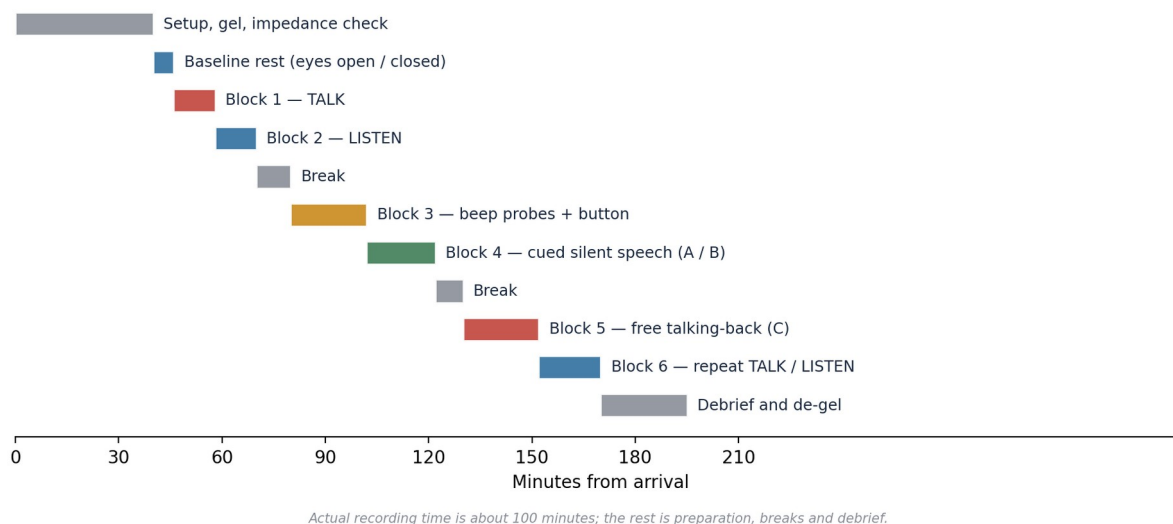


Figure 9 — The full session. Blocks 1 and 2 are repeated at the end as Block 6, which gives you a within-session reliability check: if the measurement is real, it should reproduce ninety minutes later on the same person.

That repeat block is not padding. Reliability is the question nobody asks and everybody should. If your headline measurement does not reproduce within the same person in the same session, it is noise, and you have found that out before publishing rather than after.

Who takes part

Group	Who	Why they are needed
Study group	People who report hearing transmitted voices, screened for frequency so that enough episodes occur within a recording block.	The population of interest.
Comparison group	People who do not hear voices, matched on age, sex, handedness and roughly on education.	Establishes what a normal discount score looks like in this equipment, in this room.
Clinical comparison group	People who hear voices with an established clinical explanation.	The group that makes the study decisive rather than merely suggestive. Without it, all you can say is "different from people with no voices" — which was never the interesting question.

The third group is the hard one. Recruiting it almost certainly requires a clinical or academic partner, with the ethical approval that comes with them. This is the point at which a home-built study either becomes a real one or stays a pilot. I would not pretend otherwise: without that third group the study is a feasibility exercise, worth doing and worth writing up honestly as such, but not decisive.

How many people

The differences reported in this literature are large by the standards of psychology — around 0.8 in the standard measure of effect size. Large does not mean you can get away with six volunteers.

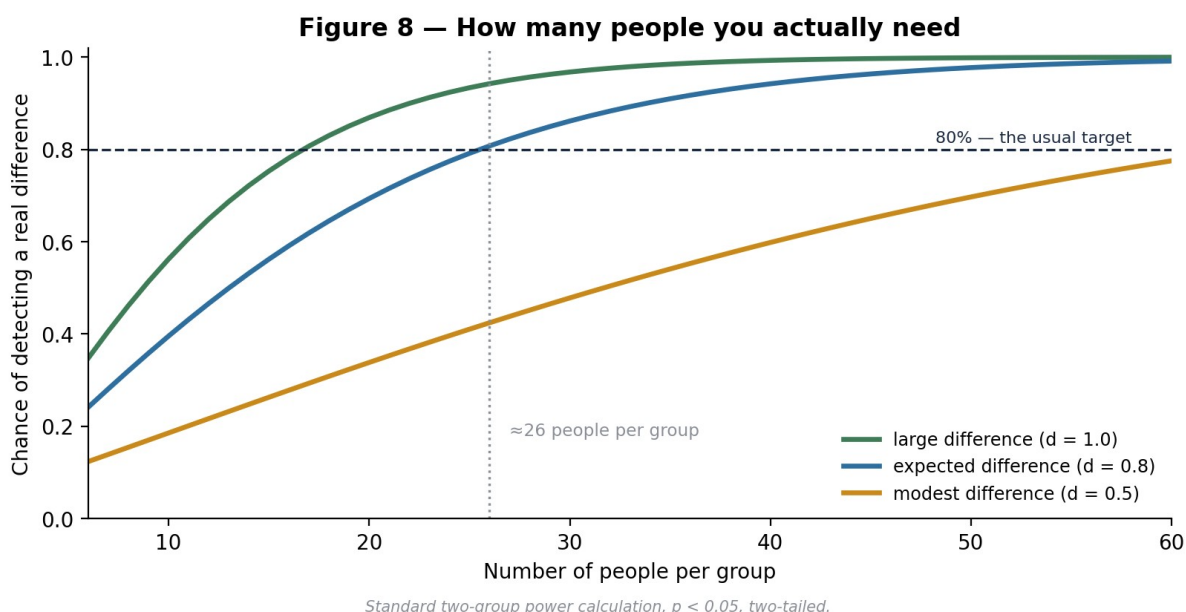


Figure 8 — With an effect of the expected size, you need roughly twenty-six people per group for a reasonable chance of detecting it. Ten people is a pilot that tests whether your equipment and protocol work — which is a legitimate thing to run, as long as you call it that.

A pilot of eight to ten volunteers tests the apparatus, the timing chain, the artifact levels and whether volunteers can actually perform the tasks. It does not test the hypothesis. Running a pilot and reporting it as a pilot is good practice. Running a pilot and reporting it as a finding is how a field acquires a reputation it then cannot shake.

Part 9 — The analysis, and the discipline around it

Figure 10 — What happens to the data after the session



Figure 10 — Every step is decided and written down before the first volunteer arrives. The group codes are broken only at step eight.



The processing itself is standard and unglamorous. What matters is not the steps but when the decisions about them are made.

Pre-register everything

Before recording a single volunteer, publish a document specifying: which electrode, which time window, which statistical test, how many trials minimum, what counts as an artifact, what the exclusion criteria are, what the primary outcome is, and which analyses are exploratory. Timestamp it publicly — the Open Science Framework is the usual venue and it is free.

The reason is not bureaucratic. If you have twenty possible measurements and you pick the one that came out significant, you will find something significant roughly every time, and it will mean nothing. Pre-registration is the mechanism that converts a search into a test. For anyone who has worked with cryptographic timestamping of documents, the logic is identical: the value comes from proving the content existed before the fact it is being used to establish.

Blind the analysis

The person who cleans the data and picks the peaks must not know which group each recording came from. Code the files. Have someone else hold the key. This is not a comment on anyone's integrity; peak-picking involves judgement calls, and judgement calls drift toward the expected answer without anyone intending it. Blinding removes the question entirely.

How to break your own study before someone else does

Before publishing anything, work through this list and satisfy yourself on each point.

- **Muscle contamination.** Volunteers in the study group may be more tense than comparison volunteers. Tension is muscle activity. Check that group differences survive strict artifact rejection, and report how many trials were rejected per group.
- **Speaking loudness.** If one group speaks louder in the TALK block, they hear a louder sound, and loudness changes the N100. Measure and match the vocal levels.
- **Trial counts.** If one group loses more trials to artifacts, their averages are noisier, which can masquerade as a smaller response. Equalise trial counts by random subsampling.
- **Medication.** Several common medications affect the N100. Record medication status and report it, even though it complicates the analysis.
- **Hearing.** Screen hearing thresholds. An undetected hearing difference will reshape every waveform in this study.
- **Expectation.** Volunteers know which group they are in and may have strong views about what the study should show. Keep the stated purpose neutral in the recruitment materials and in the room.
- **Sleep and caffeine.** Both shift ERP amplitudes. Record them, and standardise the time of day if you can.



Part 10 — What machine learning can and cannot add

Everything up to this point can be done with averaging and a t-test. That is deliberate: the primary hypotheses in this study are simple comparisons, and simple comparisons should be tested simply. But one part of the design — the cross-condition decoding in Test Three — genuinely requires a trained model, and the moment a model enters the picture, a whole category of ways to fool yourself enters with it.

So this section is written in two halves: what the models are for, and what will go wrong if you are careless. The second half is longer, which is the correct proportion.

The governing rule: classical statistics for the confirmatory questions, machine learning only where a decision boundary is genuinely needed. A study that reports its main finding as "our classifier achieved 71% accuracy" is weaker than one that reports "the discount score differed by 2.4 microvolts, 95% confidence interval 1.1 to 3.7". Accuracy figures are seductive and hard to interpret. Effect sizes with intervals are neither.

Where a model is actually needed

- **Cross-condition decoding (Test Three).** Train on Condition A versus Condition B, apply to Condition C. This is the one place a classifier is doing irreplaceable work: it is converting "does C resemble B or A" from a vague impression into a number produced by a rule fixed in advance.
- **Single-trial detection of silent replies.** Combining EEG, muscle and timing channels to estimate when a reply occurred, more precisely than a button press allows. This feeds better timestamps back into the averaging.
- **Subgroup discovery.** If the study group does not behave as one population — the third row of the outcome table in Part 3 — unsupervised clustering can propose a structure. Propose, not establish. Any cluster must then be validated on independent data or it is a Rorschach blot. Alongside clustering, pre-register regressions of the suppression score against symptom frequency, medication status and demographic variables. Treating the score as continuous extracts information a three-group comparison throws away, and it was the first analytical suggestion the design received in external review.
- **Artifact detection.** Automated flagging of bad segments is a genuinely mature application and saves enormous manual effort, with the important caveat that it must run identically and blind across all groups.

What goes into the model

EEG is not fed raw into a classifier in a study this size. It is turned into features — summary numbers that capture something meaningful — and the choice of features matters more than the choice of algorithm.

Figure 11 — From raw signal to a decision, one stage at a time

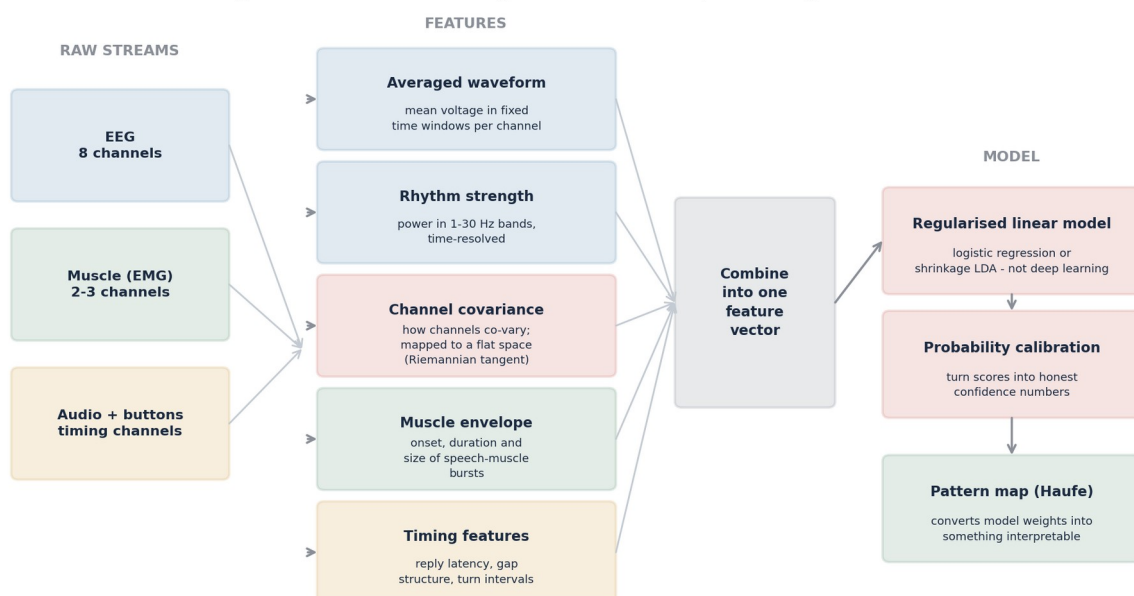


Figure 11 — The processing chain. Note that the model at the end is deliberately simple; the intelligence in this pipeline is in the feature design, not the algorithm.

Feature family	What it captures	Why include it
Time-window amplitudes	Average voltage in a handful of fixed windows on each channel.	Directly encodes the ERP shape. Interpretable, low-dimensional, robust.
Band power	How much rhythmic activity there is in each frequency band, tracked over time.	Language production and attention shift the alpha and beta rhythms. Capped at 30 Hz for the muscle-contamination reasons given earlier.
Channel covariance, mapped to tangent space	How the eight channels move together, expressed as a matrix and then flattened into a form ordinary models can use.	This is the single most effective representation for small-sample EEG classification. Covariance matrices live on a curved surface, and the Riemannian tangent-space mapping is the correct way to flatten them. It routinely beats far more elaborate methods on datasets this size.
Muscle envelope	Onset time, duration and magnitude of speech-muscle bursts.	The most direct evidence of a silent articulation event, and on a different physical channel from the EEG.
Behavioural timing	Reply latencies, gap durations, turn-interval statistics.	Carries the conversational-structure information from Test Three at essentially no cost.

Which algorithms — and which to avoid

The instinct with anything labelled "AI" is to reach for a neural network. Resist it here, for a reason that is arithmetic rather than ideological.



With seventy-five volunteers and a few hundred usable trials each, you have a small dataset by any modern standard, and the number of subjects is what limits generalisation, not the number of trials. A deep network with hundreds of thousands of parameters trained on this will achieve excellent accuracy on your data and near-chance accuracy on anyone else's. That is not a hypothetical failure mode; it is the normal outcome, and the EEG literature is full of published examples.

Approach	Verdict for this study
Regularised logistic regression (L2 or elastic net)	Primary choice. Well understood, few assumptions, gives calibrated probabilities, and the regularisation strength is the only real hyperparameter.
Shrinkage-regularised linear discriminant analysis	Excellent alternative, and the standard workhorse in brain-computer interface research for exactly these data sizes.
Riemannian methods (tangent space plus a linear model, or minimum-distance-to-mean)	Strongly recommended as the feature representation. Handles small samples gracefully and adapts across sessions and people better than most alternatives.
Linear support vector machine	Acceptable. Slightly harder to get honest probabilities out of.
Gradient-boosted trees	Usually unhelpful on EEG features; tends to latch onto noise channels. Not forbidden, but justify it.
Compact convolutional networks designed for EEG	Only worth attempting as a clearly labelled secondary analysis, and only if the study is pooled across sites. Report it alongside the linear model, never instead of it.
Large general-purpose models and foundation models	No role in analysing the signal. See the note below on where they are actually useful.

Validation, and the mistake that ruins most EEG classification papers

If you take one thing from this section, take this: never let trials from the same person appear in both the training set and the test set.

Figure 12 — The mistake that produces impressive, meaningless accuracy



Each small square is one trial. Blue = used for training, red = used for testing.

Figure 12 — Individual EEG is highly idiosyncratic. A model given some of a person’s trials to learn from can identify that person’s remaining trials with high accuracy, purely from their individual signature, while learning nothing whatsoever about the effect you care about.

The correct scheme is nested cross-validation with subject-wise splits. In plain terms: hold out an entire person, use the remaining people to both train the model and tune its settings, then test on the held-out person, and repeat with each person held out in turn. The tuning happens in an inner loop that never sees the held-out person at all.

Other ways the same problem sneaks in

- **Fitting preprocessing on everything.** If your blink-removal, your normalisation or your channel selection is computed using the full dataset before splitting, information has leaked. Every fitted step must live inside the training fold.
- **Overlapping epochs.** If consecutive trials share samples, adjacent trials are correlated and a random split scatters near-duplicates across training and test.
- **Assuming chance is 50%.** With unequal class sizes and small test sets, it is not. Establish the chance level empirically by shuffling the labels several thousand times and looking at the resulting distribution. Report your accuracy against that distribution, not against a textbook number.
- **Reporting the best run.** If you tried nine feature sets and report the one that worked, you have run nine experiments and reported one. Declare the feature set in the pre-registration, or declare the exploration honestly.

Interpreting a model without deceiving yourself

A trained linear model has a weight for every feature, and it is tempting to read those weights as a map of which brain regions mattered. This is wrong, and it is wrong in a specific, well-documented way: a classifier assigns large weights to channels that are useful for cancelling noise, not only to channels carrying signal. A weight map can therefore highlight an electrode that contains no relevant activity at all.



The fix is a standard transformation that converts classifier weights into activation patterns — what the data actually looked like — which are interpretable. It is a few lines of linear algebra and it should be applied to every weight map that appears in the write-up.

Calibration, because the number will be quoted

The output that matters in Test Three is not a label but a confidence: "Condition C was classified as B-like with probability 0.71." That number will be quoted, argued over and quite possibly screenshotted. It therefore has to be honest, and raw classifier scores are not honest probabilities — they are typically overconfident.

Fit a calibration step on the inner training folds, check it against reality on held-out data, and report a calibration curve alongside the headline figure. If the model says 0.7 and is right 70% of the time, the number means something. If it says 0.9 and is right 60% of the time, publishing the 0.9 is a form of misinformation regardless of intent.

Simulate before you record

Build the entire analysis pipeline before the first volunteer arrives, and run it on synthetic data you generated yourself, with a known effect planted in it. Two checks:

11. Feed it data containing an effect of the size you expect. Does the pipeline recover it? If not, your pipeline is broken, and you have found out for free.
12. Feed it data containing no effect at all, several hundred times. Does it report a significant result about five per cent of the time, as it should? If it reports twenty per cent, you have a leak somewhere, and you have found that out for free as well.

This costs a few days and is the single highest-return activity in the whole project. A pipeline that has never been tested against a known ground truth is an untested instrument, and nobody would accept an untested amplifier.

Where large language models genuinely help — and where they must not go

Since the question of "AI analysis" invites it, the distinction is worth stating explicitly.

Legitimate uses	Uses to refuse
Reviewing analysis code for the leakage errors listed above — a task they are genuinely good at.	Interpreting the EEG signal itself. These are not signal-analysis models, and treating their output as a measurement is a category error.
Drafting and structuring the pre-registration document.	Producing an explanation of a result after the fact. A fluent narrative for whatever came out is precisely what you do not want; it will be persuasive regardless of whether it is true.
Coding and thematic analysis of debrief interview transcripts, with human verification of every code, and literature triage across a scattered body of work.	Any processing of raw recordings through an external service. This is health data about a vulnerable population, and it should not leave the premises.



Legitimate uses	Uses to refuse
Generating the synthetic test datasets described above.	Deciding which analysis to run once the data is in. That decision was supposed to be made before.

Part 11 — The software architecture

A study like this generates perhaps three hundred gigabytes of recordings, several thousand derived files, and a chain of decisions that has to be reconstructible years later by someone who was not there. Treating that as a folder of scripts is how studies become unreproducible. Treating it as a small piece of engineering costs a couple of weeks up front and saves the entire project from a single well-aimed question at review.

The architecture below is deliberately conventional. Nothing in it is novel, and that is a feature.

Figure 13 — High-level system architecture



Figure 13 — Seven layers in the main path, plus four cross-cutting services that exist to make specific failures impossible rather than merely unlikely.

Layer by layer

1. Acquisition

Several devices produce data at once — amplifier, stimulus program, microphone, buttons — and the entire study depends on knowing their relative timing to within a few milliseconds. Two mechanisms, used together:

- **A hardware path for anything timing-critical.** The audio signal is physically split, so the same waveform that reaches the headphones also reaches a spare analogue input on the amplifier. Buttons, microphone and muscle electrodes go into the same amplifier. Those streams therefore share one clock by construction, not by synchronisation.

- **A software path for everything else.** A streaming layer that timestamps and records multiple devices into a single file is the standard tool here and works well for markers, condition labels and anything where a few tens of milliseconds do not matter.

Figure 14 — Why the sound goes into the amplifier, not just the headphones

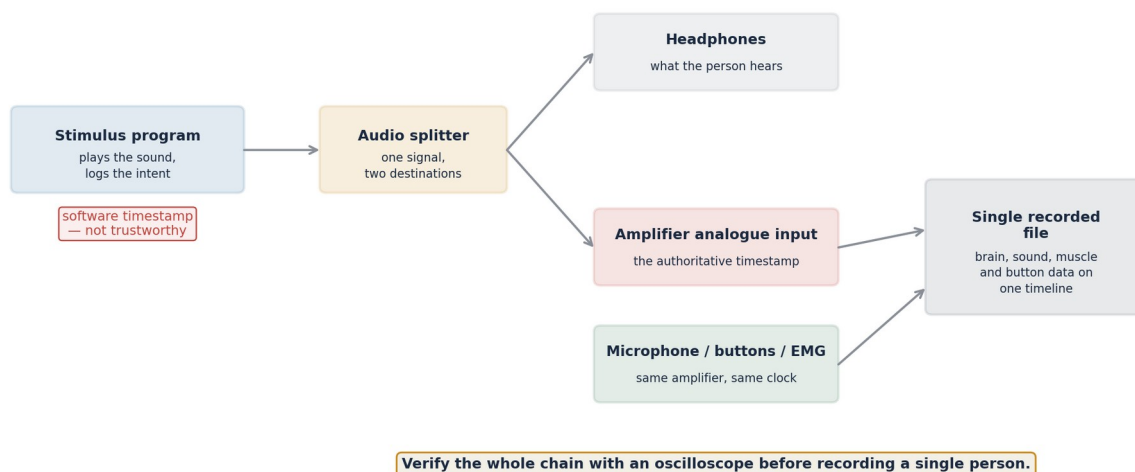


Figure 14 — The timing chain. The software timestamp records intent; the analogue input records what actually happened. Only the second one is used for the ERP analysis.

2. Ingest and quality control

Nothing goes into storage unexamined. An ingest service runs immediately after each session and does four things: converts the recording into a standard, documented folder layout; runs automated quality checks; assigns the pseudonymous subject code; and produces a one-page report the operator reads before the volunteer has left the building.

That last point matters practically. If an electrode was bad or the timing chain came loose, you want to know within ten minutes, not three weeks later when the person is unreachable.

Automatic check	Fails if...
Impedance log	any channel exceeded the threshold at any point in the session
Mains interference	power at the mains frequency exceeds a preset ceiling
Timing jitter	the measured spread between the software marker and the analogue audio onset exceeds a few milliseconds
Usable trial count	fewer than the pre-registered minimum survive artifact rejection in any condition
Channel health	any channel is flat, saturated, or uncorrelated with its neighbours
Behavioural sanity	button presses are absent, continuous, or implausibly regular

Use an established community standard for the folder layout rather than inventing one. It costs nothing, makes the dataset immediately intelligible to anyone else in the field, and means the standard analysis tools work without adaptation.



3. Storage

- **Raw recordings are written once and never modified.** Object storage with versioning switched on, or simply a filesystem with the permissions set correctly and a discipline that is actually enforced.
- **A relational database holds everything about the data.** Session registry, quality metrics, condition parameters, equipment serial numbers, software versions, protocol deviations. Small, structured, queryable — and the thing you will actually use every day.
- **Derived files live separately from raw ones,** and are treated as disposable. If you cannot delete every derived file and regenerate it from raw with one command, the pipeline is not reproducible and you should fix that before collecting more data.

4. Processing pipeline

The pipeline is code, versioned in a repository, and runs inside a container so the exact library versions are pinned. Analysis parameters live in a configuration file, not scattered through scripts, so the entire specification of a run is one readable document.

Orchestrate the steps with a workflow tool that understands dependencies, so that changing one parameter re-runs only what that parameter affects. Every output file records the hash of its inputs, the commit of the code that produced it, and the digest of the container it ran in. That triple is what makes a result traceable rather than merely plausible.

5. Feature and model layer

This is Part 10 turned into services: feature extraction, the cross-validation harness, the permutation testing, the calibration step and the pattern-map transformation. Three engineering requirements, all of which prevent specific failures described earlier:

- Every fitted transformation is expressed as a pipeline object that can only be fitted on training folds. Making the leakage error should require actively circumventing the structure.
- Every random operation takes an explicit seed, recorded in the run metadata. Two runs of the same configuration must produce byte-identical results.
- The permutation tests are expensive and embarrassingly parallel. Budget for it; a few thousand permutations across a handful of analyses is hours of compute, not minutes.

6. Confirmatory analysis

The scripts implementing the pre-registered hypotheses are written before data collection, tested on simulated data, then frozen. Freeze them properly: tag the commit, hash the tagged tree, and put that hash in the public pre-registration alongside a cryptographic timestamp.

When the data is complete, these scripts run once, on unblinding. Anything you think of afterwards is exploratory, goes in a separate directory, and is labelled as such in the paper. The separation is architectural, not just editorial — different folders, different entry points, different report templates — because a boundary that exists only in your intentions will not survive a deadline.

7. Reporting

Two automated outputs. A per-session quality report generated at ingest, and a full study report generated from the frozen scripts. Both are produced by executable documents that regenerate every figure and number from the data, so no result is ever transcribed by hand into a manuscript. Publish the report generator with the paper; anyone should be able to reproduce every figure from the deposited data with one command.

The cross-cutting services

Four things sit alongside the pipeline rather than inside it, each addressing a failure that is easy to describe and hard to prevent by good intentions.

Service	What it does	The failure it prevents
Blinding service	Holds the mapping between subject codes and group membership. Separate access control. The processing containers cannot reach it.	Unconscious bias in artifact rejection and peak-picking. Blinding enforced by architecture cannot be forgotten under deadline pressure.
Provenance and integrity	Hashes every raw file at the moment of capture. Cryptographically timestamps the pre-registration and the frozen analysis tree. Maintains an append-only audit log of protocol deviations.	The accusation — which will be made — that the data or the plan changed after the fact. The answer should be a verifiable timestamp, not a promise.
Identity vault	Holds the mapping between subject codes and actual people. Encrypted, offline, controlled by a person not involved in the analysis.	Re-identification of a vulnerable population. Also the practical basis for honouring a withdrawal request.
Local execution	All processing runs on hardware you control. No raw signal, audio or transcript is sent to any external service.	Loss of control over special-category health data, and the legal and ethical exposure that follows.

Data protection, stated plainly

Under European law, EEG recordings from identifiable participants, together with reports about their mental experiences, are health data — a special category with heightened requirements. Some practical consequences:

- **Pseudonymised is not anonymised.** If a key exists that maps codes to people, the data remains personal data and the full obligations apply. Do not describe the dataset as anonymous in the consent form unless the key has genuinely been destroyed.
- **Explicit, specific consent,** covering the analyses you intend, any future re-use, and any sharing. If you intend to deposit the data for others to reuse, say so at consent time; it cannot be retrofitted.
- **A data protection impact assessment** is appropriate here given the population and the category of data. Write it before recruitment, not after a question is asked.



- **Withdrawal has to be operationally real.** Design for it: if a participant withdraws, you must be able to locate and remove their raw files, their derived files, and their rows in the metadata database, and to say what happens to analyses already published.
- **Retention and deletion schedules** decided in advance and actually implemented, not aspirational.

What this costs to build

Two to three weeks of competent engineering, most of it configuration rather than code, before any volunteer is recruited. That is a real cost and it is worth paying, because the alternative is discovering during peer review that you cannot reconstruct how a figure was produced — and on a topic this contested, that discovery ends the project rather than delaying it.

A note on proportion. The engineering described here is not elaborate by the standards of any production system. It is deliberately boring infrastructure. The reason to invest in it is not that this study is unusually complex — it is that this study will be scrutinised far more aggressively than its size warrants, by people motivated to find a flaw, on both sides of the question. Every element above exists to convert a matter of trust into a matter of record.



Part 12 — The second half: a circulating fleet of kits

Everything up to this point describes a single-site study: one room, one operator, one set of carefully controlled conditions, seventy-five people. That is one half of the programme and it is the half that produces interpretable numbers fastest. It is not the half that answers whether a finding generalises, and on its own it will not persuade anyone who is determined not to be persuaded.

The second half did not originate with me. Mykhailo Sokolov, a fellow signatory on TI One Voice, sent me a white paper from the NeuroLaws Foundation proposing something structurally different: rather than bringing people to an instrument, mail the instrument to the people, collect recordings under fixed protocols, and analyse centrally. That idea is the backbone of this part, adapted in three ways — run through TI One Voice rather than a new platform, built around a shared circulating fleet rather than a device per person, and pointed at a pre-specified measurement rather than an open-ended anomaly search.

What the original proposal describes

- **A mail-shipped kit.** A factory-calibrated consumer headset in the fourteen-channel class, on a subset of the same international electrode layout used in this document, sampling at 256 Hz or better, wirelessly connected to the participant's own phone or tablet. No clinic, no travel.
- **A refundable deposit rather than a giveaway.** Returned when a working unit comes back. The stated purpose is not revenue but the prevention of inventory buy-out by adversarial actors — an unusual threat model, and one that tells you something about the environment the authors expect to operate in.
- **Fixed recording protocols.** Resting baseline, a shielded baseline recorded inside a Faraday pouch, event-logged sessions where the participant timestamps reported symptoms, sleep-onset recordings, countermeasure tests, and an optional cognitive-load task.
- **Centralised detection.** Filtering, artifact removal, a broad feature set, and scoring by isolation forests, autoencoder reconstruction error and changepoint detection, expressed through a structured vocabulary of deviation codes.
- **Chain of custody built in.** Device attestation at manufacture, signed consent and protocol version per session, a hash chain over every uploaded file, and scientific review before anything is exported for legal use.
- **A publication programme.** Three to five preregistered papers, ending with one on the reproducibility limits of consumer-grade detection.

What it gets right

This is a more disciplined document than most writing in this area, and the discipline is the reason it is worth building on rather than merely citing.

- **It separates observation from interpretation,** structurally. Every deviation code pairs an "observable pattern" with a "hypothesis under test", not with a finding. Most advocacy documents collapse that distinction in the first paragraph.



- **It states its own limits**, saying in plain language that the platform does not identify perpetrators and that attribution is a separate legal process. That sentence costs its authors something to write, and its presence is a reason to take the rest seriously.
- **It commits to preregistration, blinded review and independent replication** before any mechanism language. Those are the right commitments and they are adopted wholesale here.
- **It solves a real logistical problem**. This population is genuinely dispersed and genuinely unreachable by clinical neurophysiology. A mailed kit reaches people no laboratory will ever see.
- **The Faraday-pouch baseline is a good instinct**. A within-subject shielded control is the right shape of comparison, whatever one thinks of the hypothesis it addresses — provided it is blinded, which is the one change it needs.

What we would change

These are not objections to the concept. They are the places where, as originally written, the programme would generate data that could not be interpreted — which is worse than generating none, because a large corpus of uninterpretable flags is very hard to walk back once it exists.

Issue	Why it matters	What we do instead
Dry electrodes	Convenient and noisy. Adequate for resting rhythms and sleep staging; not adequate for the event-related measurements that carry the discriminating power in Parts 3 to 6.	Wet electrodes throughout. The kit is loaned rather than kept, so a fresh electrode set per participant is a consumable cost rather than a barrier.
Wireless timing	Two of the six deviation codes depend on millisecond event locking and cannot be computed reliably over a Bluetooth link.	The instrument digitises the audio on the same converter as the brain signal, as described in Part 14. Timing stops being a variable.
Open-ended anomaly detection	An anomaly is a deviation from something. Isolation forests flag at whatever rate you tune them to, and on consumer hardware in uncontrolled homes most deviations will be electrical and postural rather than neural. The white paper lists matched control recruitment as an open item; it is not an open item, it is the load-bearing wall.	Controls recruited first, on identical hardware, in comparable homes. The fleet then tests one pre-specified measurement rather than searching for anomalies.
Multiple comparisons	Six codes, a large feature set, many sessions and many participants generate an enormous number of implicit tests. Uncorrected, false positives are guaranteed — not likely, guaranteed.	A small number of pre-registered primary outcomes, correction across everything else, and the correction reported rather than only its survivors.
The narrowband code	The code for narrowband power spikes gives 10 to 12 Hz as its example. That is the alpha rhythm — the most common feature of adult EEG, strongly modulated by closing your eyes.	Exclude the alpha band from that detector, or define it against the participant's own eyes-open and eyes-closed baselines with ocular

Issue	Why it matters	What we do instead
		state recorded.
Selection effect	Kits go only to verified participants, where verification depends on the reported experience. The cohort is defined by the phenomenon under study.	Controls recruited through an entirely separate pathway, and recruitment routes reported in every paper.
Unblinded shielding	A participant recording inside a Faraday pouch knows they are inside a Faraday pouch, and expectation alone will move the measurements.	A sham pouch, identical in appearance and feel, randomised and blinded. Cheap, and it converts an anecdote into an experiment.
Legal framing ahead of results	Designing the pipeline around production of legal bundles creates pressure toward findings, and reviewers will say so.	Scientific and evidentiary tracks kept organisationally separate, with the export path opening only after independent replication.

How the two halves fit together

Figure 15 — Two instruments, two questions

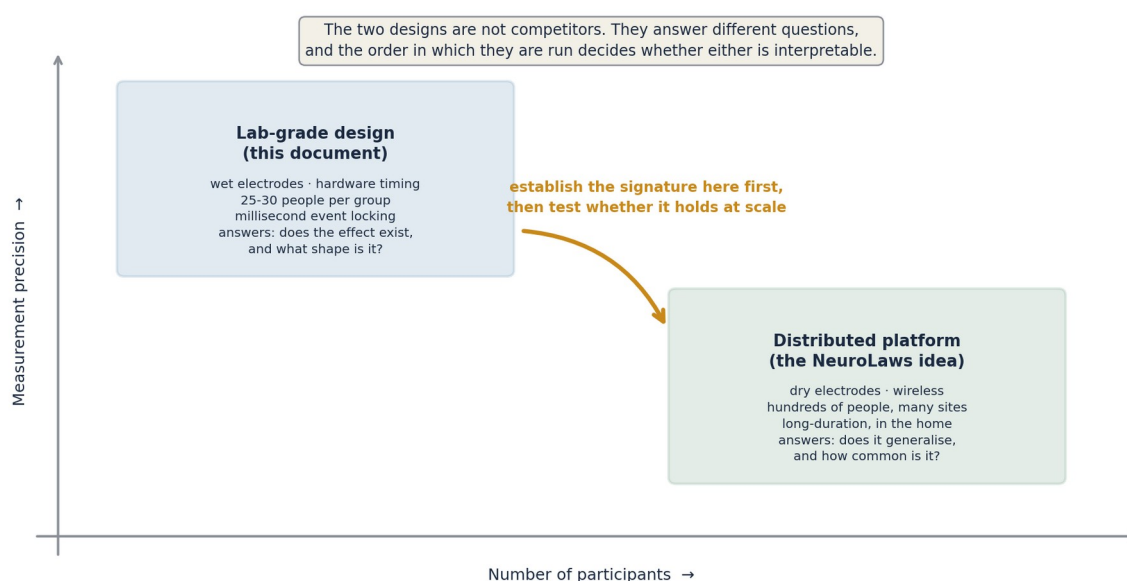


Figure 15 — A precise instrument on few people, and a coarse instrument on many. Neither substitutes for the other, and the sequence matters more than either.

The lab-grade design answers whether a measurable signature exists and what it looks like. It can answer that because it controls timing to the millisecond and conditions to the point of tedium. What it cannot do is tell you whether the finding holds across hundreds of people in ordinary homes.

The fleet answers exactly that — but only once there is a specific signature to look for. Deployed first, with an open-ended detector and no reference cohort, it produces flags with no established meaning, and every one of those flags becomes a hostage to fortune in later argument.

The sequence, stated plainly: run the precise study first, on a small number of people, to establish whether there is a defined signature and how large it is. Then take that specific pre-specified measurement — not a general anomaly detector — out to scale on the fleet. That turns the second study into a replication test with a clear prediction, which is the strongest form it can take. The reverse order produces volume without interpretability.

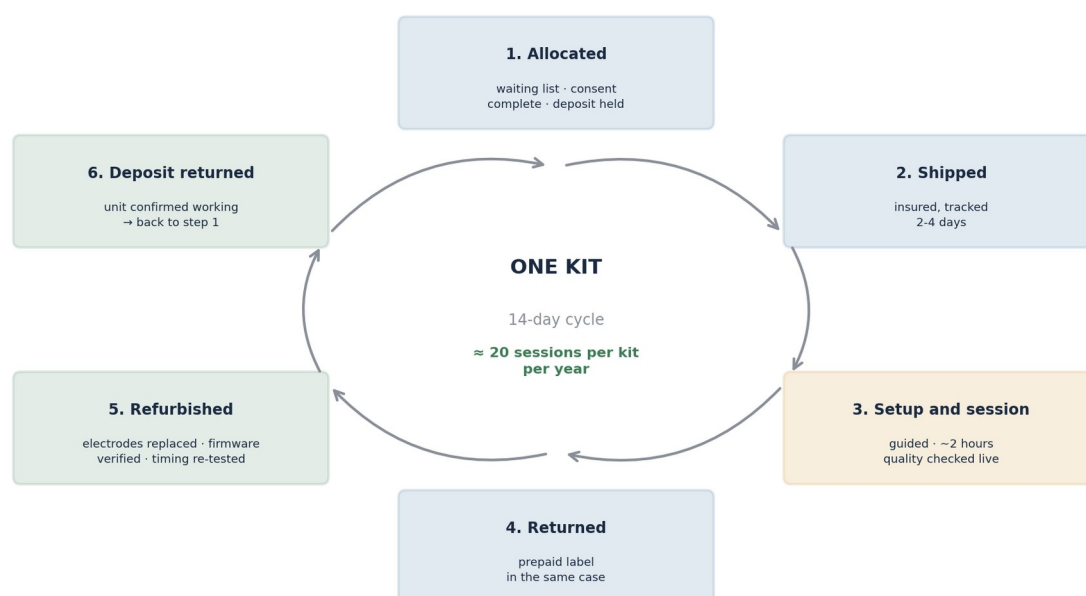
There is also a division of labour in the protocols themselves. The fleet is far better suited to work where duration matters and precision timing does not: sleep architecture across many nights, long resting recordings, symptom-timestamped sessions over weeks. Those are hard in a lab and easy at home. The event-locked tests need the timing chain. Split the protocols along that line rather than making either instrument do the other's job.

The fleet as a circulating asset

Here is where this proposal departs most sharply from the original. A device per participant means the number of participants is capped by the number of devices you can afford to give away, which for a research-grade instrument is a very small number. A shared pool changes the arithmetic entirely: the constraint becomes turnaround time rather than capital.

A participant needs the kit for a single two-hour session, or occasionally two sessions a week apart. The rest of the time it should be in transit or in someone else's hands.

Figure 19 — A kit is a circulating asset, not a giveaway



The deposit is returned at step 6. The contribution is not — it pays for the consumables and the refurbishment.

Figure 19 — The loan cycle. A kit that spends fourteen days per participant serves roughly twenty people a year; twenty kits serve the entire three-group study inside four months.

Figure 20 — Throughput of a shared kit pool

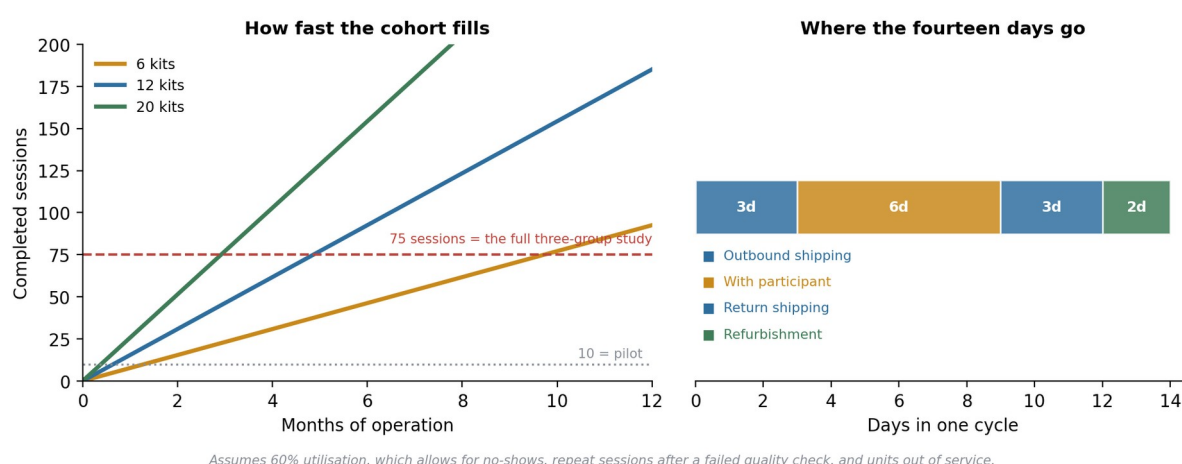


Figure 20 — Throughput. Note how few kits are actually needed: even six units complete the full study within a year, and two units are enough to prove the protocol works.

What a participant pays, and why

Because the instrument is self-built rather than bought, the whole economics of participation change. A research-grade commercial amplifier costs around €1,200; the kit described in Part 14, assembled from open hardware, costs between €230 and €300 in parts before the audio subsystem. That single fact drives everything below.

Item	Amount	What it is for
Deposit	€150, refundable	Held while the kit is out and returned in full when a working unit comes back. It is not a charge. It is set close to the €305 replacement cost of a self-built unit, so it protects the fleet without being punitive — and it is a third lower than it would have to be for commercial hardware.
Contribution	€50 or €77, not refunded	Which of the two depends on the funding model, set out below. Covers gel, saline, replaced leads, refurbishment materials, payment processing, packaging, losses, and a share of the fleet capital.
Shipping	At cost, both ways	Quoted before dispatch, insured and tracked. Excluded from the figures because it varies enormously by country — and at scale it is the single largest line in the whole programme.
Total exposure	€200-225 plus shipping, of which €150 comes back	A participant who returns the kit in working order is out the contribution and the postage, and nothing else.

Two ways to pay for it

A completed session costs the programme just over seventy-six euros, once everything is counted: the amortised share of the fleet, the consumables, the packaging and payment handling, the losses, the contingency, and the research overheads that are not really about kits at all — open-access publication fees and independent statistical review.

Figure 27 — Where the money comes from

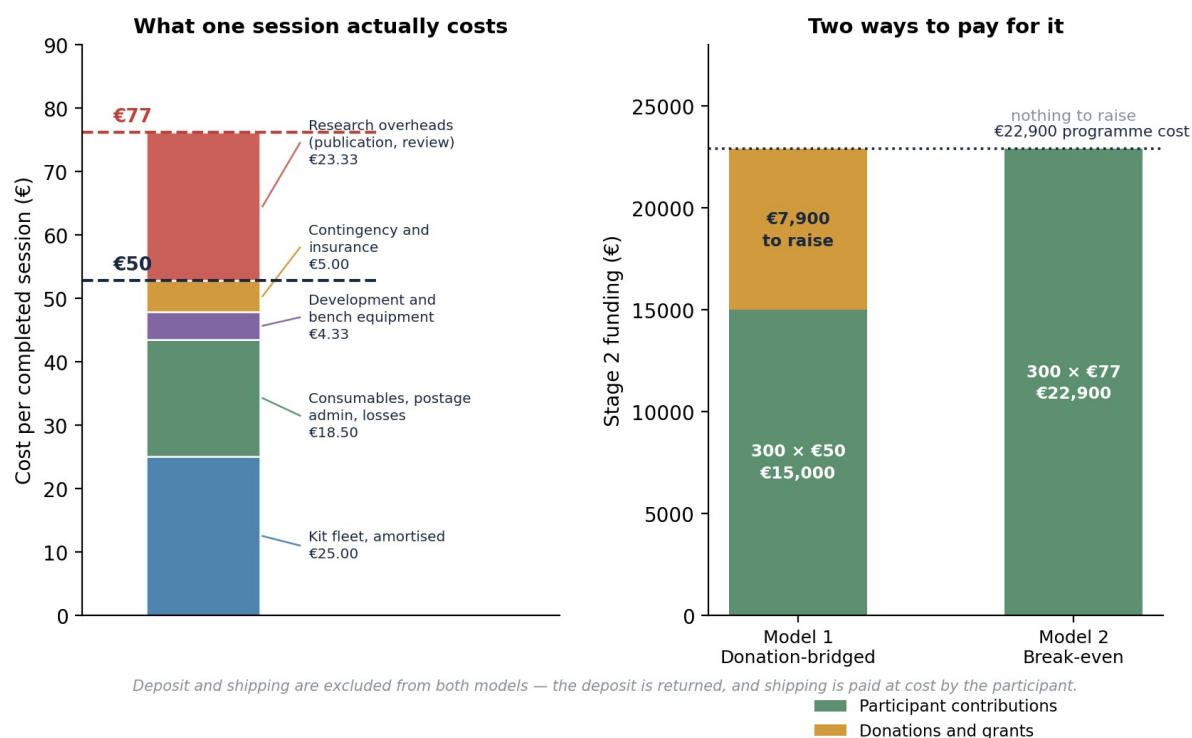


Figure 27 — The cost of a session, and the two ways of covering it. The dividing line falls in a convenient place: everything except the research overheads comes to just under fifty-three euros.

That coincidence is what makes the choice clean.

	Model 1 — donation-bridged	Model 2 — break-even
Participant contributes	€50	€77
What that covers	The full operational cost of running a session: hardware share, consumables, postage handling, admin, losses and contingency.	All of the above, plus the research overheads — publication fees and independent statistical review.
Raised from donors	€7,900 across 300 sessions	Nothing
Advantage	Lowest barrier to taking part, which matters more for the science than it first appears — see the note on selection below.	Self-sustaining. The programme does not stall if a grant application fails, and it can keep running indefinitely.
Risk	If the donations do not materialise, sessions	Twenty-five euros more is twenty-five euros

	Model 1 — donation-bridged	Model 2 — break-even
	stop or the contribution has to rise mid-study, which is an unpleasant thing to do to people who have already enrolled.	more. For some of the people this study most needs to include, that is the difference between taking part and not.

The preference is Model 1, and the reason is methodological rather than sentimental. But Model 2 exists and is stated here so that nobody is misled about what happens if the fundraising fails: the programme continues at seventy-five euros rather than quietly dying, and anyone already enrolled completes at the price they agreed to.

There is a third figure worth knowing. Once the fleet is built and the capital written off, a continuing operation costs about twenty-five euros per session in consumables, handling and replacement reserve. So the expensive part is starting; the running is cheap. A programme that reaches Stage 2 and finds something worth pursuing can carry on almost indefinitely on contributions alone.

A hardship fund is not optional, and it is not charity. Even at €50 plus a refundable deposit, two hundred euros of temporary outlay plus international postage is a serious sum, and a great many people in this population are unemployed, unwell, or estranged from the family that would otherwise lend it. If ability to pay decides who takes part, then income becomes a hidden variable running through every result, and the cohort skews toward the comparatively comfortable. That is a scientific defect before it is an ethical one. Part of any funding raised is ring-fenced to cover deposits, contributions and shipping for participants who cannot meet them, applied for privately and granted without the research team knowing who received it. Building the hardware ourselves is what makes this affordable: at commercial prices the waiver would cost four times as much per person and would have to be rationed.

Who gets a kit, and who decides

The original proposal screens applicants including a background check, risk scoring, and cross-checking of public posts. Some of that is a reasonable response to a real threat; some of it would, in this context, be corrosive. Two principles keep it on the right side of the line.

- **Screening protects the fleet, not the findings.** Checks exist to keep hardware circulating — verified address, a returned deposit, no pattern of bulk requests. They are not a judgement about whether someone's experience is real, and they must never function as one.
- **Eligibility is defined by the protocol, not by belief.** The study group needs people who report voices frequently enough to yield usable episodes within a recording block. That is a protocol requirement with a stated threshold, applied identically to everyone, and it is the only inclusion criterion that touches the phenomenon at all.
- **Rejections are explained.** A person told they cannot take part is told why, in specific terms, with a route to appeal. An opaque screening process aimed at a population whose central complaint is being dismissed without explanation would be an unforced error.



Everything runs through TI One Voice

The original proposal implies building a platform. That work is largely done. Every element the fleet needs already exists on TI One Voice or is described in Part 13: authenticated accounts with hardware-bound credentials, a Document Library with integrity proofs and evidence scoring, cryptographic timestamping, a data analysis layer, and local inference on hardware nobody else controls.

The fleet needs	On TI One Voice this is
A queue of applicants and an allocation process	An extension of the existing account and consent model, with a kit registry alongside the session registry.
Session guidance without a person in the room	The Session Runner described in Part 13 and shown in Part 15.
Recordings captured with custody intact	Streaming ingest, per-block signatures, RFC 3161 timestamping, deposit into the participant's own Document Library.
Analysis that does not pool personal data	The tiered cross-subject module in Part 13, with suppression thresholds and a privacy budget.
Findings reported without individual verdicts	The report constraint in Part 13: quality, compliance and contribution to the participant; interpretation only at cohort level.
A record that survives scrutiny	The integrity and provenance services that already underpin every other document on the platform.

The practical consequence is that the fleet is a logistics problem and a hardware problem far more than a software problem. That is an unusually good position to be starting from, and it is why the funding required is measured in tens of thousands rather than millions.

Part 13 — Running the protocol on TI One Voice

If the distributed approach is going to be attempted, the platform to run it on largely exists already. TI One Voice has the pieces: authenticated accounts with hardware-bound credentials, a Document Library with integrity proofs and evidence scoring, cryptographic timestamping, a data analysis layer, local AI inference, and a self-hosted deployment where nothing has to leave the premises. What is missing is a session runner, a signal pipeline, a report generator and a cross-subject module.

This section describes how the full 120-minute protocol from Part 8 would be delivered through the platform.

Figure 16 — The participant journey with a streaming headset



Steps 5 and 6 run concurrently: the session is being checked while it is still being recorded.

Figure 16 — The participant journey. Steps 1 to 5 happen on the participant's own device; steps 6 to 9 happen on the platform.

The session runner

The core new component is a progressive web application that acts as the operator. In a lab, a trained person sits with the participant and says "now do this, now wait, now press the button when you hear it." Remotely, the software has to be that person, and it has to be better at it than a printed instruction sheet, because there is nobody in the room to catch a mistake.

- **Guided setup with live feedback.** Electrode placement shown step by step against a head diagram, with a live impedance display per channel so the participant can see a contact improve as they adjust it. Setup does not advance until every channel is within tolerance.
- **A timing self-test before every session.** The runner emits a test tone, checks that it appears on the amplifier's analogue input at the expected offset, and refuses to start the ERP blocks if the measured jitter is out of specification. This automates the oscilloscope check from Part 7, which is the single most important pre-flight item.



- **Block-by-block narration.** Each of the six blocks is introduced in plain language, with a practice run and a comprehension check before the real trials start. The participant confirms they understand the task; the runner records that confirmation.
- **Audio playback and capture.** The TALK block records the participant's vocalisations; the LISTEN block plays those same recordings back, level-matched. Condition B of the talking-back test plays the recorded question prompts. All of this is orchestrated by the runner and captured alongside the EEG.
- **Break management.** The protocol has two scheduled breaks and the runner enforces them, tracks elapsed time, and allows a session to be suspended and resumed within a bounded window rather than lost.
- **Offline tolerance.** The session records to local storage first. Connectivity is required for upload, not for recording. A dropped connection mid-session must not cost two hours of a participant's day.

The one thing the browser cannot do. Web audio timing is not accurate enough for event-related measurement — the variability between when the code says a sound plays and when it actually reaches the ear runs to tens of milliseconds, and it is not solvable in software. The browser can instruct and orchestrate; it cannot be the clock. In the interim arrangement the kit ships a passive audio splitter and a hardware button box, around sixty euros of cable and plastic, exactly as in Figure 14. In the integrated arrangement described in Part 14 that splitter disappears into the instrument itself and the problem stops existing. Either way, without one of those two solutions the platform can deliver the resting, sleep and symptom-logging protocols but not Tests One through Four.

Part 14 shows each of these screens as a mockup in the platform's existing design language.

Getting the data in, and into custody

Two arrangements are possible, and one is much better. The session can be recorded locally and transferred when it ends, or it can be streamed to the platform as it happens. The second removes the alignment problem entirely and is the subject of Part 14. What follows is the transfer-afterwards version — what a first deployment on existing hardware would use — and its custody steps carry over unchanged to the streaming design.

13. **Client-side hashing before transmission.** The recording is hashed on the participant's device, so the integrity chain begins before the data crosses the network rather than on arrival.
14. **Resumable, chunked transfer** over an authenticated session using the existing hardware-bound token scheme. A two-hour recording on a domestic connection will be interrupted; the upload must survive that.
15. **Timestamping at ingest.** An RFC 3161 timestamp over the file hash, with the existing anchoring path, establishing when the recording existed independently of any later claim about it.

16. **Deposit into the Document Library** as an evidence artifact with its integrity proof, protocol version, device serial, firmware version and quality metrics attached as structured metadata, and scored by the existing evidence quality mechanism.

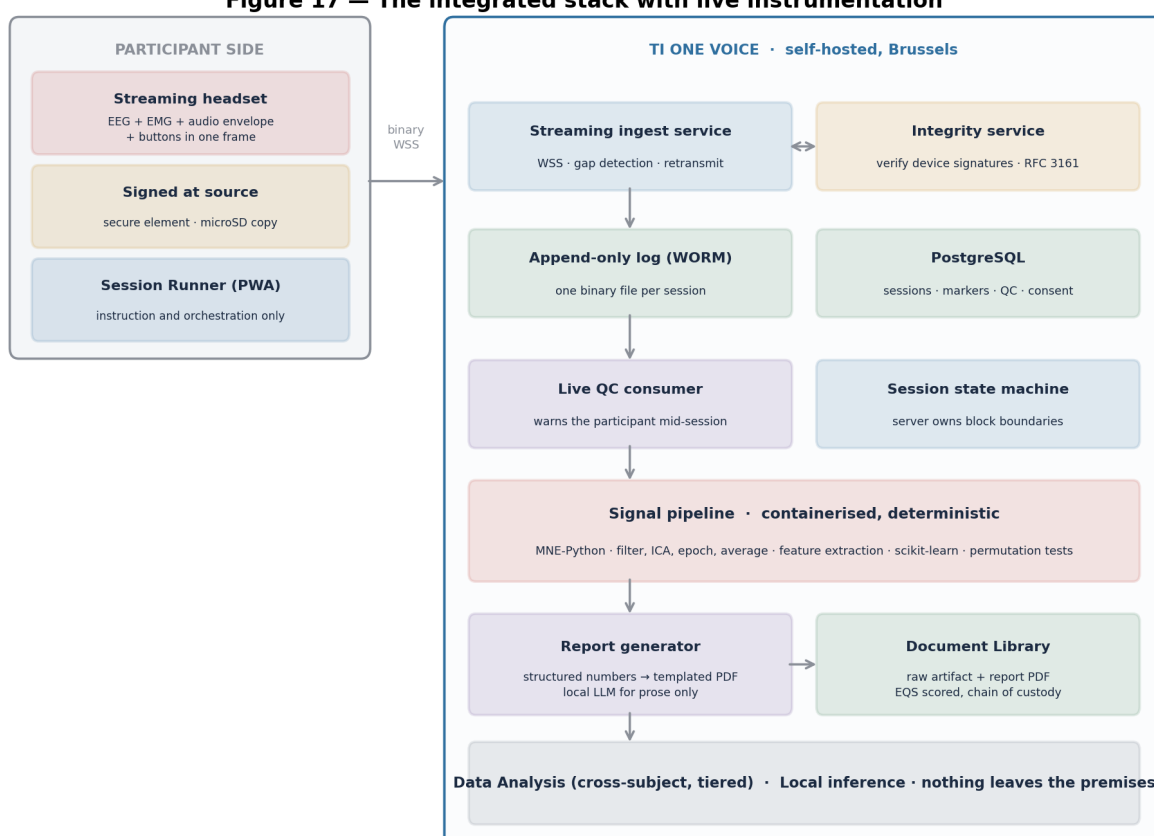
The practical benefit of using the Library rather than a separate store is that participants already understand it, the custody model is already built, and the recording sits alongside the participant's other documentation rather than in an isolated silo.

The practical weakness is that everything is discovered late. A lifted electrode, a timing chain that came loose, a block the participant misunderstood — none of it surfaces until the file is parsed, by which point the participant has taken the headset off and gone back to their life. Part 14 describes what changes when the instrument talks to the server while the session is still running.

Automated analysis, and what the report may say

Every uploaded session runs through the pipeline described in Parts 8 and 9: filtering, blink removal, epoching, artifact rejection, averaging, feature extraction. This is conventional signal processing in established scientific libraries. It is not a language model, and the distinction matters enough to state twice.

Figure 17 — The integrated stack with live instrumentation



The browser no longer sits in the data path. It instructs; the headset records; the server receives an already-coherent stream.

Figure 17 — How the new components sit against the existing stack. Local inference means the raw signal, the audio and the reports never leave the premises.



The output is a generated PDF, deposited automatically into the participant's Document Library alongside the raw recording. And here is where the design needs a constraint that will be unwelcome, so I will argue for it rather than assert it.

Why the individual report must not interpret the individual

Part 12 of this document — and every serious treatment of these measurements — states that individual variation is wide enough that one person's score is uninterpretable. That is not a formality. The overlap between groups on every measure in this design is substantial. A single person's discount score, or classifier output, or anomaly flag, carries almost no information about that person.

Generating a per-participant PDF that says "deviations detected in your session" would therefore be producing a confident-looking document that the underlying science does not support, and handing it to someone who has been waiting years for exactly that document. It would be harmful if it said yes and harmful if it said no, and it would be harmful in both directions for the same reason: the number does not mean what the reader will inevitably take it to mean.

The constraint is not a limit on ambition. It is what makes the programme credible enough to be worth running. A platform that returns individual verdicts will be dismissed by every reviewer who looks at it, and the dismissal will be correct.

The report should contain	The report must not contain
Data quality: impedance per channel, artifact rates, mains interference, measured timing jitter.	Any statement that anomalies were or were not detected in this person.
Protocol compliance: which blocks completed, trial counts surviving rejection, whether the session meets the inclusion criteria.	Any comparison of this individual against cohort norms or clinical thresholds.
Contribution record: what this session adds to the pooled dataset, and the integrity proof for the participant's own records.	Any language that could be read as a diagnosis, a confirmation, or a refutation.
Plain-language explanation of what the session measured and what the cohort study is testing.	Classifier probabilities or anomaly scores at the individual level.
A link to the current cohort-level findings, updated as the study progresses.	Predictions, extrapolations, or narrative interpretation generated by a language model.

The role of the language model, bounded

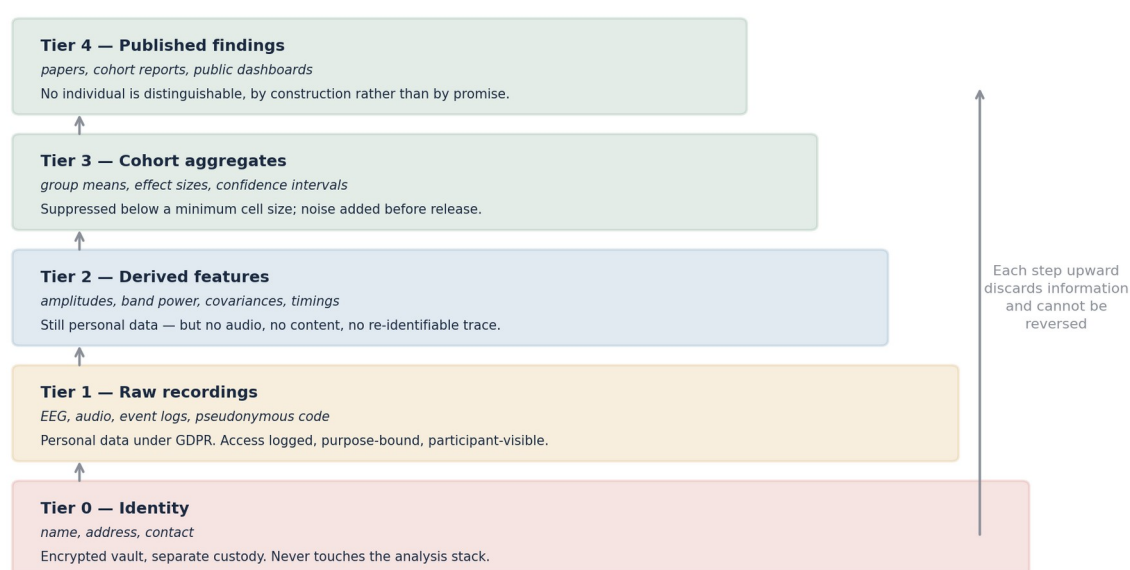
Local inference on the platform is the right way to generate readable prose, but the boundary has to be enforced architecturally rather than by prompt. Every number in the report comes from the signal pipeline and is inserted into a fixed template. The model writes connective explanation around values it is given; it does not compute, infer, estimate, or characterise. A validation step checks that every numeric token in the generated document appears in the structured input, and the report is not released if it does not.

This is a modest use of the technology and a defensible one. The alternative — letting a model read a session and describe what it sees — produces fluent text of unknown truth value, which is the worst possible output for this population and this question.

Cross-subject analysis without pooling personal data

The scientific value is in the cohort, not the session. But cohort analysis means combining data across people, and doing that on health data from a vulnerable, internationally distributed population requires more than good intentions.

Figure 18 — Privacy tiers for cross-subject analysis



Only Tier 3 and above leave the platform. The separation is enforced by access control, not by policy alone.

Figure 18 — Information is discarded at each step upward, and the boundaries are enforced by access control rather than by policy. Only Tier 3 and above leave the platform.

Three design decisions carry most of the protection:

- **Analysis operates on derived features, not recordings.** The cross-subject module never touches audio or raw signal. It sees amplitudes, band powers, covariance summaries and timings. Voice recordings — the most identifying material in the entire dataset — stay in the raw tier and are used only for level-matching within a session.
- **Aggregates are suppressed below a minimum cell size.** No statistic is released for a subgroup smaller than a pre-set threshold. This matters more than it might appear: cross-tabulating jurisdiction, age band and symptom profile can isolate a single person very quickly in a dispersed international cohort.
- **Calibrated noise is added to released aggregates.** For anything published to a dashboard or shared outside the analysis team, a formal privacy mechanism bounds how much any single participant's inclusion can shift a published number. The budget is set and recorded in advance.

A stronger option exists if the cohort grows beyond a few hundred: compute per-participant summary statistics locally and transmit only those, so raw recordings never centralise at all. That is



more engineering than a pilot warrants, but the tier structure above is deliberately compatible with moving to it later without redesigning the analysis.

Data protection obligations, concretely

The platform is established in Belgium and processes special-category health data from participants worldwide. Some specific consequences that follow, beyond the general points in Part 11:

Requirement	How it is met
Lawful basis for special-category data	Explicit consent for the research purpose, with scientific-research safeguards. Consent is per-protocol-version and re-obtained when the protocol changes materially.
Data protection impact assessment	Mandatory here on multiple grounds — large-scale special-category processing, vulnerable data subjects, novel technology. Completed and reviewed before the first kit ships, not after.
Purpose limitation	The evidentiary use and the research use are distinct purposes with distinct consents. A participant may contribute to the science without agreeing to any legal export, and the system enforces that separation rather than relying on process.
Data minimisation	No GPS. No continuous ambient audio. Environmental metadata collected only where a specific pre-registered analysis requires it, and dropped once that analysis is complete.
Withdrawal that actually works	A single operation that locates and removes the participant's raw files, derived features and metadata rows, records what happens to analyses already published, and returns a receipt. Testable, and tested.
Retention	Defined per tier and implemented as scheduled jobs. Raw recordings have the shortest life; derived features persist for the research period; aggregates are permanent.
Transparency	Participants can see which analyses their data has entered, and the access log for their own records. This is a genuine advantage of running it on a platform the participants already trust with their documentation.

What this would take to build

Component	Effort	Notes
Session runner (PWA)	The largest single piece	Guided setup, audio orchestration, block sequencing, offline buffering, timing self-test. The timing self-test is the hardest part and should be built first.
Hardware adapter kit	Sourcing, not development	Passive splitter, button box, wet electrode set for the interim arrangement. Superseded by the integrated headset in Part 14.
Signal pipeline	Moderate	Established scientific libraries in a container. Most of the work is packaging, determinism and quality gates rather than algorithms.
Report generator	Small	Templating plus a constrained local model, plus the numeric-token validator.
Cross-subject module	Moderate	Extends the existing data analysis layer with the tier boundaries, suppression thresholds and privacy budget.
Library and custody integration	Small	Mostly reuse: timestamping, integrity chain and evidence scoring already exist.
Control cohort recruitment	The real bottleneck	Not a software problem, and the thing most likely to determine whether any of the above produces interpretable results.

That last row is the honest summary of the entire section. The engineering is tractable and much of it already exists. The scientific validity depends almost entirely on something no amount of platform work can supply: a comparison group recruited independently, recorded on the same hardware, under the same conditions. Build the platform by all means — but recruit the controls in parallel, because without them the platform will produce a great deal of data and very little evidence.

Part 14 — Live instrumentation: streaming the headset to the server

Part 13 describes a session that is recorded locally and uploaded when it finishes. There is a better arrangement, and it is worth setting out in detail because it removes an entire class of problem rather than merely managing it: connect the headset to the platform directly, so that brain signal, muscle signal, sound and button presses are registered at the same moment, in the same place, and no correlation of separate sources is ever required.

That is achievable. But it needs one premise corrected first, because the correction determines almost everything about how the hardware is built.

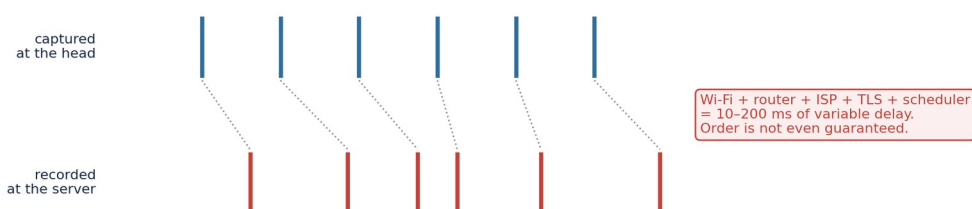
The server is not the clock

The intuitive argument is that if everything arrives at the same place, it is thereby synchronised. It is not. Between a headset in a living room and a server in Brussels sit Wi-Fi contention, a domestic router, an internet service provider, transport-layer buffering and operating-system scheduling. Together these introduce somewhere between ten and two hundred milliseconds of delay, and — the part that matters — the delay varies from packet to packet. Arrival order is not even guaranteed to match capture order.

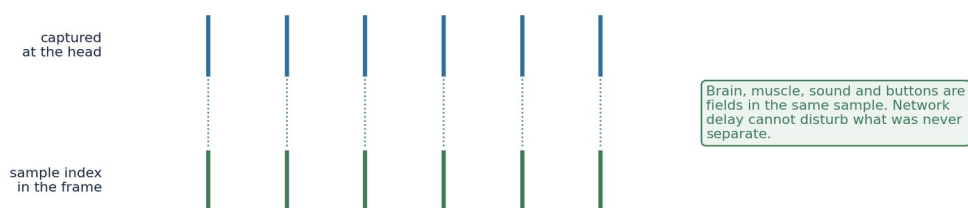
So timestamping on arrival would replace the twenty milliseconds of wireless jitter we were trying to eliminate with something larger and less predictable. For a measurement that lives in a fifty-millisecond window, that is fatal.

Figure 21 — Where the timeline is actually established

Wrong — timestamp on arrival at the server



Right — one converter, one sample counter, one frame



Co-registration happens at the point of digitisation, not at the point of storage.

Figure 21 — The distinction the whole design turns on. Co-registration must happen where the signals are digitised, not where they are stored.



The correct approach is stronger than what the split-cable arrangement in Part 7 achieves, and it is available precisely because we are building custom hardware. One converter, one sample clock, one frame. The EEG channels, the muscle channels, the audio envelope and the button states all become fields in the same sample, indexed by the same counter. There is then no correlation step at all — not because the server merged the streams successfully, but because they were never separate in the first place.

The network then carries a stream that is already internally consistent. Its jitter becomes irrelevant, because nothing downstream ever reads timing from packet arrival.

How the headset would need to be built

One point before the components: this is a self-build, not a procurement exercise. The parts list below comes to between two hundred and thirty and three hundred euros in single quantities, against roughly twelve hundred for a commercial research amplifier using the same converter chip. The difference is assembly labour, support and warranty — all of which this programme absorbs as volunteer effort. Building rather than buying is what makes a fleet of twenty-five units a four-figure decision rather than a five-figure one, and it is the single change that brings the whole thing within reach.

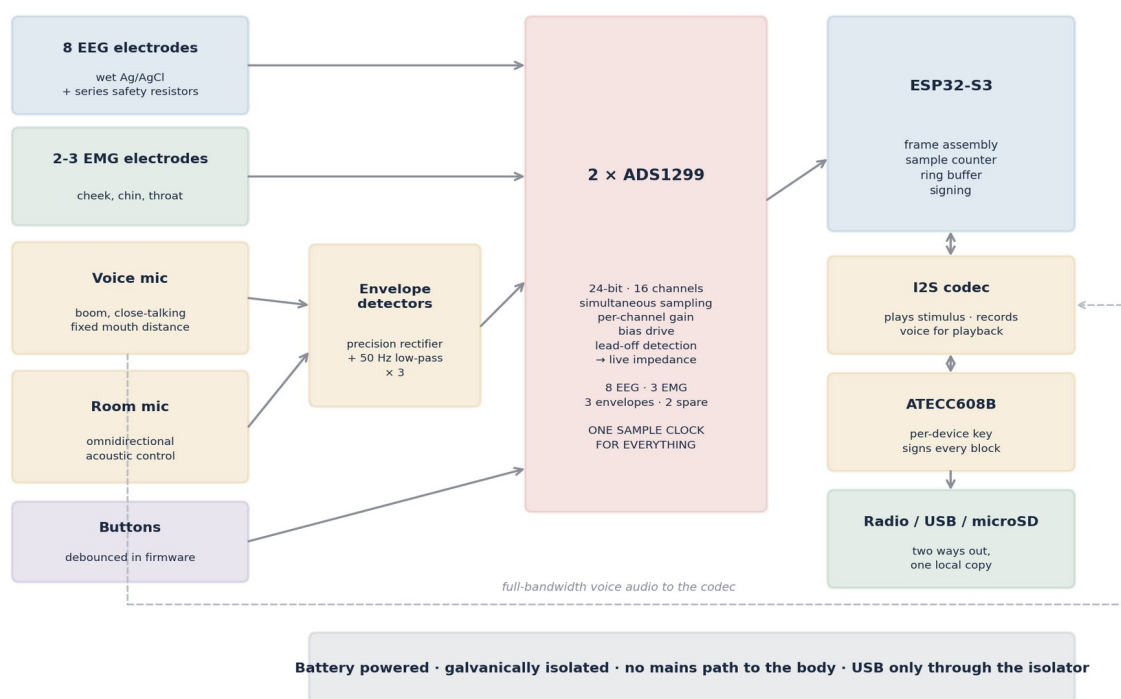
Component	Indicative cost
ADS1299 converter (single quantity)	€35-50
Four-layer PCB, five-off from a prototype house	€10
ESP32-S3 development board	€8
Passives, connectors, regulators	€15
microSD module and secure element	€5
Lithium cell and charge circuit	€12
Printed enclosure	€5
Twelve Ag/AgCl cup electrodes with leads	€70-100
Headband or cap	€25
Closed-back headphones	€25
Shipping case	€20
Subtotal per unit	€230-300

Two corrections to the earlier assumption that electrodes are consumable: they are not. Ag/AgCl cup electrodes are cleaned, disinfected and re-gelled between participants, and a set lasts on the order of twenty-five sessions before leads start failing. The genuine per-session consumable is gel, saline wipes and the occasional replaced lead — around eight euros, not fifty.

Budget three prototype iterations before a unit you would put on someone's head, and build twenty-five per cent spares. Hand-assembled units fail, and cheap hardware makes that an inconvenience rather than a crisis — which is itself an argument for building rather than buying.

Three checks that are not optional on self-built hardware. First, one qualified electronics person reviews the design before the first unit goes on a head — battery-only operation during recording, series current-limiting resistors on every lead, no USB path to a mains-powered laptop while a person is connected. Second, every unit is validated against a known signal from a function generator before first use, checking noise floor, gain accuracy and channel crosstalk; twenty-five hand-built boards are twenty-five slightly different instruments, and if device identity happens to correlate with group you have manufactured a confound. Record the device serial on every session and counterbalance units across groups. Third, borrow a commercial amplifier once and record the same person on both in the same sitting, confirming the N100 matches in latency and amplitude. Without that comparison a null result is uninterpretable, because you cannot distinguish "no effect" from "our noise floor swallowed it". All three are free. None of them can be skipped.

Figure 22 — The headset, block by block



Three envelopes — stimulus, voice and room — enter the same converter as the brain signal, so every timing relationship is fixed at the point of digitisation.

Sourcing at volume

External review raised the obvious question: why not manufacture in China and cut the hardware cost to a tenth? The direction is right and the factor is not. Volume board assembly from a Shenzhen prototype house genuinely collapses the PCB and assembly line — boards that cost tens of euros hand-built cost single euros assembled in panels. What does not shrink is the floor set by the ADS1299 itself (€35-50 even in quantity, and the one component with a serious counterfeit market)



and by wet Ag/AgCl electrodes. A realistic volume figure is €180-220 per unit rather than €305 — a meaningful saving, not an order of magnitude.

Two consequences. First, the counterfeit risk makes the per-unit validation against a known signal, already mandatory above, double as incoming inspection: a fake or re-marked converter fails the noise-floor and crosstalk checks before it ever reaches a head. Second, the planning figures in this document deliberately stay at the hand-built price. If volume sourcing delivers the saving, it widens the hardship fund or lowers the break-even contribution — it is upside, not a number the budget depends on.

The two microphones

Three separate parts of this design turn out to depend on capturing sound at the headset, and they do not all want the same microphone. Test One needs a recording of the participant's own voice, at usable fidelity, to play back level-matched in the LISTEN block. The provocation variant of Test Two needs speech onset and offset accurate to the sample. And the acoustic negative control needs to know whether a real sound occurred in the room. One capsule cannot do all three well, because the first two want to hear only the wearer and the third wants to hear only the room.

	Voice microphone	Room microphone
Type	Electret or MEMS on a short boom, close-talking, directional.	Omnidirectional MEMS, mounted on the enclosure facing outward.
Position	Fixed at roughly three centimetres from the corner of the mouth.	Away from the wearer, unobstructed.
Should hear	Only the wearer.	Only the room.
Feeds	Envelope into a converter channel; full audio through the codec.	Envelope into a converter channel; full audio gated to the scripted windows only.
Jobs it does	Speech onset and offset timing · TALK-block capture for LISTEN playback · verification that the scripted line was actually spoken.	Acoustic negative control · headphone level check · detecting speech during a silent window.

Why the boom is a measurement decision, not an ergonomic one

Test One compares the brain's response to the participant's live voice against a recording of that same voice, and the comparison only means anything if the loudness at the ear is matched between the two. If the distance from mouth to microphone varies between the recording and the playback session, the levels differ and the N100 comparison is contaminated by a loudness difference rather than an authorship difference. A fixed boom holds that geometry constant. A microphone on a table, clipped to clothing, or built into a phone does not.

Add a reference-tone calibration at setup: play a known level through the headphones, record it on both microphones, and store the resulting gains in the session metadata so that levels can be reconstructed and compared across sessions and across participants.



The channel budget still fits

Two cascaded converters give sixteen channels, and the allocation now reads:

- Eight EEG.
- Three EMG, for the speech muscles.
- Three envelopes — stimulus output, voice microphone, room microphone.
- Two spare.

Buttons consume no converter channels. They are debounced in firmware and packed as bits into the auxiliary field of the sample frame, which is what keeps them co-registered with everything else at no cost.

What it adds to the build

Item	Cost
Electret capsule with preamp, or I ² S MEMS	€4
Boom arm, mount, foam windscreen	€12
Room MEMS microphone	€3
Two further envelope detectors (precision rectifier and low-pass)	€4
Audio codec, if not already fitted for stimulus playback	€6
Total	€25-30 per unit

That takes the kit from around €280 to around €305, and the twenty-five-unit fleet up by roughly €700. It is a real increase and it does not materially move the stage figures in Part 18.

One item for the prototype checklist, alongside the timing loopback. Headphone bleed into the voice microphone. Closed-back cups and a close-talking directional capsule should keep it well down, but this needs measuring rather than assuming, because during the LISTEN block the playback will leak into the microphone and could false-trigger the speech-onset detector. The mitigation is already in the architecture rather than in the hardware: the stimulus envelope is digitised on its own channel, so known playback can be subtracted from the voice-microphone envelope and what remains is genuine vocalisation. That makes it a firmware and analysis problem, which is the cheaper kind — but it still has to be measured on the first board.

Figure 22 — The instrument. Roughly a hundred euros of components in volume, and one genuinely useful idea.

The analogue front end

The Texas Instruments ADS1299 is the chip for this and effectively the only sensible choice: eight channels, twenty-four-bit resolution, simultaneous sampling across all channels, programmable gain per channel, integrated bias drive for common-mode rejection, and built-in lead-off detection. That



last feature is what gives the live impedance display in the guided setup for free, rather than as a separate measurement circuit.

Two of them cascaded on a single serial bus gives sixteen channels: eight for EEG, two or three for muscle recording at a different gain setting, and the remainder free for the synchronisation inputs described next.

Audio into the same converter — the key idea

This is the element that removes the correlation problem, and it rests on a small insight: on the synchronisation channel you do not need speech fidelity, you need onset timing.

So a precision rectifier followed by a low-pass filter at around fifty hertz turns both the stimulus output and the microphone signal into slow envelope waveforms. Those are AC-coupled, attenuated, and fed into two spare inputs on the same converter that is digitising the brain. Sound onset is then resolved to a single sample, on the same clock as the response it evoked. If sub-sample resolution is wanted, a comparator on the envelope can drive a digital pin latched against the sample counter.

Buttons are handled the same way: debounced in firmware, sampled on every data-ready interrupt, packed as bits into an auxiliary field of the frame. A button press is therefore inherently co-registered with the brain signal rather than separately timestamped and later aligned.

The microcontroller

An ESP32-S3 covers the requirement comfortably: a serial bus for the converter, Wi-Fi for transport, USB for the fallback path, and enough external memory to buffer several minutes of samples. If the device is also to play the stimulus audio itself — removing the participant's phone from the timing path entirely — add an audio codec and pre-download each block's audio to the device before the block begins. That is the fully self-contained version, and it is the one to aim at.

Local storage is the authoritative copy

A microSD card writes the identical stream throughout. The network stream exists for liveness, quality control and custody; the card is the recording. A two-hour session must not be destroyed by a router rebooting at minute ninety, and no amount of buffering discipline fully protects against that if the card is absent.

Device identity in hardware

A secure element such as the ATECC608B, holding a per-device private key provisioned at manufacture, signs each block of samples at source. This is materially stronger than hashing at upload: the integrity chain begins inside the instrument, before the data has been anywhere. It is also exactly the property the NeuroLaws white paper asks of "device attestation" without quite specifying how to obtain it.

Safety is not optional here. This is a device connected to a person's head. Battery power plus wireless transport gives galvanic isolation by construction, which is one of the strongest

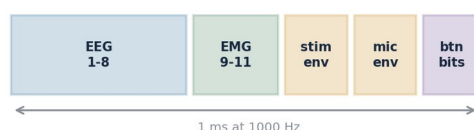


arguments for the direct topology. The moment a USB cable connects the device to a mains-powered laptop, a certified isolation barrier becomes necessary. Series current-limiting resistors belong on every electrode lead. And under European medical device regulation, an instrument used in research on human subjects carries real obligations even when it is not being placed on the market as a medical device. Have that conversation with someone who does it professionally before building a batch, not after.

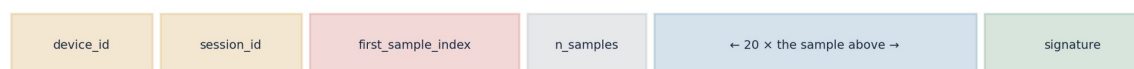
What goes over the wire

Figure 23 — The frame format

ONE SAMPLE — every field shares the same index



ONE NETWORK FRAME — about 20 ms of samples



The sample index is the only timeline. Nothing downstream reads timing from packet arrival, so network jitter, reordering and reconnection are transport problems rather than data problems.

16 channels × 4 bytes × 1000 Hz = 64 kB/s. Roughly a low-bitrate podcast.

Figure 23 — The frame. The sample index is the only timeline in the entire system.

Bandwidth is a non-issue: sixteen channels at four bytes and a thousand samples per second is sixty-four kilobytes per second, roughly a low-bitrate podcast. The interesting design decisions are elsewhere.

Concern	How it is handled
Timeline	A monotonic sample counter, assigned at the converter and never recomputed. Every frame carries the index of its first sample. Wall-clock time is metadata, never the basis of any analysis.
Loss and reordering	Sequence numbers with server-side gap detection. The server requests retransmission from the device's ring buffer, which holds several minutes.
Disconnection	The device buffers and backfills on reconnect. A dropped connection costs continuity of the live view, not the session.
Clock drift	A periodic exchange estimates the offset and drift between device and server clocks, stored as metadata. Crystal drift of thirty parts per million is about two hundred milliseconds over two hours — irrelevant within a session, since every channel drifts together, and relevant only for wall-clock alignment.
Authentication	Two credentials, both required. The device certificate proves which instrument; the participant's hardware-bound session token proves which person. The server will not

Concern	How it is handled
	open a session without both.
Integrity	Per-block signatures verified on arrival, and the whole chain re-verified at finalisation against the device's own copy.

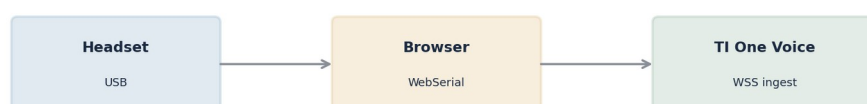
Figure 24 — Two transport topologies

A — Direct: the headset holds its own credentials



Works on any device the participant owns, including iOS. The one to build toward.

B — Tethered: the browser relays



Chrome and Edge on desktop only — useful for setup and as a fallback, not as the primary path.

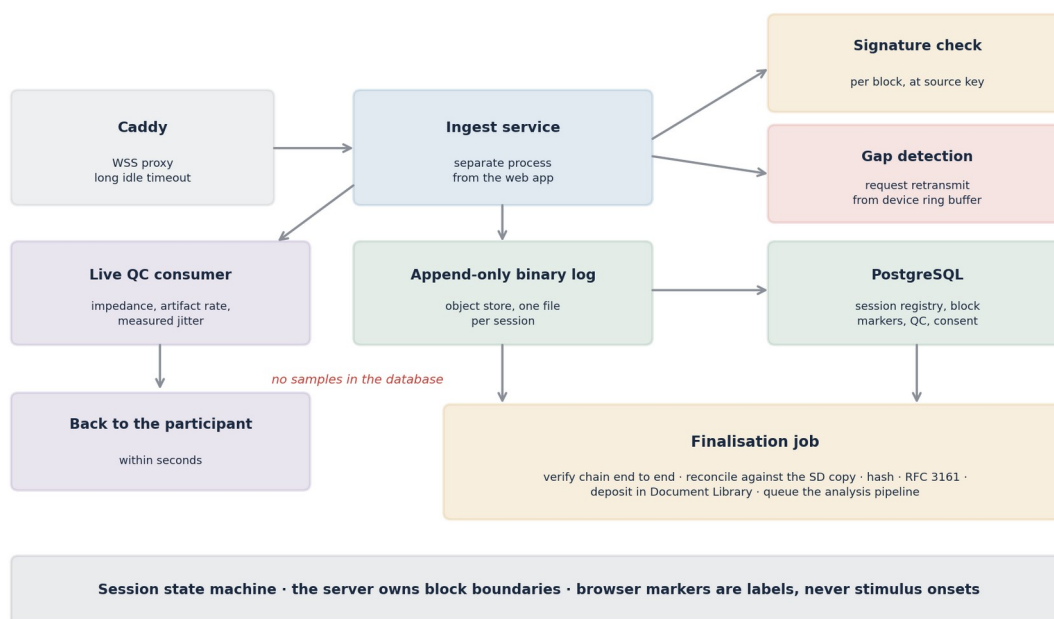
In both cases the device certificate proves which instrument; the participant token proves which person. The server requires both.

Figure 24 — Two topologies. Build both, but treat the direct path as the destination.

The direct topology — headset holding its own network credentials and its own certificate, browser reduced to a screen — is the one worth building toward. It works on any device the participant owns, including devices where browser hardware access is restricted, and it keeps the mains-powered computer out of the electrical path to the body. The tethered topology over a serial connection from the browser is useful for provisioning and as a fallback, but it is limited to certain desktop browsers and cannot be the primary route.

What the platform would need

Figure 25 — The server-side write path



About 500 MB per session raw, or roughly 1.5 GB if full-rate audio is retained.

Figure 25 — The write path. Note what is absent: any step that aligns one data source against another.

- **A dedicated streaming ingest service.** The existing application framework handles this well, but the long-lived connections should live in a separate process from the web application so that a two-hour socket is not occupying an application worker. The reverse proxy needs websocket forwarding with generous idle timeouts.
- **An append-only binary log per session, written straight to object storage.** Samples do not belong in the relational database. The database holds the session registry, block markers, event index, quality metrics and consent state — structurally similar to what it already holds for documents.
- **A session state machine owned by the server.** The server decides when a block begins and ends and pushes that to both the runner and the device. But note the asymmetry: markers originating in the browser inherit network delay, so anything precision-critical must originate on the device. Browser markers are fine for labelling "block three started"; they are not fine for stimulus onsets.
- **A live quality-control consumer reading the stream in parallel.** Rolling impedance, artifact rate, measured audio-to-sample jitter, dropped-frame count. This is the real payoff of streaming: the participant learns at minute four that an electrode has lifted, instead of learning three weeks later that the session was unusable.
- **A finalisation job.** Verify the signature chain end to end, reconcile the streamed copy against the device's card, hash, timestamp, deposit into the Document Library, queue the analysis pipeline.

Storage lands at roughly five hundred megabytes per session for the signal data, or around one and a half gigabytes if full-rate audio is retained rather than discarded after level-matching.



What this buys, concretely

Problem in the upload-later design	What live streaming does to it
Separate audio, EEG and button files to align afterwards	Eliminated. They were never separate — they are fields of one sample.
Browser audio timing unreliable for event-locked measurement	Eliminated. The device plays and digitises the stimulus on its own clock.
Session found to be unusable weeks later	Caught within seconds, while the participant is still wearing the headset.
Integrity chain starts at upload	Starts inside the instrument, at the moment of digitisation.
External audio splitter and button box to ship and explain	Absorbed into the device. Fewer parts, fewer instructions, fewer failure modes.
Participant must remember to upload	Nothing to remember.

The full stack, end to end

Everything described so far in this part is one layer of a chain that runs from a disc of silver chloride on someone's scalp to a hashed file in Brussels. It is worth seeing the whole of it at once, because the design decisions only make sense in relation to each other.

Figure 28 — The full stack, from electrode to stored evidence

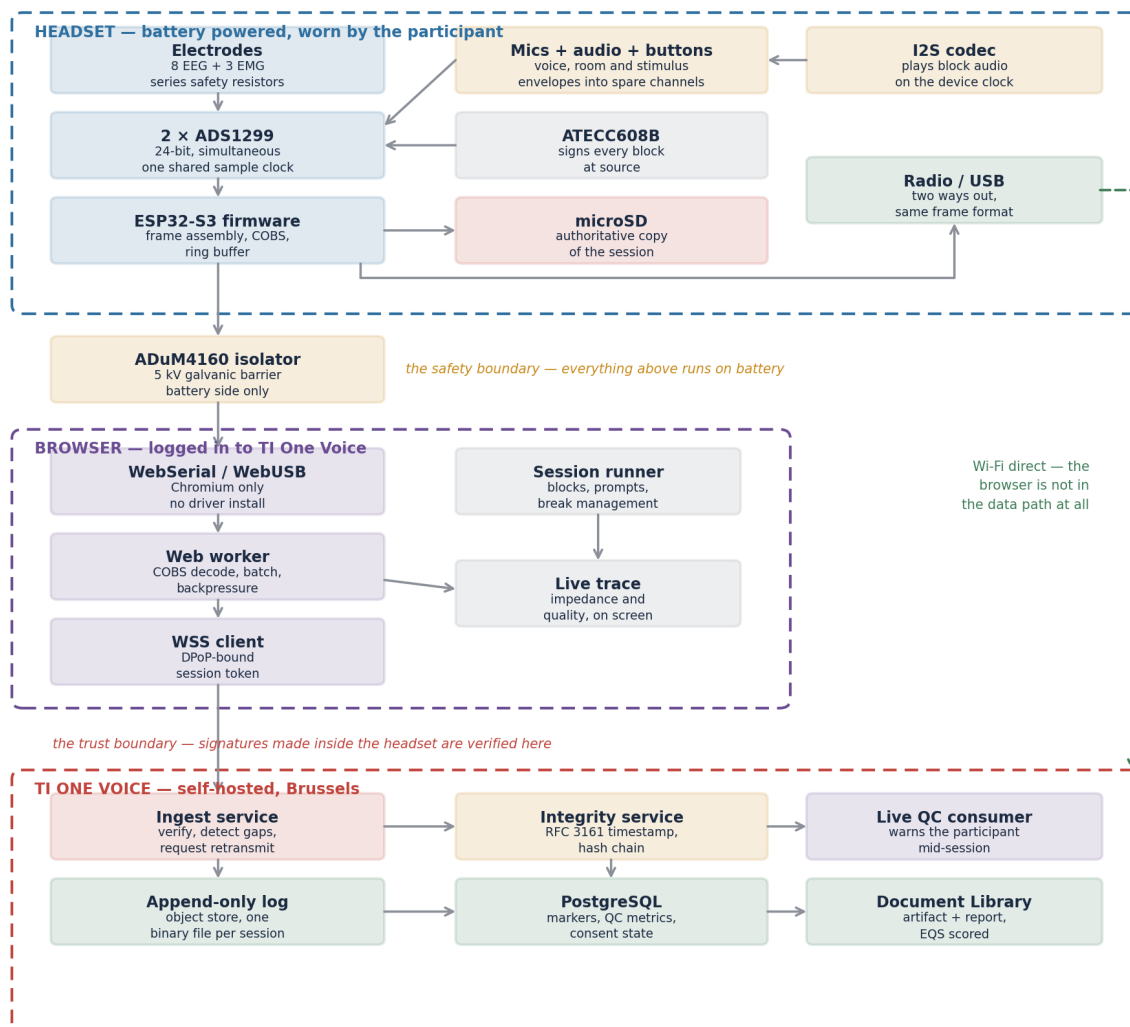


Figure 28 — Every component from electrode to stored evidence. Three boundaries matter: the isolator separates battery from mains, the browser sandbox separates the page from the device, and the ingest service separates what is claimed from what is verified.

Three properties of this arrangement are worth stating explicitly, because each one removes a category of problem rather than managing it.

- **Timing is established once, at the converter.** The audio envelope, the button states, the muscle channels and the brain channels are fields of a single sample. Nothing downstream — not the browser, not the network, not the server — can disturb a relationship that was fixed before any of them saw the data.
- **Integrity is established once, inside the instrument.** The secure element signs each block at source. The browser is therefore trusted for liveness but not for integrity, which is what makes it acceptable to route data through a general-purpose web page at all.
- **One frame format crosses every boundary.** The bytes leaving the microcontroller over USB and the frames arriving at the ingest service over a websocket have the same structure. One parser, one test suite, and swapping the USB path for the Wi-Fi path replaces two components without touching anything else.



Getting from a USB socket to a web page

The participant plugs the headset in, opens TI One Voice, and the site finds the device. That is the goal. It is achievable on a majority of desktops and on no iPhone at all, and the reason is worth understanding before committing to an architecture.

Browsers deliberately restrict access to attached hardware. Two of the relevant APIs have been formally declined by two of the four major engines, on the grounds that a web page able to talk to arbitrary USB devices is a fingerprinting surface and a route to reflashing hardware. Those are position statements rather than backlog items, and the right planning assumption is that they will not change.

Interface	Chromium desktop	Chromium Android	Firefox	Safari
WebSerial	Yes	No	Declined	Declined
WebUSB	Yes	Yes	Declined	Declined
WebHID	Yes	No	Declined	Declined
Web Bluetooth	Yes	Yes	Declined	Declined
Websocket to the device over Wi-Fi	Yes	Yes	Yes	Yes

The last row is the reason the direct Wi-Fi topology is the destination rather than a fallback. It is the only transport that works everywhere, and it happens also to be the one that keeps a mains-powered laptop out of the electrical path to the participant's head.

What plug and play can actually mean

A page cannot silently enumerate USB devices — the first connection always requires a click. But it can come very close to invisible after that. On load, the page asks the browser for devices the participant has already authorised; if the headset is among them it connects with no interaction at all. A connect event fires when a previously authorised device is plugged in. Only the very first session shows a picker, and that picker can be filtered to the device's own vendor and product identifiers so it contains exactly one entry.

So the honest description is: one click, once, ever. Every subsequent session is genuinely automatic. If neither interface is available — Safari, Firefox, an iPad — the page detects that and routes the participant to Wi-Fi provisioning instead, which is a slightly longer setup but works.

What this requires of the firmware

- **A composite USB device.** One serial-class interface and one vendor-specific interface, so the same hardware serves both the WebSerial path and the WebUSB path without a mode switch.
- **Descriptors that make Windows bind automatically.** A platform capability descriptor advertising Microsoft OS 2.0 support, so Windows loads a generic driver on its own. Without this, the WebUSB path on Windows requires the participant to run a driver installation tool, which for this population is where the setup ends.



- **A serial number that means something.** Populated from the secure element, so the server can bind a stream to a registered instrument rather than to whatever claims to be one.
- **Byte-stuffed framing.** Consistent overhead byte stuffing gives an unambiguous frame delimiter, so a decoder that joins a stream mid-flight or loses a byte resynchronises at the next boundary instead of producing plausible nonsense.
- **A ring buffer and an explicit gap marker.** Serial flow control cannot be relied on. If the host stalls, the device drops the oldest samples and says so in the stream. Silent loss is the one failure mode that would corrupt the science rather than merely interrupt it.
- **A command channel on the same interface.** Start and stop block, load block audio, set per-channel gain, run an impedance check, exchange clock offsets, begin a signed firmware update.

Bandwidth is not a consideration. Sixteen channels at four bytes and a thousand samples per second is sixty-four kilobytes per second, against a full-speed USB budget of roughly one and a half megabytes. The link runs at a few per cent of capacity.

What it requires of the browser code

The read loop belongs in a dedicated worker thread, not on the thread that renders the interface. If the participant scrolls, or the live trace repaints, or the operating system decides to garbage-collect, a read loop on the main thread will drop samples. The worker decodes frames, batches them into roughly twenty-millisecond chunks, and pushes them over the websocket; the interface receives a downsampled copy for display at around thirty frames per second.

Backpressure is handled by buffering rather than dropping. If the uplink slows, the worker accumulates and the session continues, because the device is writing its own copy to the memory card regardless. A slow connection degrades the live view, not the recording.

The safety conflict that has to be resolved before any USB connection is made. A USB cable to a mains-powered laptop places a galvanic path between the electrical supply and electrodes on a person's head. That directly contradicts the battery-only rule stated earlier in this part. The resolution is an inline isolator on the device side of the cable — a full-speed-capable digital isolator providing several kilovolts of separation, with the isolated side powered from the headset's own battery. It costs ten to fifteen euros per unit and takes the kit from around \u20ac265 to around \u20ac280. It belongs on the board from the first revision, because retrofitting it means a new layout.

Build order

WebSerial first: it is a few hundred lines, needs no descriptor work beyond a standard serial class, and covers most desktop participants for Stage 0 and Stage 1. Add the vendor interface and the Windows descriptors when Android tablets become a recruitment constraint. Build Wi-Fi for Stage 2, when the cohort includes people on iPads and Firefox and the browser route starts costing enrolments rather than convenience.

The order to build it in

One recommendation, and it runs against the instinct to start soldering.

Do not build the fleet first. One or two boards, or a borrowed OpenBCI Cyton — which contains the same ADS1299, exposes the analogue inputs, and streams over Wi-Fi — will prove the entire architecture. Ingest service, sample-indexed timeline, live quality control, backfill, finalisation: all of it can be validated on two units in a matter of weeks, and the protocol problems found while they cost an evening rather than a production run.

Then build the fleet, once the design is frozen. Twenty-five units at around €305 is roughly €7,500 of parts and a great many evenings of assembly. That is entirely achievable with volunteers, and nothing in the streaming design above changes when the instrument underneath it is replaced. What would sink it is building twenty-five boards to a design that turns out to need revising.

Phase	Instrument	What it proves
1	Off-the-shelf amplifier with analogue inputs, tethered over USB to the browser	The frame format, the ingest service, the sample-indexed timeline, live quality control. Runs the lab study in Part 8 as well.
2	Same amplifier with its wireless module, streaming directly	The direct topology, backfill on disconnection, device authentication, session state machine over the network.
3	Self-built fleet units	Integrated audio envelope and buttons, hardware-signed frames, on-device stimulus playback, a form factor a participant will actually wear for two hours.

The scientific programme does not wait on phase three. Everything in Parts 3 to 6 can be run on phase-one hardware, and phase one is where you discover whether the timing chain behaves the way the design says it does.



Part 15 — The screens

Everything in Part 13 is architecture. This part is what the participant actually sees, because in a remote study the interface is not a presentation layer over the protocol — it is the protocol. There is no researcher in the room to notice a misplaced electrode, to repeat an instruction that did not land, or to see that someone has become distressed. The screens have to do all of that.

The module sits under Documenting in the left-hand navigation, alongside the Document Library, Interview-Me and the Evidence Timeline. That placement is deliberate rather than convenient: a recording session is a form of documentation, it produces an artifact with an integrity proof, and it belongs with the other things a participant has documented rather than in a separate scientific annex.

The mockups below use the platform's existing design system — the same navigation, palette, card treatment and typography as the rest of TI One Voice — so that nothing about the module feels like a bolted-on experiment.

14.1 Module landing

The entry screen has to answer three questions before anyone commits two hours: is my equipment ready, what is going to happen to me, and what happened to my previous sessions.



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EEG Sessions

Guided recording protocol · contributes to the cohort study

Start guided session

Your kit

Device	OBCI-8CH · SN 4471-A
Firmware	v3.1.2 · verified
Link	Wi-Fi direct · paired
Electrode gel	reorder soon
Device certificate	valid · signs at source
Last timing check	0.4 ms · passed

What this session involves

About two hours in total: roughly 40 minutes of setup, 100 minutes of recording split into six blocks, and two breaks. You can stop at any point without giving a reason, and a stopped session is not held against you.

You will be asked to speak aloud, to listen to recordings of your own voice, to press a button when you hear a voice, and to reply silently in your head. Each block is explained before it starts.

Your sessions

DATE	PROTOCOL	QUALITY	REPORT
2026-08-14	Full 120 min	Usable	PDF
2026-07-30	Full 120 min	Usable	PDF
2026-07-11	Full 120 min	Rejected	PDF
2026-06-28	Setup only	Practice	—

Rejected sessions stay in your Document Library with their quality report. Nothing is deleted on your behalf.

Cohort study

Findings are published at cohort level only. Your individual recordings are never interpreted for you, because a single person's result cannot be read reliably — the current findings page explains why.

Read current cohort findings

Screen 1 — EEG Sessions landing. Kit status including the last timing check, a plain-language description of what the session involves, session history with quality outcomes, and a route to the cohort findings.

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TI One Voice · EEG module

Three questions, answered up front

ACT 1 · BEFORE YOU START

proposal walkthrough · nothing recorded yet

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Documenting

01 Setup 02 Timing 03 Baseline 04 Blocks 1-2 05 Block 3 06 Blocks 4-5 07 Block 6 08 Finish

EEG Sessions

Kit status, what a session involves, and your history

Your kit

Headset	OneVoice 8ch · sn 0142	Linked
Last timing check	0.4 ms · 26 Aug	Pass
Electrode gel	2 of 5 sachets left	Low

What a session involves

About 2 hours 10 minutes in six blocks with two breaks. You will be asked to speak a syllable aloud, to listen to a recording of your own voice, to press a button while a voice is present, and to reply silently in your head.

You can stop at any point. An unfinished session still counts.

Session history

26 Aug 2026	2h 08m	Usable
12 Aug 2026	2h 11m	Usable
28 Jul 2026	0h 46m	Rejected
14 Jul 2026	2h 09m	Usable

Rejected sessions stay visible with their reports.

Cohort so far

71

usable sessions contributed

Findings are reported for the whole cohort. You will be told when there are any.

Start a session

The screen answers three questions before anyone commits two hours: is my kit ready, what is going to happen to me, and what happened to my previous sessions.

Sarah
Participant 1 · hears voices

0:00:00 of ~2h10

Animation frame for Screen 1 — the module landing as drawn in the walkthrough.

Two details in this screen carry more weight than they appear to. The first is that the description of the session says out loud that the participant will be asked to speak, to listen to their own voice, to press a button and to reply silently — before they start, not when each block arrives. The second is that rejected sessions remain visible in the history with their reports. A platform that quietly deletes failed attempts is a platform whose record cannot be audited, and this population has excellent reasons to be suspicious of records that tidy themselves.

14.2 Setup: electrode placement

Fitting eight wet electrodes without help is the single hardest thing being asked of a remote participant, and it is where most sessions will be lost if the interface does not carry it. The screen gives live per-channel impedance next to a placement diagram, and refuses to advance until every channel is in range.

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← EEG Sessions

01 Setup

02 Timing

03 Baseline

04 Blocks 1-2

05 Block 3

06 Blocks 4-5

07 Block 6

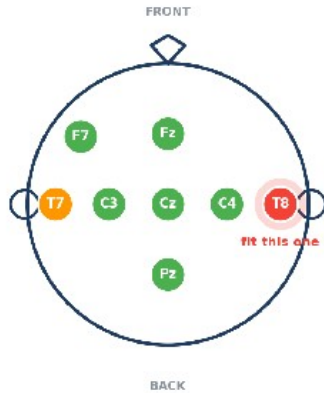
08 Finish

● Streaming · linked

Stop session

Fit the electrodes

Step 1 of 8 · nothing is recorded yet

Placement guide — electrode 6 of 8 (T8)


Part the hair, wipe the skin, apply a pea-sized amount of gel and press the disc until the reading on the right turns green. Take your time — a good contact now saves an hour later.

Live contact quality

CHANNEL	IMPEDANCE	STATUS
Fz	4.1 kΩ	Good
Cz	3.6 kΩ	Good
Pz	5.2 kΩ	Good
C3	4.8 kΩ	Good
C4	6.0 kΩ	Good
T7	7.9 kΩ	Adjust
T8	22.4 kΩ	Too high
F7	5.5 kΩ	Good
EMG cheek / chin	6.8 / 7.1 kΩ	Good

Two channels still need work. The session cannot start until every channel is below 10 kΩ — this is what stops a wasted two hours.

Skip channel

Continue

Screen 2 — Guided placement with live contact quality. The channel currently being fitted is highlighted on the head diagram, and Continue stays disabled while any channel is out of tolerance.



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01
Setup

02
Timing

03
Baseline

04
Blocks 1-2

05
Block 3

06
Blocks 4-5

07
Block 6

08
Finish

Fit the electrodes

Step 1 of 8 · nothing is recorded yet

Placement guide — electrode 6 of 8 (T8)

FRONT

BACK

Part the hair, wipe the skin, apply a pea-sized amount of gel and press the disc until the reading on the right turns green.

Live contact quality

CHANNEL	IMPEDANCE	STATUS
Fz	4.1 kΩ	Good
Cz	3.6 kΩ	Good
Pz	5.2 kΩ	Good
C3	4.8 kΩ	Good
C4	6.0 kΩ	Good
T7	4.9 kΩ	Good
T8	6.2 kΩ	Good
F7	5.5 kΩ	Good
EMG cheek / chin	6.8 / 7.1 kΩ	Good

All eight channels in range. The session can start.

Skip channel

Continue

0:14:05 of ~2h10

Every channel shows its impedance live. T8 reads twenty-two kilo-ohms — too high. Part the hair, wipe the skin, apply gel, and press until it turns green.

Sarah

Participant 1 · hears voices

Animation frame for Screen 2 — all channels in range, Continue enabled.

The hard gate is the point of the screen. It is tempting to make it a warning that can be dismissed, on the grounds that participants are adults who can judge for themselves. That would be a mistake: a participant cannot judge the consequence, because the consequence is invisible until the data is analysed weeks later and found to be unusable. Enforcing it here converts a silent loss into thirty seconds of visible annoyance.

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14.3 Setup: the timing self-test

This screen automates the oscilloscope check argued for in Part 7, and it is the reason the remote version of this protocol is credible at all.

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01 Setup

02 Timing

03 Baseline

04 Blocks 1-2

05 Block 3

06 Blocks 4-5

07 Block 6

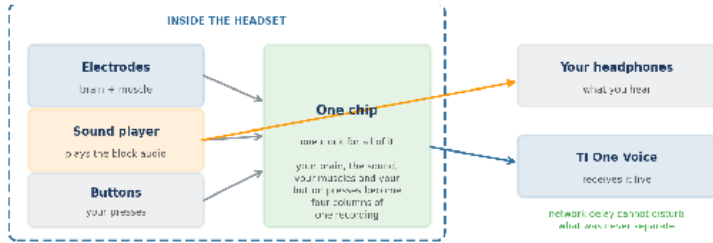
08 Finish

Timing self-test

Step 2 of 8 · checking that sound and brain data share one clock

Why this runs before every session

The measurements in blocks 1 to 5 depend on knowing exactly when each sound reached your ears — to within a few thousandths of a second. Neither your browser nor the network can promise that. So the headset plays the sound itself and records its own copy of it on the same chip, at the same instant, as your brain signal. Sound, muscle activity, button presses and EEG are not four recordings that get lined up afterwards; they are four columns of one recording.



Result

0.4 ms

measured spread across 40 test tones, on the headset's own clock

PASSED — limit is 5 ms

Checks

Audio envelope on channel 12	OK
Buttons latched to sample counter	OK
Link to server · backfill tested	OK
Microphone level	OK
Headphone output level	OK
Mains interference (50 Hz)	Low

Run again

Continue to baseline

Screen 3 — Automated timing verification. Forty test tones are played and the spread between the software marker and the amplifier's own copy of the audio is measured. Out of specification, and the event-related blocks do not run.



TI One Voice · EEG module

Nought point four milliseconds

ACT 3 · THE CLOCK

proposal walkthrough · nothing recorded yet

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01 Setup 02 Timing 03 Baseline 04 Blocks 1-2 05 Block 3 06 Blocks 4-5 07 Block 6 08 Finish

Timing self-test

Step 2 of 8 · checking that sound and brain data share one clock

Why this runs before every session

The measurements depend on knowing exactly when each sound reached your ears — to within a few thousandths of a second. Neither your browser nor the network can promise that. So the headset plays the sound itself and records its own copy of it on the same chip, at the same instant, as your brain signal.

Electrodes
brain + muscle

Sound player
plays the block audio

Buttons
your presses

One chip
one clock for all of it

Network delay cannot distort what was never separate.

Result

0.4 ms

measured across 40 test tones, on the headset's own clock

PASSED — limit is 5 ms

Checks

Audio envelope on channel 12 OK

Buttons latched to sample counter OK

Link to server · backfill tested OK

Microphone level OK

Headphone output level OK

Mains interference (50 Hz) Low

Streaming · linked Stop session

Sound, muscle, button presses and brain signal are four columns of one recording on one clock — not four recordings lined up afterwards. Out of specification, and the blocks do not run.

Sarah

Participant 1 · hears voices

Animation frame for Screen 3 — the timing self-test, passed.

It also explains why the check exists, in two sentences, with a diagram. That explanation is not decoration. A participant who understands why the timing check exists will take it seriously when it fails — and the rejected session in the library screen later in this part is exactly that failure.

14.4 Recording: the speaking block

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Session in progress · started 10:42

01 Setup

02 Timing

03 Baseline

04 **Blocks 1-2**

05 Block 3

06 Blocks 4-5

07 Block 6

08 Finish

Block 1 — say “ah”

Recording · about 12 minutes · block 1 of 6

● Streaming · linked

Stop session

Say “ah” out loud, briefly and comfortably.

Leave about two seconds between each one. Do not rush, and do not try to be exact.

MICROPHONE



Level is good. Your voice is being recorded so it can be played back in block 2.

63 / 100

vocalisations recorded

Roughly 4 minutes left in this block

Live signal quality

Channels in range	8 / 8
Trials kept so far	59 / 63
Blink rate	Normal
Jaw tension	Slightly high

Try to keep your jaw loose between each “ah”. Clenching adds noise we have to throw away.

Next: block 2 plays these same recordings back to you. Nothing is asked of you except to listen.

Screen 4 — Block 1. A single clear instruction, a microphone level meter, a trial counter, and live feedback on signal quality including a specific, actionable prompt about jaw tension.



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← EEG Sessions

01 Setup02 Timing03 Baseline04 Blocks 1-205 Block 306 Blocks 4-507 Block 608 Finish

Block 1 — say the syllable

Recording · about 18 minutes · block 1 of 6

Say “ah” when the circle turns blue

Speak at a comfortable volume. Then just listen.

WAIT

Microphone level

Progress

30

of 100 trials recorded

Quality

Channels in range	8 / 8	OK
Trials kept	24 / 30	OK
Blink rate	normal	OK
Jaw tension	raised	Relax jaw

0:35:52 of ~2h10

Blocks one and two. Sarah says a syllable aloud. Later she hears a recording of that same syllable played back to her.

Sarah

Participant 1 · hears voices

Animation frame for Screen 4 — the speaking block.

Note what is on the screen and what is not. There is one instruction, in one sentence, in large type. There is no live EEG trace — a scrolling waveform is fascinating and it invites the participant to watch their own brain activity and react to it, which is precisely what you do not want during a recording. The quality panel reports states rather than signals: channels in range, trials kept, blink rate, jaw tension. The one piece of feedback that asks for a behaviour change ("keep your jaw loose") is the one the participant can act on.





14.5 Recording: marking the voices

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Session in progress · started 10:42

01
Setup

02
Timing

03
Baseline

04
Blocks 1-2

05
Block 3

06
Blocks 4-5

07
Block 6

08
Finish

Block 3 — mark the voices
Recording · about 22 minutes · block 3 of 6

● Streaming · linked

Stop session

Hold the button down for as long as a voice is present. Release it when the voice stops.

You will also hear occasional short beeps. Ignore them — there is nothing to do about the beeps.

BUTTON HELD
6.4 s

Press the physical button on the box, not the screen.

THIS BLOCK SO FAR

Shaded segments are periods you marked. This is a record of what you reported, not an assessment of it.

14:08
elapsed in this block

Collected

Periods marked	11
Beeps during marked periods	31
Beeps during quiet periods	44
Target per group	60

The block extends automatically if either group is short at the end.

If nothing happens during this block, that is a valid result and you should not try to make something happen. Sit quietly and let it run.

Screen 5 — Block 3. The button state is shown large, with a timeline of marked periods accumulating below and running counts of probe tones falling inside and outside them.

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Session in progress · started 10:42

01 Setup

02 Timing

03 Baseline

04 Blocks 1-2

05 Block 3

06 Blocks 4-5

07 Block 6

08 Finish

Block 3 — mark the voices

Recording · about 22 minutes · block 3 of 6

Hold the button down for as long as a voice is present. Release it when the voice stops.

You will also hear occasional short beeps. Ignore them.

4.3 s

Press the physical button on the box, not the screen.

THIS BLOCK SO FAR

Shaded segments are periods you marked. This is a record of what you reported, not an assessment of it.

14:15

elapsed in this block

Collected

Periods marked6

Beeps during marked periods19

Beeps during quiet periods26

Target per group60

If nothing happens during this block, that is a valid result and you should not try to make something happen. Sit quietly and let it run.

Streaming · linked

Stop session

1:13:29 of ~2h10

The timeline shows Sarah her own button presses. The wording underneath is the most carefully chosen text in the module: this is a record of what you reported, not an assessment of it.

Sarah

Participant 1 · hears voices

Animation frame for Screen 5 — marking the voices.

The wording underneath the timeline is the most carefully chosen text in the entire module: this is a record of what you reported, not an assessment of it. The screen is showing the participant their own button presses. It is not showing them a finding, and it must not be readable as one — a coloured timeline is exactly the kind of graphic that gets screenshotted and captioned as evidence.

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The note in the corner is doing similar work in the opposite direction. If nothing happens during a twenty-minute block, some participants will feel they have failed, or that the study will judge them. The block explicitly tells them that a quiet block is a valid result and that they should not try to produce one. Without that instruction, the data from this block is worth considerably less.

14.6 Recording: replying without speaking

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Session in progress · started 10:42

01
Setup

02
Timing

03
Baseline

04
Blocks 1-2

05
Block 3

06
Blocks 4-5

07
Block 6

08
Finish

Block 5 — reply without speaking

● Streaming · linked

Stop session

Recording · about 22 minutes · block 5 of 6

If a voice says something and you answer it in your head, press the reply button the moment you begin your answer.

Do not move your lips and do not make any sound. Answer exactly as you normally would — do not compose anything special for the recording.

VOICE BUTTON
not held

REPLY BUTTON
pressed 1.2 s ago

Earlier in block 4 you practised this twice: once counting silently to yourself, and once answering a recorded question in your head. Those two practice blocks are what make this one readable.

18
replies marked

Target 30 · about 9 minutes left

Recorded alongside

Speech-muscle channels

Active

Frontal channels

Clean

Microphone

Silent

The microphone confirms you stayed silent. It is not used for anything else in this block.

Feeling pressured or distressed? Stop the session. It does not have to be finished, and an unfinished session is still a contribution.

Screen 6 — Block 5, the talking-back condition. Two hardware buttons with distinct states, a reminder of the two practice conditions completed in block 4, and a visible route out of the session.



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← EEG Sessions

01 Setup

02 Timing

03 Baseline

04 Blocks 1-2

05 Block 3

06 Blocks 4-5

07 Block 6

08 Finish

Block 5 — reply without speaking

Recording · about 20 minutes · block 5 of 6

Streaming · linked Stop session

Reply in your head. Do not say it aloud.

Press START as you begin the reply, END when you finish.

START

END

WAITING FOR A VOICE

PRACTICE CONDITIONS COMPLETED IN BLOCK 4

- ✓ Say the reply out loud **done**
- ✓ Imagine hearing the reply **done**
- Reply silently to the voice **now**

Silence check

Cheek EMG	below threshold	Silent
Chin EMG	below threshold	Silent
Microphone	no speech detected	Silent
Replies marked	15	

This is a quality check, not surveillance.

You can stop at any time

This block asks you to engage with something many people find frightening, for twenty minutes, alone. You can stop now and the session still counts as a contribution.

Sensors on the cheek and chin confirm the mouth stayed still, and the microphone confirms silence. That is a quality check rather than surveillance, and the screen says which.

Sarah
Participant 1 · hears voices

1:40:20 of ~2h10

Animation frame for Screen 6 — replying without speaking, with the silence check.

This is the block where distress is most likely, because it asks people to engage deliberately with an experience many of them find frightening, for twenty minutes, alone. The stop control is on every screen in the module; on this one there is also an explicit invitation to use it, and the statement that an unfinished session still counts as a contribution. Both are there because the alternative — someone pushing through a block they want to stop because they think stopping wastes everyone's time — is a harm the design can prevent for the cost of two sentences.



The panel confirming that the microphone registered silence is a quality check rather than surveillance, and the screen says so. In a study asking people to do something in their heads while wired to a machine in their own home, the difference between a check and a watch has to be stated rather than assumed.



14.7 Upload and custody

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Finish

Session complete

Recorded 10:41–12:39 · streamed live throughout · nothing left to send

The stream

Frames received356,412 of 356,412

Gaps detected and backfilled3 · all recovered

Reconciled against headset cardidentical

Your Wi-Fi dropped for eleven seconds at 11:02. The headset kept recording and sent the missing stretch when it reconnected, so no part of the session was lost.

Integrity

Signed inside the headset, per blockVerified

SHA-256a3f1...9c02

RFC 3161 timestampDone

Blockchain anchorQueued

Each block was signed by the headset's own key at the moment it was digitised, so the chain of custody starts inside the instrument rather than at the server.

What happens next

1. Quality was checked continuously during the session — nothing to wait for.

2. The recording is deposited in your Document Library with its integrity proof.

3. A session report is generated and filed alongside it.

4. De-identified measurements join the cohort analysis.

Because the session was checked while it ran, you already know it is usable. Anything that had gone wrong would have been raised at the time, not weeks later.

Your consent for this session

Research use · protocol v2.1Granted

Evidentiary exportNot granted

These are separate choices and you can change either at any time.

Screen 7 — Resumable upload with the integrity chain shown as it is built, what happens next in four steps, and the two consents for this session displayed separately.



TI One Voice · EEG module

Two consents, shown separately

ACT 9 · CUSTODY

proposal walkthrough · nothing recorded yet

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← EEG Sessions

01 Setup 02 Timing 03 Baseline 04 Blocks 1-2 05 Block 3 06 Blocks 4-5 07 Block 6 08 Finish

Finalising the session

Step 8 of 8 · the recording has been arriving all along

Streaming as it happened

Headset
one clock, one frame

16 kB/s

TI One Voice
append-only ingest

Sustained rate

16 kB/s

Session so far

97 MB

Dropouts backfilled

3

Gaps outstanding

0

The headset holds several minutes in a ring buffer, so a dropped connection costs the live view — not the session.

Sealed at finalisation

Hashed on the participant's device

a41f 90c2 7e18 ...

Signed by the headset key

sn 0142 · ES256

Timestamp token

RFC 3161 · 11:07:44

Anchored

Consent for this session

Research use in the cohort analysis

Granted

Evidentiary export

Not granted

2:07:36 of ~2h10

Research use in the cohort analysis is granted. Evidentiary export is not. She can see, per session, exactly what she has agreed to.



Sarah
Participant 1 · hears voices

Animation frame for Screen 7 — finalising the session, with both consents shown separately.

Showing the hash being computed on the participant's own device before transmission is not a technical flourish. It is the visible form of a claim the platform makes elsewhere in prose: that the chain of custody begins with the participant rather than with the server. Anyone who has read the Document Library documentation will recognise the pattern, and seeing it applied to a new artifact type is what makes the new module feel like part of the same system.



The two consents shown side by side — research use granted, evidentiary export not granted — make the purpose separation from Part 13 concrete at the moment it matters. A participant can see, per session, exactly what they have agreed to.

14.8 The session report

This screen is the one that required the most restraint to design, and it is where the constraint argued for in Part 13 becomes visible.

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← EEG Sessions

Session report · 26 August 2026
Quality and protocol compliance · filed in your Document Library

Download PDF

Result
All quality gates passed. This session enters the cohort analysis.

USABLE

Quality gates

CHECK	MEASURED	GATE
Impedance, worst channel	8.1 kΩ	Pass
Mains interference	low	Pass
Timing jitter	1.8 ms	Pass
Trials kept — block 1	91 / 100	Pass
Trials kept — block 2	88 / 100	Pass
Probes, voice periods	64	Pass
Probes, quiet periods	71	Pass
Silent replies marked	29	Marginal
Channel health	8 / 8	Pass

Protocol compliance
All six blocks completed in the prescribed order, both breaks taken, comprehension checks passed before each block. Protocol v2.1, no deviations recorded.

Your contribution

Sessions contributed	3
Usable sessions	3
Cohort size to date	71

What this report does not say. It does not tell you whether anything was detected in your recording. Individual results in this kind of measurement overlap far too much between people to be read one at a time, and a number that looked like an answer would mislead you in whichever direction it pointed. Findings are reported for the whole cohort, and you will be told when there are any.

Screen 8 — The generated session report. Quality gates with measured values, protocol compliance, a contribution record — and an explicit statement of what the report deliberately does not say.



TI One Voice · EEG module

And nothing about the person

ACT 10 · THE REPORT

proposal walkthrough · nothing recorded yet

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Finish

Session report · 26 August 2026

Quality and protocol compliance · filed in your Document Library

Result

All quality gates passed. This session enters the cohort analysis.

USABLE

Quality gates

CHECK	MEASURED	GATE
Impedance, worst channel	8.1 kΩ	Pass
Mains interference	low	Pass
Timing jitter	1.8 ms	Pass
Trials kept — block 1	91 / 100	Pass
Trials kept — block 2	88 / 100	Pass
Probes, voice periods	64	Pass
Probes, quiet periods		
Silent replies marked		
Channel health		

Protocol compliance

All six blocks completed in the prescribed order, both breaks taken.
Protocol v2.1, no deviations recorded.

Your contribution

Sessions contributed	3
Usable sessions	3
Cohort size to date	71

What this report does not say.

It does not tell you whether anything was detected in your recording.
Individual results in this kind of measurement overlap far too much
between people to be read one at a time.

No individual verdict is given. At the level of one person the measurement genuinely does not support one, and giving her one anyway would be the most damaging thing the platform could do.



Sarah

Participant 1 · hears voices

2:10:00 of ~2h10

Animation frame for Screen 8 — the session report and what it does not say.

Everything here is about the recording: impedance, interference, timing jitter, trials surviving rejection, probe counts, protocol compliance. Nothing here is about the person. The note in the corner states that directly, explains why in one sentence, and does not apologise for it.



I expect that note to be unpopular, and I think it should stay exactly as written. A participant who has waited years for an instrument to take them seriously will open this report hoping for a verdict. Giving them one would be the single most damaging thing the platform could do, in either direction, because the measurement genuinely does not support it at the level of one person. Saying so plainly, on the screen, at the moment they are looking for it, is more respectful than a hedge — and it is the only version of this screen that a reviewer would accept.

The contribution counter is the honest substitute for a verdict. It tells the participant something true and something that matters: that their two hours went into a dataset, that the dataset is growing, and that findings will come from it.



14.9 Deposit in the Document Library

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Document Library > EEG

Document Library

Filtered: EEG session artifacts

Export all

Session 2026-08-26 — raw recording

Timestamped EQS 94

eeg_recording.bdf · voice_recordings.wav · events.tsv · session.json · 312 MB

SHA-256	a3f1c8d2...77b49c02
RFC 3161 token	issued 26 Aug 2026 12:51:07 UTC · verified
Blockchain anchor	block 21,884,102 · confirmed
Protocol	guided-eeg v2.1 · device OBCI-8CH SN 4471-A · fw 3.1.2

Session report 2026-08-26

Usable

Generated PDF · quality, compliance and contribution record · 4 pages

Open

Share with counsel

Session report 2026-07-11

Rejected

Timing jitter 41 ms — splitter was not connected. Kept for completeness; excluded from analysis.

Session report 2026-08-14

Usable

Generated PDF · 4 pages

Every artifact here carries its own integrity proof and can be exported independently. Sharing is per-document and per-recipient — nothing is shared by default, including with the research team, which receives only de-identified measurements.

Screen 9 — The session artifacts filed alongside everything else the participant has documented, with hash, timestamp token, blockchain anchor, protocol and device provenance attached.

(The Document Library view is not shown in the animation; the version 2 mockup applies.)



The rejected July session is visible in this view with its reason recorded: timing jitter of 41 milliseconds after the headset fell back to the tethered path. That entry is worth more than it looks. It demonstrates that the quality gates are real, that failures are recorded rather than hidden, and that the reason is specific enough to prevent a repeat. A library containing only successes would be a weaker evidentiary record, not a stronger one.

14.10 Interface rules that follow from the science

The screens above are one arrangement of the module, and the arrangement will change. The rules underneath them should not, because each one exists to prevent a specific failure described earlier in this document.

Rule	What it prevents
One instruction per screen, in plain language, in large type.	Task misunderstanding, which in a remote session is invisible and unrecoverable. Comprehension checks before each block for the same reason.
Hard gates on impedance and timing, not dismissible warnings.	Two hours of unusable data, discovered weeks later when the participant cannot be reached.
No live EEG trace during recording.	Participants watching and reacting to their own signal, which contaminates the very measurement being taken.
Button timelines are labelled as records of what was reported, never as findings.	A screenshot of the interface being circulated as instrumented proof of something the interface never claimed.
A stop control on every screen, with no justification required.	Someone continuing through distress because the interface made stopping feel like failure.
No individual interpretation anywhere in the module.	A confident-looking verdict the underlying measurement cannot support, handed to the person least able to discount it.
Failed sessions stay visible with their reasons.	A self-tidying record, which is worth nothing evidentially and which this community has every reason to distrust.
Quality feedback names an action or says nothing.	Anxiety-inducing readouts the participant cannot act on. "Jaw tension high" is useful; a coherence value is not.

The test for any change to these screens. Ask whether the new version could produce, in a participant's mind, a belief the data does not support — in either direction. Most interface decisions in this module are ordinary usability decisions. The handful that are not are the ones where a well-



Rule	What it prevents
	intentioned improvement, such as showing someone their own results, would quietly convert a research instrument into a machine for generating false certainty.

Part 15b — The walkthrough, screen by screen

Part 15 argues for each participant screen. This part shows the whole flow, in the order the animated walkthrough presents it, as still frames — first the participant side, then the administrator side. The frames are mockups; every number on them is illustrative; nothing has been recorded. The cast is fictional: Sarah reports voices, Marcus is a control, Priya administers the study, Dr Okafor is the neurologist and Dr Lindqvist the signal-processing and statistics lead. They stand for the roles Part 19 asks for.



Frame 1 — Title. The walkthrough opens on the same disclaimer this document opens on: every screen is a mockup, no study has been run.

TI One Voice · EEG module

Who appears in this walkthrough

Five people appear. Two participants, a study administrator, a neurologist, and a specialist in signal processing and statistics.



Sarah

Participant 1 · hears voices



Marcus

Participant 2 · control



Priya

Study administrator

Frame 2 — Who appears. Two participants, an administrator, a neurologist and a statistician.



The participant side (Acts 1 to 10)

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EEG Sessions

Kit status, what a session involves, and your history

Your kit

Headset	OneVoice 8ch - sn 0142	Linked
Last timing check	0.4 ms - 26 Aug	Pass
Electrode gel	2 of 5 sachets left	Low

What a session involves

About 2 hours 10 minutes in six blocks with two breaks. You will be asked to speak a syllable aloud, to listen to a recording of your own voice, to press a button while a voice is present, and to reply silently in your head.

You can stop at any point. An unfinished session still counts.

Session history

26 Aug 2026	2h 08m	Usable
12 Aug 2026	2h 11m	Usable
28 Jul 2026	0h 46m	Rejected
14 Jul 2026	2h 09m	Usable

Rejected sessions stay visible with their reports.

Cohort so far

71 usable sessions contributed

Findings are reported for the whole cohort. You will be told when there are any.

Start a session

The screen answers three questions before anyone commits two hours: is my kit ready, what is going to happen to me, and what happened to my previous sessions.

Sarah
Participant 1 · hears voices

0:00:00 of ~2h10

Act 1 — The landing screen. Kit status with the last timing check, a plain-language description of what the session involves, session history including a rejected session with its reason, and the cohort-level findings link. Corresponds to Screen 1 of Part 15.

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Fit the electrodes

Step 1 of 8 · nothing is recorded yet

Placement guide — electrode 6 of 8 (T8)

FRONT

BACK

Part the hair, wipe the skin, apply a pea-sized amount of gel and press the disc until the reading on the right turns green.

Live contact quality

CHANNEL	IMPEDANCE	STATUS
Fz	4.1 kΩ	Good
Cz	3.6 kΩ	Good
Pz	5.2 kΩ	Good
C3	4.8 kΩ	Good
C4	6.0 kΩ	Good
T7	7.9 kΩ	Adjust
T8	22.4 kΩ	Too high
F7	5.5 kΩ	Good
EMG cheek / chin	6.8 / 7.1 kΩ	Good

Two channels still need work. The session cannot start until every channel is below 10 kΩ — this is what stops a wasted two hours.

Skip channel Continue

0:01:09 of ~2h10

Step one is the hardest thing the study asks of a remote participant: fitting eight wet electrodes without help.

Sarah
Participant 1 · hears voices

Act 2 — Fitting the electrodes, alone. Live impedance per channel; T8 reads twenty-two kilo-ohms and is flagged. Corresponds to Screen 2.



TI One Voice · EEG module

Live contact quality

ACT 2 · SETUP

proposal walkthrough · nothing recorded yet

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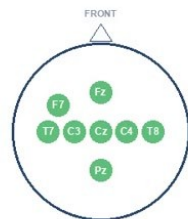
← EEG Sessions

- 01 Setup
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- 07 Block 6
- 08 Finish

Fit the electrodes

Step 1 of 8 · nothing is recorded yet

Placement guide — electrode 6 of 8 (T8)



Part the hair, wipe the skin, apply a pea-sized amount of gel and press the disc until the reading on the right turns green.

Live contact quality

CHANNEL	IMPEDANCE	STATUS
Fz	4.1 kΩ	Good
Cz	3.6 kΩ	Good
Pz	5.2 kΩ	Good
C3	4.8 kΩ	Good
C4	6.0 kΩ	Good
T7	4.9 kΩ	Good
T8	6.2 kΩ	Good
F7	5.5 kΩ	Good
EMG cheek / chin	6.8 / 7.1 kΩ	Good

All eight channels in range. The session can start.

Skip channel

Continue

Every channel shows its impedance live. T8 reads twenty-two kilo-ohms — too high. Part the hair, wipe the skin, apply gel, and press until it turns green.



Sarah

Participant 1 · hears voices

Act 2 — All eight channels in range. Continue was disabled until every channel was under ten kilo-ohms. The gate is deliberate: a bad contact is invisible until the data is analysed weeks later.



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Finish

Streaming · linked

Stop session

Timing self-test

Step 2 of 8 · checking that sound and brain data share one clock

Why this runs before every session

The measurements depend on knowing exactly when each sound reached your ears — to within a few thousandths of a second. Neither your browser nor the network can promise that. So the headset plays the sound itself and records its own copy of it on the same chip, at the same instant, as your brain signal.

Electrodes
brain + muscle

Sound player
plays the block audio

Buttons
your presses

One chip
one clock for all of it

Network delay cannot distort what was never separate.

Result

3.2 ms

measured across 40 test tones, on the headset's own clock

Checks

Audio envelope on channel 12

Buttons latched to sample counter

Link to server · backfill tested

Microphone level

Headphone output level

Mains interference (50 Hz)

OK

0:17:57 of ~2h10

Sarah

Participant 1 · hears voices

Step two. Forty test tones are played, and the spread between the software marker and the amplifier's own copy of the audio is measured.

Act 3 — The timing self-test. Forty tones are played and the spread between the software marker and the amplifier's own copy of the audio is measured. The first run reads 3.2 ms. Corresponds to Screen 3.

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01
Setup

02
Timing

03
Baseline

04
Blocks 1-2

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Block 3

06
Blocks 4-5

07
Block 6

08
Finish

Streaming · linked

Stop session

Timing self-test

Step 2 of 8 · checking that sound and brain data share one clock

Why this runs before every session

The measurements depend on knowing exactly when each sound reached your ears — to within a few thousandths of a second. Neither your browser nor the network can promise that. So the headset plays the sound itself and records its own copy of it on the same chip, at the same instant, as your brain signal.

Electrodes
brain + muscle

Sound player
plays the block audio

Buttons
your presses

One chip
one clock for all of it

Network delay cannot distort what was never separate.

Result

0.4 ms

measured across 40 test tones, on the headset's own clock

PASSED — limit is 5 ms

Checks

Audio envelope on channel 12	OK
Buttons latched to sample counter	OK
Link to server · backfill tested	OK
Microphone level	OK
Headphone output level	OK
Mains interference (50 Hz)	Low

0:20:47 of ~2h10

Sound, muscle, button presses and brain signal are four columns of one recording on one clock — not four recordings lined up afterwards. Out of specification, and the blocks do not run.

Sarah
Participant 1 · hears voices

Act 3 — Passed at 0.4 ms. Sound, muscle, button presses and brain signal are four columns of one recording on one clock — not four recordings lined up afterwards. Out of specification, and the event-related blocks do not run.

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Baseline

Step 3 of 8 - eight minutes, nothing asked of you

Sit still and breathe normally

Four minutes with your eyes open, then four with them closed. This records what this room and this headset look like when nothing is happening, so everything measured later has something to be compared against.

EYES OPEN
6:17 remaining

There is no live brain trace during the recording blocks. Watching your own signal invites you to react to it, which is precisely what the measurement cannot allow.

Signal quality

Fz

Cz

Pz

C3

Good

Good

Good

Good

Streaming · linked

Stop session

0:23:26 of ~2h10

Four minutes with eyes open, four with them closed. This records what the room and the headset look like when nothing at all is being asked.

Sarah

Participant 1 · hears voices

Act 4 — Baseline, eyes open. Four minutes eyes open, four eyes closed: what the room and the headset look like when nothing at all is being asked. (Not described in Part 8 of version 2; see the change list.)



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← EEG Sessions

01 Setup02 Timing03 Baseline04 Blocks 1-205 Block 306 Blocks 4-507 Block 608 Finish

Block 1 — say the syllable

Recording · about 18 minutes · block 1 of 6

Say “ah” when the circle turns blue

Speak at a comfortable volume. Then just listen.

WAIT

Microphone level

Progress

30

of 100 trials recorded

Quality

Channels in range	8 / 8	OK
Trials kept	24 / 30	OK
Blink rate	normal	OK
Jaw tension	raised	Relax jaw

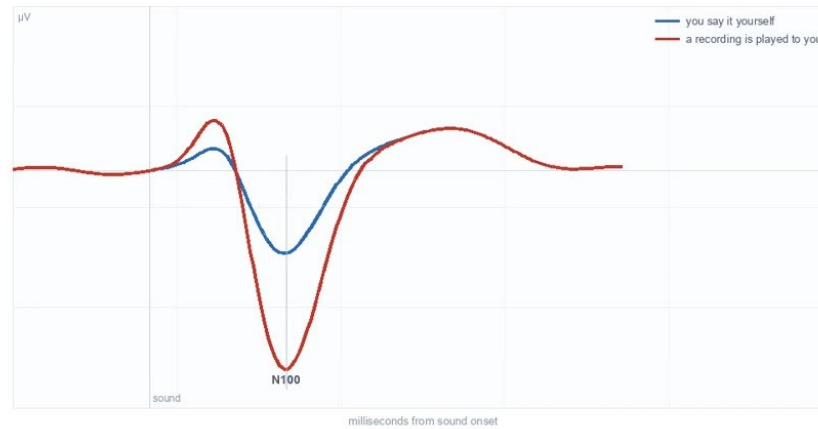
0:35:52 of ~2h10

Blocks one and two. Sarah says a syllable aloud. Later she hears a recording of that same syllable played back to her.

Sarah
Participant 1 · hears voices

Act 5 — Block 1, say the syllable. One instruction, a microphone level, a trial counter, and quality feedback that names an action ("jaw tension raised — relax jaw"). Corresponds to Screen 4.

The N100 — and why it gets smaller



Identical sound. Identical ears. The only difference is whether your own brain caused it — and the brain discounts what it caused itself.

What each result would mean

Suppression normal

Self-monitoring intact. The simple internal-speech account gets harder to hold.

Suppression reduced

Looks like the pattern already known from clinical voice-hearing.

No usable signal

The method failed, and that is reported as a failure.



Dr Lindqvist
Signal processing & statistics

Act 5 — The explainer shown between blocks. Identical sound, identical ears; the only difference is whether your own brain caused it. Three readings: suppression normal, suppression reduced, no usable signal. This is the Part 3 outcome table in the statistician's words.



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Session in progress · started 10:42

01 Setup

02 Timing

03 Baseline

04 Blocks 1-2

05 Block 3

06 Blocks 4-5

07 Block 6

08 Finish

Block 3 — mark the voices

Recording · about 22 minutes · block 3 of 6

Hold the button down for as long as a voice is present. Release it when the voice stops.

You will also hear occasional short beeps. Ignore them.

BUTTON HELD

4.3 s

Press the physical button on the box, not the screen.

THIS BLOCK SO FAR

Shaded segments are periods you marked. This is a record of what you reported, not an assessment of it.

14:15

elapsed in this block

Collected

Periods marked	6
Beeps during marked periods	19
Beeps during quiet periods	26
Target per group	60

If nothing happens during this block, that is a valid result and you should not try to make something happen. Sit quietly and let it run.

1:13:29 of ~2h10

The timeline shows Sarah her own button presses. The wording underneath is the most carefully chosen text in the module: this is a record of what you reported, not an assessment of it.

Sarah
Participant 1 · hears voices

Act 6 — Block 3, mark the voices. Button state shown large, a timeline of marked periods, running counts of probes inside and outside them. The wording under the timeline: this is a record of what you reported, not an assessment of it. Corresponds to Screen 5.



TI One Voice · EEG module

A quiet block is a valid block

ACT 6 · BLOCK 3

proposal walkthrough · nothing recorded yet

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Session in progress · started 10:42

01 Setup

02 Timing

03 Baseline

04 Blocks 1-2

05 Block 3

06 Blocks 4-5

07 Block 6

08 Finish

Block 3 — mark the voices

Recording · about 22 minutes · block 3 of 6

Hold the button down for as long as a voice is present. Release it when the voice stops.

You will also hear occasional short beeps. Ignore them.

BUTTON HELD

6.2 s

Press the physical button on the box, not the screen.

THIS BLOCK SO FAR

Shaded segments are periods you marked. This is a record of what you reported, not an assessment of it.

19:50

elapsed in this block

Collected

Periods marked	9
Beeps during marked periods	27
Beeps during quiet periods	39
Target per group	60

If nothing happens during this block, that is a valid result and you should not try to make something happen. Sit quietly and let it run.

If nothing happens, that is a valid result, and the screen says so — because a participant who feels they have failed will try to produce something, and then the data is worth much less.



Sarah
Participant 1 · hears voices

1:19:37 of ~2h10

Act 6 — A quiet block is a valid block. The screen says so, because a participant who feels they have failed will try to produce something, and then the data is worth much less.

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← EEG Sessions

01 Setup 02 Timing 03 Baseline 04 Blocks 1-2 05 Block 3 06 Blocks 4-5 07 Block 6 08 Finish

Block 5 — reply without speaking

Recording · about 20 minutes · block 5 of 6

Reply in your head. Do not say it aloud.
Press START as you begin the reply, END when you finish.

START END

REPLY IN PROGRESS

PRACTICE CONDITIONS COMPLETED IN BLOCK 4

- ✓ Say the reply out loud done
- ✓ Imagine hearing the reply done
- Reply silently to the voice now

Silence check

Cheek EMG	below threshold	Silent
Chin EMG	below threshold	Silent
Microphone	no speech detected	Silent
Replies marked	9	

This is a quality check, not surveillance.

You can stop at any time
This block asks you to engage with something many people find frightening, for twenty minutes, alone. You can stop now and the session still counts as a contribution.

1:36:40 of ~2h10



Sarah
Participant 1 · hears voices

Blocks four and five. Think the reply, do not say it. Two buttons mark where the reply begins and ends.

Act 7 — Block 5, reply without speaking. Two hardware buttons mark where the reply begins and ends; the three practice conditions from Block 4 are listed. Corresponds to Screen 6.



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← EEG Sessions

01 Setup 02 Timing 03 Baseline 04 Blocks 1-2 05 Block 3 06 Blocks 4-5 07 Block 6 08 Finish

Block 5 — reply without speaking

Recording · about 20 minutes · block 5 of 6

Reply in your head. Do not say it aloud.
Press START as you begin the reply, END when you finish.

START END

WAITING FOR A VOICE

PRACTICE CONDITIONS COMPLETED IN BLOCK 4

✓ Say the reply out loud done

✓ Imagine hearing the reply done

● Reply silently to the voice now

Silence check

Cheek EMG	below threshold	Silent
Chin EMG	below threshold	Silent
Microphone	no speech detected	Silent
Replies marked	15	

This is a quality check, not surveillance.

You can stop at any time

This block asks you to engage with something many people find frightening, for twenty minutes, alone. You can stop now and the session still counts as a contribution.

Streaming · linked Stop session

1:40:20 of ~2h10

Sensors on the cheek and chin confirm the mouth stayed still, and the microphone confirms silence. That is a quality check rather than surveillance, and the screen says which.



Sarah
Participant 1 · hears voices

Act 7 — The silence check. Cheek and chin EMG confirm the mouth stayed still; the microphone confirms silence. A quality check, not surveillance — and the screen says which.

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Block 5 — reply without speaking

Recording · about 20 minutes · block 5 of 6

Reply in your head. Do not say it aloud.
Press START as you begin the reply, END when you finish.

START END

REPLY IN PROGRESS

PRACTICE CONDITIONS COMPLETED IN BLOCK 4

✓ Say the reply out loud done

✓ Imagine hearing the reply done

● Reply silently to the voice now

Silence check

Cheek EMG	below threshold	Silent
Chin EMG	below threshold	Silent
Microphone	no speech detected	Silent
Replies marked	26	

This is a quality check, not surveillance.

You can stop at any time

This block asks you to engage with something many people find frightening, for twenty minutes, alone. You can stop now and the session still counts as a contribution.

Streaming · linked Stop session

This is the block where distress is most likely. The stop control is on every screen, and an unfinished session still counts as a contribution.

1:48:17 of ~2h10

Sarah

Participant 1 · hears voices

Act 7 — The way out. This is the block where distress is most likely. The stop control is on every screen; an unfinished session still counts as a contribution.



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Streaming · linked Stop session

Block 6 — the surprise test

Recording · about 9 minutes · block 6 of 6

Just listen. There is nothing to do.

Most of the tones are identical. Occasionally one is different. You do not have to react to it or count them — the automatic change detector in the brain responds whether or not you are paying attention.

standard tone oddball

If this response is normal, the hearing system's alarm for unexpected sound is working — which matters for interpreting everything measured earlier.

Mismatch response

µV

— standard — oddball

sound

milliseconds from sound onset

Difference wave peaks near 170 ms

This is the only screen in the module that shows a brain response while the session is running, and it is shown after the block, not during it.

Block six, run identically for Marcus, who does not hear voices. Mostly identical tones with occasional oddballs, testing whether the automatic change detector is working normally.

Marcus
Participant 2 · control

1:57:08 of ~2h10

Act 8 — Block 6 in the animation: the passive oddball block, run identically for Marcus. This is the one screen in the module that shows a brain response, and it is shown after the block. It conflicts with version 2, where Block 6 is the reliability repeat; see the change list.

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TI One Voice · EEG module

Two consents, shown separately

ACT 9 · CUSTODY

proposal walkthrough · nothing recorded yet

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Streaming · linked Stop session

Finalising the session

Step 8 of 8 · the recording has been arriving all along

Streamed as it happened

Headset
one clock, one frame

16 kB/s

TI One Voice
append-only ingest

Sustained rate	16 kB/s
Session so far	97 MB
Dropouts backfilled	3
Gaps outstanding	0

The headset holds several minutes in a ring buffer, so a dropped connection costs the live view — not the session.

Fallback path

On off-the-shelf hardware the session is recorded locally and uploaded afterwards — resumable, hashed first. Same custody steps, discovered late.

Sealed at finalisation

✓ Hashed on the participant's device
a41f 90c2 7e18 ...

✓ Signed by the headset key
sn 0142 · E5256

✓ Timestamp token
RFC 3161 · 11:07:44

● Anchored

Consent for this session

Research use in the cohort analysis
Granted

Evidentiary export
Not granted

Research use in the cohort analysis is granted. Evidentiary export is not. She can see, per session, exactly what she has agreed to.



Sarah
Participant 1 · hears voices

2:07:36 of ~2h10

Act 9 — Finalising. The headset has been streaming all along at sixteen kilobytes a second; the ring buffer backfilled three dropouts. Hashed on the participant's device, signed by the headset key, timestamped, anchored. Research use granted; evidentiary export not granted. Corresponds to Screen 7.



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← EEG Sessions

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Session report · 26 August 2026

Quality and protocol compliance · filed in your Document Library

Result

All quality gates passed. This session enters the cohort analysis.

Quality gates

CHECK	MEASURED	GATE
Impedance, worst channel		
Mains interference		
Timing jitter		
Trials kept — block 1		
Trials kept — block 2		
Probes, voice periods		
Probes, quiet periods		
Silent replies marked		
Channel health		

Protocol compliance

All six blocks completed in the prescribed order, both breaks taken.
Protocol v2.1, no deviations recorded.

Your contribution

Sessions contributed	3
Usable sessions	3
Cohort size to date	71

What this report does not say.

It does not tell you whether anything was detected in your recording.
Individual results in this kind of measurement overlap far too much
between people to be read one at a time.

The session report. Impedance, interference, timing jitter, trials surviving rejection, protocol compliance.

Sarah
Participant 1 · hears voices

2:10:00 of ~2h10

Act 10 — The session report. Impedance, interference, timing jitter, trials surviving rejection, protocol compliance. Corresponds to Screen 8.



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Session report · 26 August 2026

Quality and protocol compliance · filed in your Document Library

Result

All quality gates passed. This session enters the cohort analysis. **USABLE**

Quality gates

CHECK	MEASURED	GATE
Impedance, worst channel	8.1 kΩ	Pass
Mains interference	low	Pass
Timing jitter	1.8 ms	Pass
Trials kept — block 1	91 / 100	Pass
Trials kept — block 2	88 / 100	Pass
Probes, voice periods	64	Pass
Probes, quiet periods		
Silent replies marked		
Channel health		

Protocol compliance

All six blocks completed in the prescribed order, both breaks taken. Protocol v2.1, no deviations recorded.

Your contribution

Sessions contributed	3
Usable sessions	3
Cohort size to date	71

What this report does not say.

It does not tell you whether anything was detected in your recording. Individual results in this kind of measurement overlap far too much between people to be read one at a time.

No individual verdict is given. At the level of one person the measurement genuinely does not support one, and giving her one anyway would be the most damaging thing the platform could do.

Sarah
Participant 1 · hears voices

2:10:00 of ~2h10

Act 10 — What the report does not say. No individual verdict is given; the contribution counter is what replaces it.

The administrator side (Acts 11 to 16)

Nothing in Part 15 shows what the study team sees, and that omission matters, because the analysis side is where the discipline argued for in Parts 9 and 10 either becomes architecture or stays a promise. Every screen below is aggregated across the cohort. No individual is identifiable on any of them, and the banner saying so is fixed to the bottom of every screen.



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Data Analysis · administrator

Cohort overview

De-identified. No screen in this section shows an individual result.

Usable sessions
142
of 158 recorded

Participants
71
25 voice-hearing · 25 control · 21 clinical

Sites
6
4 countries

Protocol
v2.1
no deviations

Group composition — matched

Report voices (TI)		10
No voices (control)		10
Clinical comparison		8

Matching

Age band	matched	OK
Sex	matched	OK
Handedness	matched	OK
Headset model	identical	OK
Analyst blinding	group labels masked	Active

ADMIN ONLY

A different set of screens. From here on nothing is about any individual: seventy-one people, a hundred and forty-two usable sessions, group labels masked.

Priya
Study administrator

Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Act 11 — Cohort overview. Usable sessions against recorded, participants by group, sites, protocol version, group composition against target, matching checks — and one line that matters more than the rest: analyst blinding, group labels masked, active.

Step 1 — Does the instrument work at all?

The first thing checked is the control group, not the study group. Their own-speech N100 must be markedly smaller than the played-back N100. This is the decades-old effect the whole design rests on; if the controls do not show it, the rig is wrong and nothing downstream can be believed. It is the software form of the commercial-amplifier comparison in Part 14: a null result on an instrument that has not demonstrated the reference effect is uninterpretable. Only after this gate passes are the groups unmasked.



Sanity gate	Illustrative value	Gate
Suppression present in controls	yes	pass
Effect size (Cohen's d), own speech vs playback	large	pass
Split-half reliability	high	pass
Timing jitter, cohort median	a few milliseconds	pass

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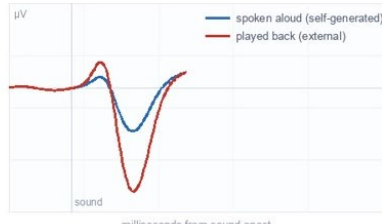
Data Analysis · administrator

Step 1 — does the instrument work at all?

The control group is checked before the study group is unmasked.

ADMIN ONLY

Controls · own speech versus the same sound played back



The dip about a tenth of a second after a sound is markedly smaller when the person caused the sound themselves. This is the decades-old effect the whole design rests on.

Sanity gate

Suppression present in controls	yes	Pass
Effect size (Cohen's d)	1.42	Large
Split-half reliability	0.81	Pass
Timing jitter, cohort median	1.6 ms	Pass

If the controls had not shown this, the rig would be wrong and nothing else in the dataset could be believed. This gate is checked first, and it is the one that can stop the analysis.

They show it clearly. The response to a self-spoken sound is much smaller than to the same sound played back. Only now is it worth unmasking the groups.



Dr Okafor
Neurologist

Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Act 12 — The sanity gate, passed. The dip a tenth of a second after a sound is markedly smaller when the person caused it. Only now is it worth unmasking the groups.

Step 2 — The pre-registered comparison

Three planned contrasts on the N100 suppression ratio, each reported as an effect size with an interval: study group against no-voices controls; study group against the clinical comparison group; controls against clinical. The outcome table in Part 3 covers what each pattern would mean. The animation illustrates one of them — the study group resembling the controls rather than the clinical group — and the screen text is worth keeping as written: a real result and an unwelcome one for a simple internal-speech account; it says nothing yet about where the experience comes from.

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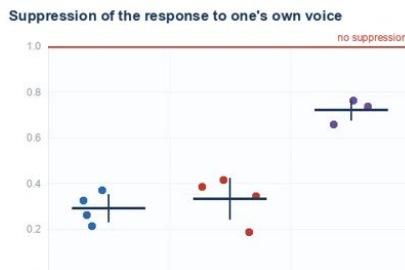
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Data Analysis · administrator

Step 2 — the pre-registered comparison

N100 suppression ratio · lower means more suppression



Suppression of the response to one's own voice

The group who report voices look like the controls, not like the clinical comparison group. Their self-monitoring appears intact.

Pre-registered test

TI vs control	$d = 0.11$	n.s.
TI vs clinical AVH	$d = 1.28$	$p < .001$
Control vs clinical	$d = 1.36$	$p < .001$

This on its own is a real result and an unwelcome one for a simple internal-speech account. It says the mechanism that mislabels inner speech in psychosis does not look broken in this group. It says nothing about where the experience comes from.

The group who report voices look like the controls, not like the clinical comparison group. Their self-monitoring appears intact.



Dr Okafor
Neurologist

Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Act 13 — The pre-registered comparison. Three groups on one axis, three named contrasts with effect sizes, and the illustrative reading.

Step 3 — Is the hearing system already busy?

Test Two's probe-tone contrast, inside against outside marked periods, with the same test run in the controls against their button-only block. Before the contrast is read, a fixed list of alternatives is modelled and regressed out, and the list is displayed. These are the confounds Parts 4 and 9 already name; what is new is that they are one pipeline step with one screen, so a reviewer can see what was removed and in which order.

- Jaw and neck muscle, from the EMG envelope.

- Eye movement and blinks, from the EOG channel.
- Attention and expectation — the slow anticipatory shift described in Part 4.
- Button-press motor activity, from the button-only control block.

TI One Voice · EEG module

The alternatives go first

ACT 14 · A SCENARIO

proposal walkthrough · nothing recorded yet

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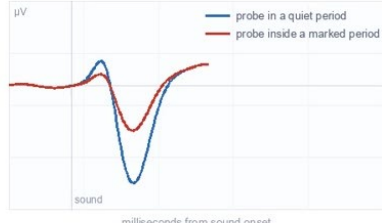
Data Analysis · administrator

Step 3 — is the hearing system already busy?

Response to probe tones inside marked periods versus quiet periods

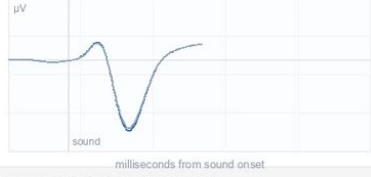
ADMIN ONLY

Reports voices · probe tone response



In this scenario the response to an identical probe is smaller while the participant reports a voice — as though the auditory system already has something to do.

Controls · same test



Alternatives modelled and removed

- Jaw and neck muscle
- Eye movement and blinks
- Attention and expectation
- Button-press motor activity

Before that means anything, the ordinary explanations are modelled and subtracted. Muscle tension. Eye movement. Attention. Expectation. The button press itself.



Dr Lindqvist

Signal processing & statistics

Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Act 14 — The scenario, with the alternatives modelled and removed before it is allowed to mean anything.



Step 4 — The cross-subject test that would matter

This test is absent from version 2 and should be pre-registered. A shared frequency is the first thing anyone would check for and the easiest thing to fake. What a genuine external event would look like across people is time-locking: a response at a consistent latency after the reported onset, in people who have never met. The display is a heatmap — one row per participant, one column per time offset from that person's marked onset — and a band forming in the same column across rows is the pattern that would be hard to explain by suggestion or by chance.

Reported	Meaning
Participants showing the band	a count out of the group size, with the threshold pre-registered
Median latency	milliseconds after the marked onset
Spread across people	plus or minus milliseconds; a narrow spread is the signature
Same test in controls	expected: no band; a band in controls means the test is picking up the button press

The same screen states what this is not: not a decoded word, not a location, not a transmitter. The headset cannot receive radio and no screen claims it can. This test belongs in the cross-subject module of Part 13 and in the pre-registration of Part 9. Its main weakness is the one Part 6 names: the marked onset lags the experience by an unknown, variable amount, so the band will be smeared. The provocation design of Part 4, which time-locks to speech offset, and the EMG timestamps of Part 5 are the two ways to sharpen it.

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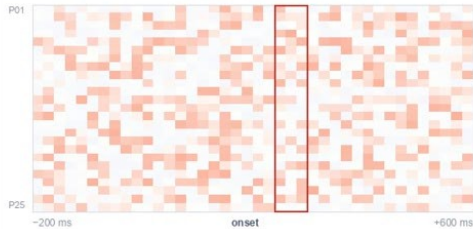
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Step 4 — the cross-subject test that would matter

Each row is one participant · each column a time offset from a marked onset

Probe response, time-locked to the reported onset



A band forming in the same column across people who have never met is the pattern that would be hard to explain by suggestion or by chance. Not a shared frequency — a shared latency.

What this is not

It is not a decoded word, not a location, and not a transmitter. An EEG headset is a voltmeter taped to a scalp. It cannot receive radio.

Consistency of latency

Participants showing the band	2 / 25
Median latency	116 ms
Spread across people	± 14 ms
Same test in controls	2 / 25

ADMIN ONLY

What would a genuine external signal look like across people? Not a shared frequency — that is the first thing anyone would check and the easiest thing to fake.



Dr Lindqvist
Signal processing & statistics

Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Act 15 — The cross-subject test as the animation draws it: rows are participants, columns are time offsets from the marked onset.

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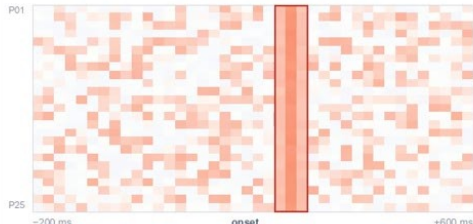
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Step 4 — the cross-subject test that would matter

ADMIN ONLY

Each row is one participant · each column a time offset from a marked onset

Probe response, time-locked to the reported onset



A band forming in the same column across people who have never met is the pattern that would be hard to explain by suggestion or by chance. Not a shared frequency — a shared latency.

What this is not

It is not a decoded word, not a location, and not a transmitter. An EEG headset is a voltmeter taped to a scalp. It cannot receive radio.

Consistency of latency

Participants showing the band	14 / 25
Median latency	116 ms
Spread across people	± 14 ms
Same test in controls	2 / 25

Even then: not a decoded word, not a location, not a transmitter. An EEG headset is a voltmeter taped to a scalp. It cannot receive radio, and no screen here will claim it can.



Dr Okafor
Neurologist

Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Act 15 — What it would not be. Even a clean band is not a decoded word, a location or a transmitter.

Step 5 — The discipline around the number

A checklist rendered as a screen, each item timestamped: hypotheses registered before recording (Open Science Framework, with the date); analysis run with group labels masked, unmasked only at Step 1; time windows fixed in advance, 80 to 130 milliseconds; every test in the family corrected, with the corrected q reported alongside the raw p; exploratory results labelled as such. The headline is an effect size with a 95 per cent interval, never a p-value alone. Twenty-five per group is enough to see a large effect and not enough to settle a small one, and the screen says so.



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Step 5 — the discipline around the number

Pre-registration, blinding and correction, applied before anything is claimed

How this analysis was constrained

- ☒ Hypotheses registered before recording OSF · 14 Mar 2026
- ☒ Analysis run with group labels masked unmasked at step 2
- ☒ Time windows fixed in advance 80-140 ms
- ☒ Every test in the family corrected Benjamini-Hochberg
- ☐ Exploratory results labelled as such

Reported as

d = 0.83
95% CI [0.31, 1.34]
q = .011 after correction

An effect size with an interval, not a headline. Twenty-five and twenty-five is enough to see a large effect and not enough to settle a small one.

And if it comes out the other way.

The most likely outcome remains the ordinary one: that the pattern matches internally generated speech. That result gets published too. The method was fixed in advance precisely so it cannot be bent once the numbers are in.

Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Reported as an effect size with a confidence interval. Twenty-five and twenty-five is enough to see a large effect and not enough to settle a small one.

Dr Lindqvist
Signal processing & statistics

Act 16 — The discipline around the number. Every constraint on the analysis, with a date, next to the number it constrains.

Then the sentence that has to survive every redesign of this screen: the most likely outcome remains the ordinary one — that the pattern matches internally generated speech — and that result is published too. The method was fixed in advance precisely so that it cannot be bent once the numbers are in.



Part 16 — What this design cannot do

I have said versions of this throughout, but it belongs in one place, stated without softening.

- **It cannot detect a transmission.** An EEG amplifier is not a receiver. Whatever this study finds, it will not be evidence about the presence or absence of anything arriving from outside.
- **It cannot decode content.** Eight electrodes on a scalp cannot recover words. Anyone claiming otherwise with this equipment is mistaken or selling something.
- **It cannot locate the source in the brain.** Eight sensors do not support source localisation. Statements about which brain region is responsible are not available from this design.
- **It cannot tell any individual anything about themselves.** This is entirely a group-level comparison. Individual variation in these measures is wide enough that one person's score is uninterpretable. Offering individual testing on this basis would be misleading, and the demand for it would be considerable, which is exactly why the line has to be drawn in advance and held.

Version 3 note: Part 16c re-examines this line with the arithmetic and defines what, bounded, can be said to one person after three sessions. The verdict — Outcome A or Outcome B for one individual — remains unavailable, for the reasons given there.

- **It cannot rule out a disconnected internal generator.** The external reviewer of this design put the alternative in one sentence: "there is a brain region X generating intrusive voices, this brain region is not wired sufficiently with other regions so that when it generates voices, it is perceived by the other parts of the brain as not self-generated and also a surprise." Such a generator would produce an undiscounted N100, no readiness potential, and surprise responses — from inside the head. Every in-rig discriminator in this design is blind to the difference between that and an external input. Only the lower-pathway tests of Part 16b, on the second instrument, can separate them, and only in one direction: see the known-explanations section there.
- **No amount of modelling repairs any of the above.** A classifier trained on data that does not contain the answer will still output a number. The limits in this list are limits of the instrument, and no analysis, however sophisticated, adds information that was never recorded.

What it can do is narrow the space. At present, the debate about this population runs almost entirely on assertion. A properly run version of this study would replace one piece of that assertion with a number. That is a smaller contribution than anyone on either side wants, and a larger one than the field currently has.

Part 16b — The other branch: what Outcome B would require

Parts 3 to 6 are written to survive either outcome, but the document spends almost all of its words on the branch where the voices behave like inner speech. The walkthrough spends its last five acts on the other branch, and the walk exposes two things version 2 does not say: which measurements would separate the two outcomes, and where the two-thousand-euro instrument stops.

The fork, decided in advance

Measure	Outcome A — behaves like inner speech	Outcome B — behaves like an external sound
N100 to the reported voice	reduced — the brain discounts it	full size — not discounted
The second before the reported onset	a motor build-up is present	no build-up; it arrives cold
Order of activation	frontal leads temporal	temporal leads frontal
Reading	internally generated	not generated by the ordinary speech-production route

TI One Voice · EEG module
Two outcomes, decided in advance

ACT 17 · THE FORK
proposal walkthrough · nothing recorded yet

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Two outcomes, and the design has to survive both
The measurements below are what separates them — decided before the data came in

OUTCOME A
The voice behaves like inner speech

OUTCOME B
The voice behaves like an external sound

So consider the other branch. Suppose the voices do not behave like inner speech at all. The design has to survive that outcome too, and the measurements that separate the two were fixed before any data came in.


Dr Okafor
Neurologist

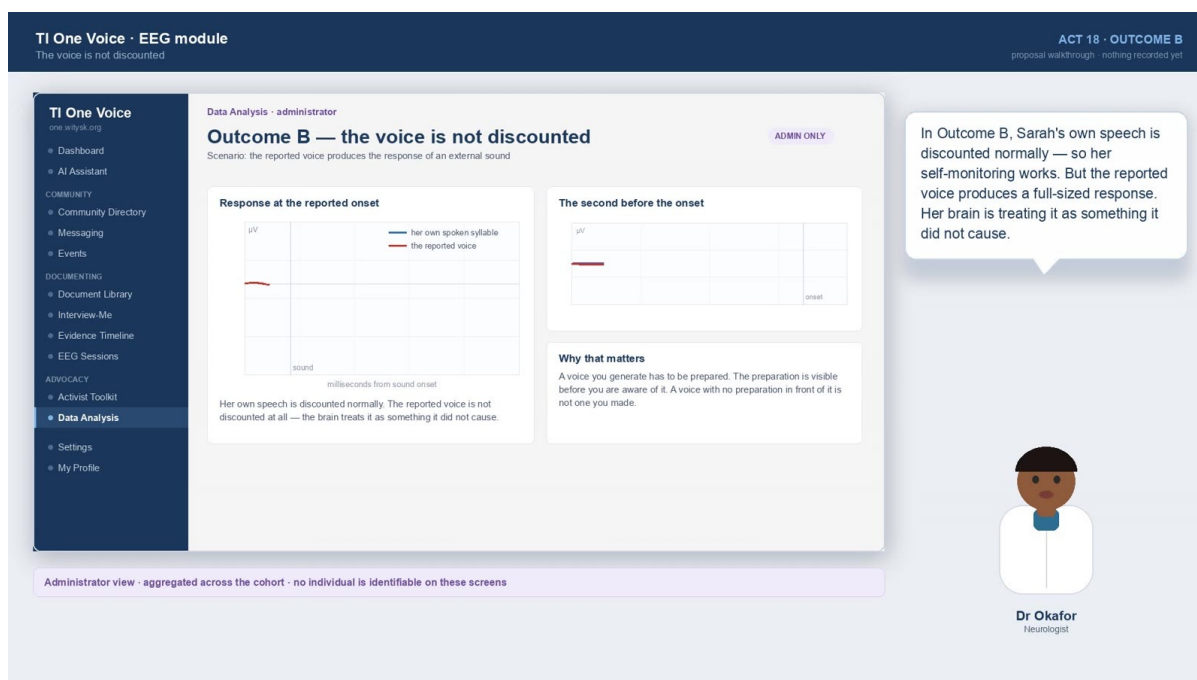
Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Act 17 — The fork. Two outcomes, and the measurements that separate them, decided before the data came in.

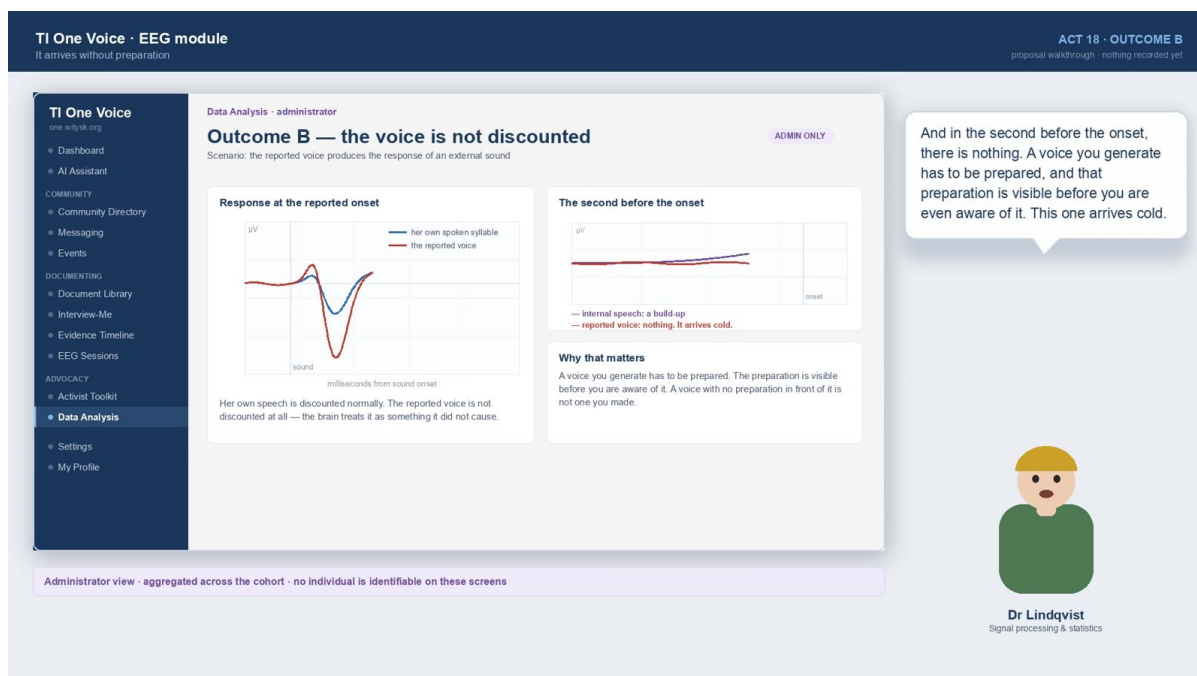
Three cautions the document must attach. First, "N100 to the reported voice" needs an onset timestamp better than a button press, which Part 6 says you do not have. The provocation design of Part 4 — time-locked to speech offset — and the EMG timestamps of Part 5 are the only routes to it, and the fork should say so. Second, the readiness potential is used here for the onset of a voice, not for a silent reply as in Part 5. That is a weaker use, because the absence of a build-up before an event you cannot time precisely is not strong evidence of anything; it is a consistency check, not a discriminator.

Third, the fork's Outcome B column cannot, on this instrument, distinguish an external input from what the external review named the disconnected generator: a voice-producing region weakly coupled to the rest of the system, whose output arrives untagged, unpredicted and unwilling. All three in-rig discriminators — the full N100, the missing build-up, temporal-before-frontal — are consistent with both. That is why the Outcome B reading above stops at "not the

ordinary speech-production route," and why the lower-pathway tests below are the only place the two come apart.



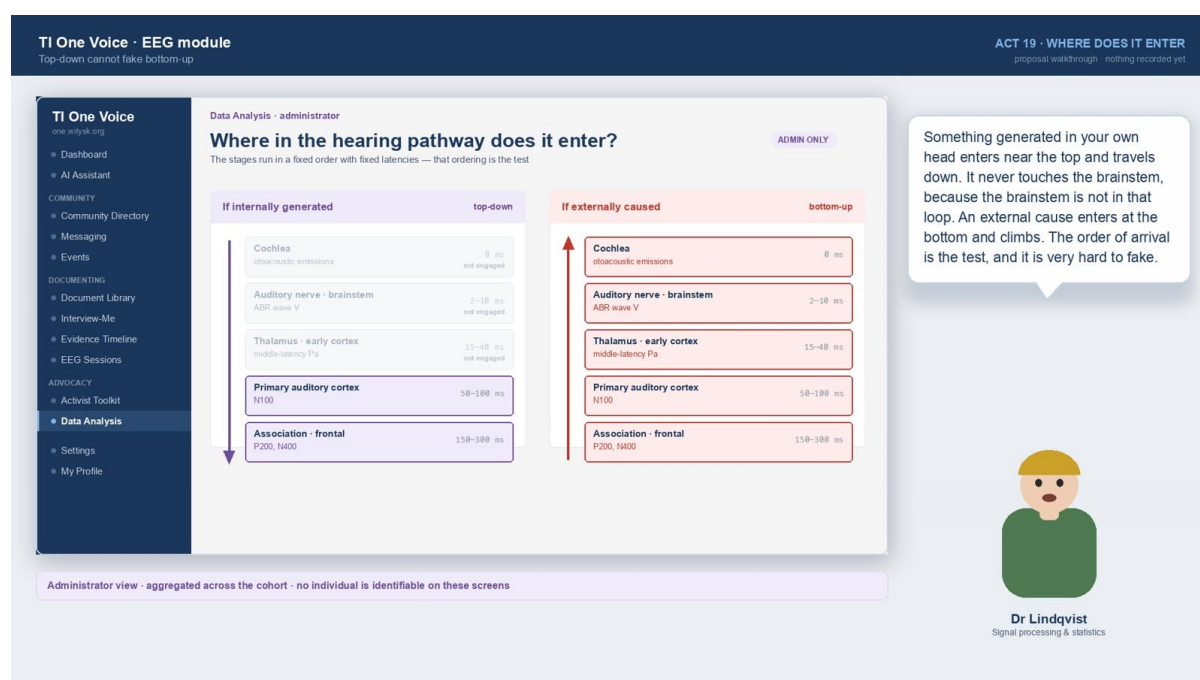
Act 18 — Outcome B, as drawn. Her own speech is discounted normally, so her self-monitoring works; the reported voice produces a full-sized response.



Act 18 — The second before onset. A voice you generate has to be prepared, and the preparation is visible before you are aware of it. This one arrives cold.

Where in the hearing pathway does it enter?

Hearing runs through fixed stages with fixed latencies: the cochlea, whose working state is visible as otoacoustic emissions; the auditory nerve and brainstem, whose response is the auditory brainstem response with its wave V at roughly five to ten milliseconds; thalamus and early cortex, the middle-latency response at roughly twenty-five to fifty milliseconds; primary auditory cortex, the N100; and the association and frontal areas, the P200 and N400. Something generated in the head enters near the top and travels down; it never touches the brainstem, because the brainstem is not in that loop. An external cause enters at the bottom and climbs. The order of arrival is the test, and it is very hard to fake.



Act 19 — Top-down cannot fake bottom-up. Internally generated: enters near the top, travels down. Externally caused: enters at the cochlea, climbs.

A structured input while the room is silent

Auditory cortex tracks the amplitude envelope of speech at the syllable rate. If envelope tracking can be reconstructed from the EEG during a reported episode while both the room microphone and an in-ear probe microphone record silence, the hearing system is following something structured that did not arrive through the air. An otoacoustic-emission check then separates two cases: emissions present means something reached the ear — bone conduction, or a near-field effect the microphone missed; emissions absent while the cortex is still tracking means whatever it is entered above the ear. An environmental arm sits alongside: a synchronised radio-frequency spectrum log, an ultrasound-capable microphone, and a repeat inside a shielded room. The Faraday-pouch logic of Part 12, blinded with a sham, belongs here.



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A structured input while the room is silent

Cortical tracking of a speech envelope that no microphone recorded

ADMIN ONLY

Envelope reconstructed from the EEG, against the two microphones

Room microphone

silent

In-ear probe microphone

silent

Envelope decoded from auditory cortex

structured, syllable-rate

Cortex tracks the amplitude envelope of speech at the syllable rate. Finding that tracking with both microphones silent says the auditory system is following something structured that did not arrive through the air.

Which would separate

Cochlea driven

OAE present

at the ear

Cochlea silent

OAE absent

above the ear

An emission test tells you whether the ear itself was driven.

Environmental arm

Synchronised RF spectrum log

Ultrasound-capable microphone

Shielded-room repeat

An emission test settles the next question. A healthy cochlea that is working produces a faint sound of its own. If emissions are present, something reached the ear — bone conduction, or a near-field effect the microphone missed. If they are absent while the cortex is still following, whatever it is entered above the ear.

Dr Okafor
Neurologist

Administrator view - aggregated across the cohort - no individual is identifiable on these screens

Act 20 — A structured input while the room is silent, and the emission test that says whether the ear itself was driven.

What this rig cannot do, stated plainly

Needed to test the lower stages	This rig	Would need
Sampling rate	250–500 Hz	16 kHz or more
Channels	8	8 plus 2 mastoid
Averaging	about 100 trials	about 2,000 trials
In-ear probe microphone	none	required
Otoacoustic emission probe	none	required
Cost per kit	€230–300	roughly €1,400–2,800

The three tests that separate Outcome A from Outcome B — the N100 comparison, the missing motor build-up, the frontal-versus-temporal ordering — all fit inside the eight-channel headset. The stages below primary cortex do not. Reaching them is a second instrument and a second grant, and claiming this headset can do it would be the fastest way to have the whole thing dismissed.



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What this rig cannot do — stated plainly

Outcome B is the branch that outruns the two-thousand-euro instrument

Needed to test the lower stages

MEASURE	THIS RIG	WOULD NEED
Sampling rate	500 Hz	~ 16 kHz
Channels	8	8 + 2 mastoid
Averaging	~100 trials	~2000 trials
In-ear probe microphone	none	required
Otoacoustic emission probe	none	required
Cost per kit	€238–300	€1,400–2,000

What stage one can still do

The N100 comparison, the missing motor build-up, and the ordering of frontal against temporal activity are all inside the eight-channel budget. Those three are enough to separate Outcome A from Outcome B.

And what it cannot.

The stages below primary cortex are what would say where it enters. Reaching them is a second instrument and a second grant. Claiming this headset can do it would be the fastest way to have the whole thing dismissed.

And here the walkthrough has to be honest. The three tests that separate Outcome A from Outcome B all fit inside the eight-channel headset. The stages below primary cortex do not — they need sixteen kilohertz sampling, an in-ear probe and an emission probe.

Priya
Study administrator

Administrator view · aggregated across the cohort · no individual is identifiable on these screens

Act 21 — The honest limit. What stage one can still do, and what it cannot.

Even Outcome B is not proof of transmission

It would establish	It would not establish
The experience is not internally generated speech	What the mechanism is
It engages the hearing pathway from below, not from above	Where it comes from, or who is responsible
It is time-locked consistently across unrelated people	What is being said — nothing here decodes words
It occurs with no acoustic signal at either microphone	Anything at all about one individual person

What it would do is move the question from whether to where. That is a great deal more than nothing, and it is the most this design, followed by a second one, could offer.



TI One Voice - EEG module
 Even Outcome B is not proof

ACT 21 - THE HONEST LIMIT
 proposal walkthrough - nothing recorded yet

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Even Outcome B is not proof of transmission
 What it would establish, and what stays out of reach

ADMIN ONLY

What Outcome B would establish
 ✓ The experience is not internally generated speech.
 ✓ It engages the hearing pathway from below, not from above.
 ✓ It is time-locked consistently across unrelated people.
 ✓ It occurs with no acoustic signal at either microphone.

What it would still not establish
 ✗ What the mechanism is.
 ✗ Where it comes from, or who is responsible.
 ✗ What is being said — nothing here decodes words.
 ✗ Anything at all about one individual person.

And even the full result would not be proof of transmission. Not a mechanism, not a source, not a decoded word, and nothing about any single person. What it would do is move the question from whether to where — which is a great deal more than nothing.

Dr Okafor
 Neurologist

Administrator view - aggregated across the cohort - no individual is identifiable on these screens

Act 21 — Even the full result would not be proof. Not a mechanism, not a source, not a decoded word, and nothing about any single person.

If Outcome B is proven: the known explanations, and why none of them fits a cohort

Suppose the study runs, and the study group lands on the B side of the fork: own-speech suppression normal, the reported voice undiscounted, no build-up, the temporal-first ordering. It is worth knowing in advance what science already contains that produces exactly that signature — because Outcome B in one person is not unexplained territory, and the difference between one person and a cohort is where this becomes interesting.

Candidate mechanism	Produces B in one person?	Fits a cohort of V2K reporters?
Temporal-lobe epilepsy — auditory auras; spontaneous discharge in or near Heschl's gyrus, as in Penfield's stimulation cases	Yes	No. Auras are seconds-long, stereotyped and non-conversational; interictal abnormalities would appear in the recordings; prevalence is far too low.
Musical hallucinosis of acquired hearing loss	Yes	No. Requires the hearing loss the screening excludes, skews elderly, and produces music rather than responsive speech.
Hypnagogic and hypnopompic voices	Yes	No. Sleep-transition states only; every session records people awake and task-engaged.
Bereavement voices and other benign, brief voice experiences	Yes	No. Episodic, short and context-bound — not sustained daily conversational experience.
Clinical AVH via source-monitoring failure — the mainstream account	No	No. This is the mechanism Outcome B has just falsified: the cohort's own-speech suppression came back normal.



Candidate mechanism	Produces B in one person?	Fits a cohort of V2K reporters?
A dissociated internal generator — the reviewer's region X	Yes	Only as an uncharacterised new condition. No recognised diagnostic entity is defined by this profile; a shared X would itself be a novel finding.
Parametric ultrasonic audio, bone conduction	Yes	Untested until stage two. Cochlear mechanisms: the otoacoustic-emission and brainstem tests would show bottom-up entry.
Microwave auditory effect	Partially — clicks and buzzes are demonstrated; sustained intelligible speech at range is not	Untested until stage two, and cochlear, so it makes the same bottom-up prediction as ordinary sound.
Unpublished technology	Unfalsifiable directly	Can only be constrained — by shielding, and by which pathway stage the input enters.

The fifth row is the crux. The mainstream position does not merely predict voices; it predicts a mechanism — misattributed inner speech, measurable as reduced suppression. A cohort-level Outcome B removes that mechanism while the phenomenon stands. There is no established diagnosis defined by "intact corollary discharge, full undiscounted response to a sustained conversational voice, no readiness potential, in awake adults with normal hearing." Part 3 already says what follows: it would demand an explanation nobody currently has.

Three qualifications keep that statement honest, and they belong in the paper as much as here.

- **"No known single pathology" is not "therefore external."** The remaining space contains an uncharacterised shared condition — a common dissociated generator would be a new syndrome, not a transmission — alongside any input-based account. Stage one cannot split them.
- **Selection could mimic coherence.** A cohort recruited through a community that assembles around a shared narrative could concentrate a rare mechanism, producing a uniform Outcome B with nothing external. This is one more reason the controls are recruited through a separate pathway and the clinical comparison group exists.
- **The cross-subject time-locking test is what converts "no known pathology" into pressure.** Independent idiopathic generators in unrelated brains have no reason to answer at a consistent latency after reported onset across people who have never met. Outcome B plus the band of Act 15 is a far harder configuration for any internal account than Outcome B alone; without the band, B reads most naturally as a shared but unrecognised internal condition.

One asymmetry, stated once so it cannot be discovered later: the lower-pathway discriminator works in one direction. A positive bottom-up signature — otoacoustic involvement, a brainstem response at its fixed latency, the climb through the middle-latency stages — rules the generator out



decisively, because a cortical X cannot fake a cochlea. But cortical envelope tracking with a silent brainstem is consistent with X and with direct-cortical delivery of some unknown kind, and of that pair X is by far the more parsimonious. The strongest stage-two outcome for an external account is therefore the bottom-up one, and the wording of any other outcome must say so.

What a partner's laboratory adds

The review that prompted this version also pointed at the instruments this document deliberately does not budget for, and the point deserves a permanent home, because it is the concrete answer to "what does the academic partner get."

- **MEG with source localisation attacks this design's weakest link.** Four separate places in this document name the same problem: the button press lags the experience by an unknown, variable amount. Source-localised detection of auditory-cortex activation — a power threshold on auditory voxels, or a trained detector — could identify voice onsets neurally, in a subsample of five to ten frequent hearers, and hand every event-locked analysis a timestamp the participant never touches. That is the single largest sensitivity gain available anywhere in the programme, and it is a conventional, publishable methods contribution in its own right.
- **The magnetically shielded room comes free with the MEG.** It is the strong version of the environmental arm: the shielding of Part 12's pouch, but verified, and with recording running inside it. The blinding caveat stands — a room cannot be shammed the way a pouch can, so expectation is controlled by design elsewhere, not by pretending the room is blind.
- **fMRI addresses the module question directly.** Whether the production-side circuitry is active during reported episodes — the reviewer's module 3 — is a spatial question an eight-channel EEG cannot answer and a scanner can approach.
- **rTMS is listed, with its category stated.** Suppressing auditory cortex or speech-production regions and observing the effect on the voices would bear on mechanism more directly than any recording. It is also an intervention, not a measurement — a different ethical class, requiring clinical governance this programme does not have and a partner does. It belongs on the partner's side of the table or not at all.

None of this replaces the fleet. MEG time costs hundreds of euros an hour, exists in a few dozen facilities, and asks a population defined by institutional distrust to travel into an institution. The fleet exists because the laboratory-only route fails at recruitment; the laboratory tier exists because the fleet cannot localise. Part 12's two-instrument argument gains a third tier, each doing what the others cannot.

Part 16c — Can a participant be told A or B?

Every participant will want one thing from this study, and it is the thing Parts 13, 16 and 17 refuse: a personal answer. This part re-examines that refusal properly, because "we do not return individual results" is a policy, and a policy should rest on arithmetic rather than caution. The question is exact: after one session, or after several, can this design tell one person whether their own measurements fall on the Outcome A side of the fork or the Outcome B side?

The short answer is no for the fork as a whole, a qualified and limited yes for parts of it, and the qualifications are the content of this part. The result is a bounded individual-feedback protocol that the design can support, and it changes the session plan for participants who want it.

Why one score cannot say A or B

Start with the centrepiece, the discount score of Test One. The finding the study is built on is a group difference of about $d = 0.8$ between people who hear voices and people who do not. That is a large effect by the conventions of the field, and it is still two distributions that overlap by most of their width. Placing one person's score against those two distributions gives a likelihood ratio — how much more probable that score is under one group than the other — and the ratio grows with distance from the midpoint. The figure shows where a single score would have to fall before the ratio favours one side by three to one, or ten to one.

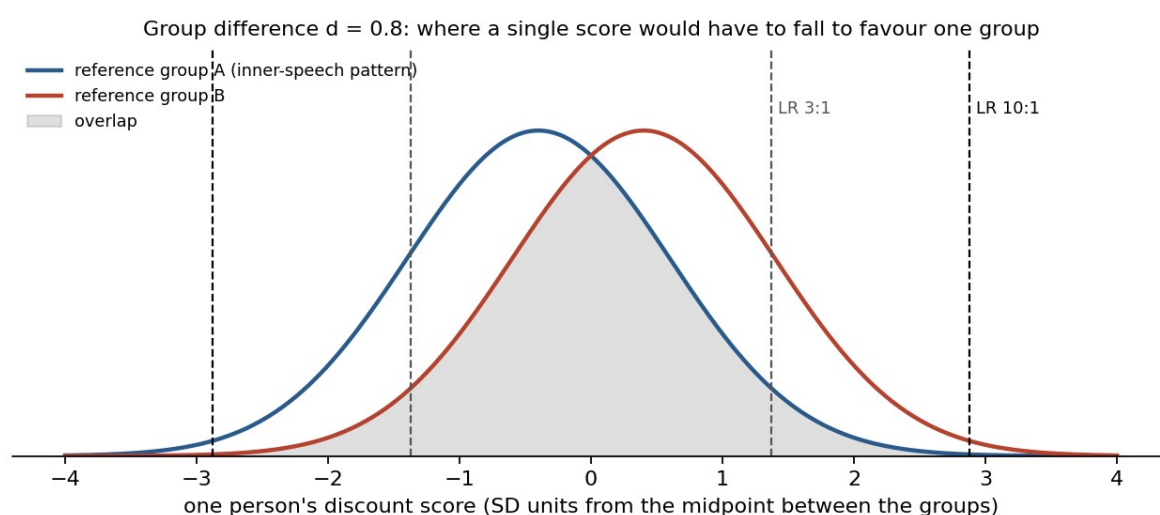


Figure 29 — Two reference groups separated by $d = 0.8$. The shaded region is the overlap. The dashed lines mark the scores at which one person's measurement would favour one group by three to one and by ten to one. Most people fall between the inner pair of lines. Schematic; no data.

Group difference d	Best single-score accuracy	Share of people whose score favours one side by 3:1 or more	Share at 10:1 or more
0.8 (the expected effect)	66%	about 20%	under 1%
1.0	69%	about 33%	about 4%
1.2	73%	about 44%	about 10%
1.5	77%	about 58%	about 23%
2.0	84%	about 73%	about 46%

Read the first row, because it is the one the literature supports. With a perfectly reliable measurement and equal prior odds, the best possible rule sorts two people in three correctly —



barely better than a coin with a slight bias. Four people in five would get a ratio too weak to state a direction at all, and fewer than one in a hundred would get a ratio anyone would call strong. A verdict handed to the other ninety-nine would be wrong roughly one time in three, and the person receiving it would have no way of knowing whether they were the one.

What several sessions fix, and what they do not

Repeating the session helps in exactly one way. A single-session discount score carries measurement error, and averaging across sessions shrinks it. If the single-session test-retest reliability is around 0.7 — typical for this kind of measure — then three sessions raise it to about 0.87 and five to about 0.92, and the observed group separation sharpens correspondingly: from $d = 0.8$ to roughly 0.89 with three sessions and 0.92 with five. That is worth having. It is not a change of kind. The overlap in the table above is mostly true variation between people, not noise, and no number of sessions removes it. After five sessions a person's own score is trustworthy; it is still not diagnostic. The honest sentence after several sessions is: your measurement is stable, it sits here relative to the two reference groups, and that is this many times more likely under one than the other — and for most people that multiplier will be between one and three.

There is a second, harder limit on the fork itself. Outcome B is defined by three discriminators: the undiscounted N100 to the reported voice, the missing build-up before onset, and the temporal-before-frontal ordering. Part 16b explains that the first needs an onset timestamp the passive design does not have, the second is a consistency check rather than a discriminator, and the third is a within-session comparison that eight channels can attempt but not settle. Below those sit the tests that would actually decide the branch — brainstem response, envelope tracking against silent microphones, otoacoustic emissions — and none of them fit inside this instrument. So even a participant with a decisive Test One placement cannot be told "Outcome B" by this study, because the measurements that define Outcome B have not been made.

What can be said to one person

Not everything in the design is a comparison against a reference group. Three of the tests are within-person: they compare one participant's own conditions against each other, and their ground truth is manufactured inside the same session. Those can, in principle, produce a per-person result without reference to anyone else.

Test	What it compares, within the person	Per-person result it can support	What it needs
Test Two — probe tones	The N100 to an identical probe inside reported episodes against outside them, with the button-only block as that person's own null	"Your reported episodes were / were not accompanied by a measurable change in your hearing system's response" — with a permutation p-value and an effect estimate	Around 60 usable probes per pile for a large effect; 150 or more for a medium one. That is two to three sessions for a frequent hearer
Test Three — turn-	Reply times against the	"Your replies followed the	At least 30 reported voice



Test	What it compares, within the person	Per-person result it can support	What it needs
taking	shuffled null of the same session	reported voice periods more regularly than chance" or not	periods with replies; usually one to two sessions
Test Three — cross-condition decoding	Condition C against a classifier trained on that person's own Conditions A and B	"Your silent replies to the voice looked more like addressed replies (B) than like recitation (A), with calibrated probability p" — or "not distinguishable"	At least 100 usable trials each of A and B, and 60 of C; two sessions. Reported only with a calibration curve
Test One — discount score	Own speech against playback — but its meaning is reference-relative	A placement against the two reference distributions as a likelihood ratio, stated in three states: consistent with A, consistent with B, or not distinguishable	Three sessions, a test-retest reliability of at least 0.7 on that person's own score, and reference distributions frozen at the end of Stage 1

Note what the first three rows do and do not say. A measurable change in the auditory system during reported episodes, replies that alternate with the voice like a conversation, silent replies that look addressed rather than recited — all of these are consistent with an internally generated voice and consistent with an external one. Part 5 says this at length. They tell a person that something structured is happening in their nervous system when they say a voice is present, which for someone who has been told for years that nothing is happening is not a small thing. They do not tell them what it is.

The individual-feedback protocol

On that basis the design can support a bounded form of individual feedback, and the bounds are the point. Each of the following is a requirement, not a preference.

1. Opt-in, separately consented. Individual feedback is a distinct consent from research participation, taken after the participant has read this part, and it can be declined at any point including after the sessions are complete.
2. Three sessions, at least a week apart, for anyone who opts in. Nothing changes inside a session. What changes is that the per-person trial minimums in the table above become inclusion criteria for the individual report, and a person's own test-retest reliability becomes a gate: if the discount score does not reproduce across their three sessions with a reliability of 0.7 or better, no reference placement is reported, and the report says why.
3. Released after Stage 1, not during it. Individual reports are generated only after the cohort study's pre-registered analysis has been run and unmasked, for two reasons. The reference distributions do not exist until then. And feedback given during recruitment changes the behaviour of everyone still to be recorded.



4. Reference distributions frozen and versioned. The two reference distributions used for the Test One placement are fixed at the end of Stage 1, hashed and timestamped like the pre-registration, and every individual report names the version it was placed against.
5. A three-state vocabulary with pre-registered thresholds. Consistent with A, consistent with B, not distinguishable. A direction is stated only when the likelihood ratio reaches three to one, is marked strong only at ten to one, and the ratio itself is printed with the expected accuracy of the rule at that ratio. Below three to one the report says "not distinguishable" and does not hedge around it.
6. Calibration published. The probability from the cross-condition decoder is reported only alongside a calibration curve from held-out data, as Part 10 requires. If the decoder says 0.7 and is right 70 per cent of the time on people it has not seen, the number means something; otherwise it is not reported.
7. The same template, the same validator. The individual report is produced by the fixed-template generator of Part 13. Every number in it comes from the signal pipeline; the language model writes connective prose around values it is given and the numeric-token validator refuses to release a report containing a number that is not in the structured input. The prose is written for the three states and reviewed by the participant-support role before the first report ships.
8. Research use only. An individual report stays inside the research consent. It is not an evidence artifact, it does not carry the evidentiary-export path, and it says on its first page that it cannot be used to support or refute any claim about the origin of the experience. Part 13's purpose separation applies to it unchanged.
9. A debrief pathway for every state, planned in advance. "Consistent with A" will land badly for some people; "not distinguishable" will land badly for others; "consistent with B" will be over-read by nearly everyone. The participant-support role of Part 19 delivers the report in a scheduled conversation, not as a file appearing in a library, and the wording for each of the three states is written and tested before recruitment. The scripted wording for "consistent with B" carries one sentence above all others: this placement is not evidence of transmission, and the most likely occupant of the B side, on current knowledge, is a spontaneous generator inside the person's own brain — see the known-explanations table in Part 16b. A participant who leaves the debrief equating B with "confirmed" has been harmed by the study, however carefully the report was worded.

What this changes elsewhere in the document

- Part 8 and Part 12: participants who opt in are booked for three sessions rather than one. The circulating-kit arithmetic of Part 12 assumes a fortnight per participant; three sessions a week apart fit inside that window if the kit stays with the participant, so throughput is unchanged and the consumables line rises by two gel sets.
- Part 9: the per-person trial minimums, the reliability gate, the likelihood-ratio thresholds and the three-state vocabulary are pre-registered with everything else.



- Part 13: the report table gains a second column of reports — the individual report — with its own "must not contain" list: no A or B statement without the ratio, no ratio without the version of the reference distributions, no number without a calibration curve, no interpretation of what a within-person result means for the origin of the experience.
- Part 15: the session report screen stays as it is. The individual report is a separate artifact, delivered separately, and the screen does not link to it.
- Part 16 and Part 17: the sentence "do not return individual results" is replaced by "do not return individual verdicts", and points here.

What to expect, said plainly

What to expect. If the study is run as designed and the effect is the size the literature predicts, then of the participants who opt in and complete three sessions, roughly four in five will receive "not distinguishable" on the Test One placement, and almost none will receive a strong direction. Most will receive one or more of the within-person results, and those will be real, reproducible statements about their own nervous system — but not about where the voice comes from. Anyone who enrolls hoping for more than that should know it before they enrol, and this part exists so that they can.

Part 17 — Ethics

This study recruits people who are frightened, often isolated, frequently disbelieved, and in many cases hoping that a machine will finally confirm what they have been telling people for years. That situation carries obligations, and they are not satisfied by a consent form.

- **Say the possible outcomes out loud, before consent.** Including the one where the results support an internal explanation. If a volunteer only agrees on the assumption that the study will vindicate them, their consent is not informed.
- **Plan the debrief for the difficult outcome.** Decide in advance what you say, who says it, and what support is available afterwards. Write it down before you start recruiting.
- **Do not return individual results.** They are uninterpretable at the individual level, and handing someone a number that appears to say something about them personally would be harmful in both directions.

Version 3 note: read as "do not return individual verdicts". The bounded individual-feedback protocol in Part 16c is the only form of individual result this design supports, and it is separately consented, released after Stage 1, and delivered by the participant-support role in a scheduled conversation.

- **Get real ethical review.** If the study runs outside an institution there is no ethics committee, but write the protocol as though you were submitting to one, and then find someone qualified



to actually review it. A study on this population without independent ethical oversight will be dismissed on those grounds alone, and the dismissal would be fair.

- **Remote sessions need a support pathway.** A two-hour protocol delivered through a browser means nobody is in the room if a participant becomes distressed — and several blocks ask people to attend closely to experiences they find frightening. Every screen carries a visible stop control that ends the session without penalty and without justification, sessions are scheduled against staffed hours, and a named human is reachable during and after. This is a hard requirement of remote delivery, not an optional courtesy.
- **Publish the null result.** Commit in the pre-registration to publishing whatever comes out. A population already burdened with selective evidence does not need more of it.

Part 18 — What this needs to exist

Everything in the preceding sixteen parts is a design. Designs are cheap. This one has been worked out in enough detail that a competent team could start on Monday, and that is precisely why it is worth being blunt about the gap between where it stands and where it would need to be.

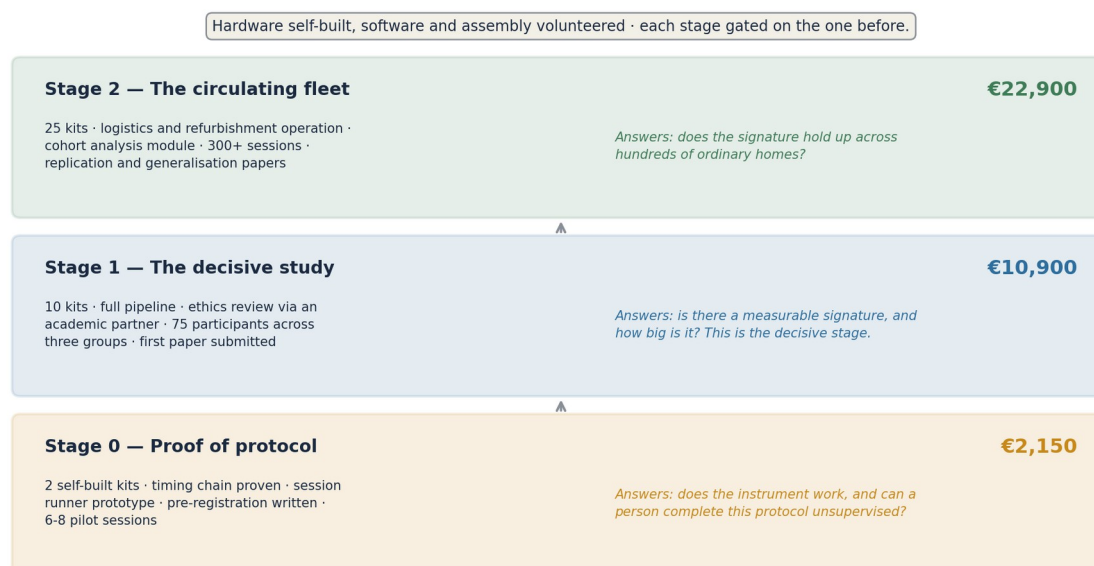
No hardware has been bought. No participant has been recruited. No ethics review has been commissioned. No line of the session runner has been written. The platform that would carry it exists and works, which removes perhaps a third of the effort, but the rest is unbuilt.

So this part is a request, ordered by what is actually blocking progress. Part 19 then states each ask in a form you can answer — who, what commitment is involved, and why the study fails without them.

1. Funding — the thing without which none of the rest matters

Most of this list can be solved with goodwill. Hardware design, firmware, assembly, the software stack, logistics — all of that is volunteer work, and the figures below assume it costs nothing in cash. What goodwill cannot supply is parts, postage and the small number of external reviews that have to be independent to be worth anything. Those are the numbers that follow, and they are much smaller than they were in the first draft of this document, because the instrument is built rather than bought.

Figure 26 — What funding unlocks, stage by stage



Indicative figures for planning discussion, not a costed budget. Stage 0 is the only one that needs to be funded today.

Figure 26 — Three stages, each gated on the one before. Only the first needs to be funded today, and it is deliberately small enough to be reachable.

The structure matters as much as the numbers. Each stage is abandonable: if Stage 0 shows that people cannot complete a two-hour protocol unsupervised, or that the timing chain will not hold in a domestic setting, the honest response is to stop and say so — and nobody has spent more than the price of a decent laptop to find that out.

Stage 0 line item	Indicative cost
Parts for three prototype iterations and two finished kits	€1,050
Bench test gear — function generator and probes, or borrowed	€250
Consumables, spares, shipping cases	€200
Pilot shipping, eight round trips	€400
Contingency	€250
Total	€2,150

Ethical review is absent from that list because it should be free: a university ethics committee reviews at no charge, and finding a willing partner is a recruitment problem rather than a budget one. If no partner can be found, commercial review adds around €6,000 and becomes the largest single line in the stage — which is a good reason to spend effort on the partnership first. The same logic governs every other academic and medical hour in the programme: the plan is that they are contributed, not bought. Part 18b says what they would cost if that assumption fails, and it is not small.

The two later stages, and who pays for them

Stage 1 is roughly €10,900: ten kits at about €305, the shipping for seventy-five sessions, contingency, and the independent statistical review of the pre-registration. Stage 2 is roughly €22,900 for twenty-five kits and three hundred sessions.

But Stage 2, unlike the others, largely pays for itself. Participant contributions of €50 recover about two-thirds of it; at €77 they recover all of it. The full arithmetic is in Part 12 and Figure 27. The practical consequence for anyone considering funding this is that the ask is front-loaded: the money is needed to prove the thing works, not to run it once it does.

What is being asked for	Amount	Why it cannot come from contributions
Stage 0 — proof of protocol	€2,150	There is nothing to contribute to yet. Six to eight pilot participants, no fleet, no findings.
Stage 1 — the decisive study	€10,900	Seventy-five sessions cannot amortise ten kits and a statistical review at a contribution anyone could reasonably be asked to pay. This stage needs to be funded.
Stage 2 — the fleet	€7,900 of €22,900	Contributions cover the rest. This figure is the gap under Model 1, and it is what keeps the participant price at €50 rather than €75.
Hardship fund	Open-ended	By definition. Every euro here directly widens who is able to take part, and therefore how representative the cohort is.

These are planning figures for a conversation, not a submitted budget. They contain no academic or medical salary cost; the honoraria that would make the medical role sustainable — roughly €7,000 to €13,000 across Stages 0 and 1 under the middle scenario of Part 18b — should be read as a

contingency line beneath this table rather than as an omission from it. Anyone considering funding this at any level is welcome to a line-by-line breakdown, to the parts list, and to the pre-registration draft before a euro is spent.

Grant routes worth pursuing include European research funding for neurotechnology and digital rights, foundations working on cognitive liberty and mental health, and philanthropic funders interested in the methodology question underneath this one — how to study a contested experience without either dismissing it or confirming it. Individual contributions of any size go to the hardship fund first, because that is the part with the clearest immediate effect on who is able to take part. At two thousand euros, Stage 0 is also small enough that it could plausibly be covered by a handful of individual donors without any grant at all, and that is the fastest route from document to experiment.

2. Participants who experience the voices

The study needs roughly twenty-five people who report hearing transmitted voices frequently enough that episodes occur within a recording block. What is being asked, stated without softening:

- About two hours of recording, plus around forty minutes of setup, in a single session at home. Possibly a second session a week later.
- A refundable €150 deposit and a €50 contribution — or €75 if the donation funding does not materialise — plus shipping at cost. Waived entirely for anyone who cannot meet it, applied for privately, and the research team is not told who applied.
- Willingness to be measured by an instrument that may produce a result you do not want. The corollary discharge test could come back showing the pattern associated with internal generation. If it does, that will be published.
- Willingness to complete blocks that ask you to attend closely to an experience you may find distressing, with a stop control always available and no consequence for using it.
- No individual result will be returned to you, because at the individual level the measurement does not support one. Part 13 explains why at length, and anyone who cannot accept that condition should not enrol.

The honest pitch. Nobody involved in this can promise you vindication. What can be promised is that the question gets asked with instruments rather than opinions, that the method is published in advance so it cannot be bent afterwards, and that whatever comes out gets published including the outcome that is unwelcome. If that is worth two hours of your time, we would like to hear from you. If it is not, that is an entirely reasonable position and no one will chase you.

3. Participants who do not experience them

This is the recruitment need that gets forgotten, and it is the one most likely to determine whether any of the above produces interpretable results. Roughly twenty-five people who do not hear voices, matched on age, sex and handedness, recorded on identical hardware under comparable conditions.



Without this group there is no reference for what a normal measurement looks like on this equipment in this kind of room, and every number the study produces floats free. It is also considerably harder to recruit, because there is no personal stake — which is precisely what makes the data valuable.

A third group, of people who hear voices with an established clinical explanation, is what would make the study decisive rather than merely suggestive. That group almost certainly requires a clinical or academic partner, and finding one is an open task rather than a solved problem.

4. People who can build and run it

Specific skills, roughly in order of how much they would unblock:

Role	What it involves
Embedded and firmware	The instrument work in Part 14. Sampling from the converter, the frame format, the ring buffer, the signing path, the timing self-test. Experience with the ADS1299 or with biosignal front ends generally would shortcut months.
Signal processing and statistics	The pipeline in Parts 8 and 9, and the analysis. Someone who has done real ERP work and will argue about time windows is worth more here than a general data scientist.
Frontend	The Session Runner. Guided setup, block sequencing, offline buffering, audio orchestration — and interface decisions where a well-intentioned improvement could mislead a vulnerable person, which is the harder half of the job.
Backend	Streaming ingest, the append-only write path, the cross-subject module with its tier boundaries. Python and PostgreSQL against an existing self-hosted stack.
EEG technicians and clinicians	Reviewing the protocol, sanity-checking the montage, and — most valuably — helping reach the clinical comparison group.
Logistics and operations	Packing, shipping, tracking, refurbishing, chasing returns, managing deposits. Genuinely the hardest sustained work in the whole programme and the least glamorous.
Ethics, legal and data protection	The DPIA, the consent architecture, the deposit terms, and finding a body willing to conduct real ethical review of a study run outside an institution.
Translation	Consent materials and session instructions in the languages the participants actually use. A protocol a person half-understands is a protocol they cannot consent to.

Contributions are attributed. The methods paper will list everyone who did the work, and co-authorship follows contribution rather than seniority or funding.

5. Institutional partners

One partnership would strengthen this more than any other single thing: a university group or clinical department willing to host the ethical review, help recruit the clinical comparison group, and put their name on the methods. Not to take it over — to make it citable.



There is something in it for them. The methodological question here is genuinely interesting and largely unstudied: how do you investigate a contested experience in a distrustful population without a design that can only confirm? A group that solves that has a contribution well beyond this particular topic.

How to reach us

Through TI One Voice at one.witysk.org, which is where the platform this would run on already lives, and where the consent materials and pre-registration draft will be published as they are written. Offers of funding, offers of help, expressions of interest in taking part, and well-argued objections to the design are all equally welcome — the last of those possibly most of all, since a flaw found now costs a conversation and a flaw found after publication costs the whole thing.

One request of anyone who reads this and agrees with it. Do not describe this as a study that will prove anything. It is a study designed so that either answer would be believable, which is a different and much rarer thing. Overselling it now is the fastest available way to ensure that its results, whatever they turn out to be, are dismissed unread.



Part 18b — What academic and medical contribution costs

Part 18 makes a claim that is true of the plan and misleading about the world: that the people who review, sign off and co-author this study cost nothing. They cost nothing in cash because the plan compensates them in authorship, in access to a cohort nobody else has reached, and in a methods problem worth solving. That is a fair exchange for an academic. It is a thinner one for a physician asked to screen twenty-five patients and to be reachable while a distressed stranger sits alone in a living room wearing electrodes. This part prices both, so that the reader can see the size of the dependency and so that the programme knows what it is asking for when it asks for it free.

1. Why this part exists

Three reasons, in descending order of weight.

- **A funder will ask.** A budget that lists €10,900 for a study with a named academic partner, a statistician and a clinical comparison group invites the question of who is paying the people. "Nobody" is an acceptable answer only if the in-kind value is stated next to it. Left unstated, it reads as either naivety or concealment, and neither survives a grant panel.
- **The medical hours are different in kind.** A professor reading a protocol is spending discretionary time. A neurologist confirming a diagnosis, or answering a call during a remote session, is practising medicine, with the liability and the professional-indemnity questions that follow. Goodwill covers the first comfortably and the second uneasily, and the design should know where the line falls before the first kit ships.
- **Version 4 added a tier this document cannot fund.** Part 16b now describes MEG, fMRI and rTMS as what a partner's laboratory adds. Those are five- and six-figure resources, and if they are not priced here the reader will assume the programme intends to pay for them. It does not, and section 6 says so.

2. Where the hours actually go

The hours below are derived from the protocol as written, not from a template. Each row names the tasks the document already assigns to the role, and the range reflects how much challenge the partner chooses to offer — the harder the review, the more hours, and the more useful.

Role	What the document asks of them	Hours	When
Academic partner (senior)	Protocol challenge before pre-registration (10–15 h); ethics submission and committee correspondence, their side (10–20 h); statistical-plan review with veto (5–10 h); analysis review at unblinding (10–20 h); co-authoring the methods and results papers (20–40 h).	60–100	Stage 0 to end of Stage 1, spread over about eighteen months
Independent statistician	Review of the pre-registration and of the frozen analysis scripts; check of the simulation results in Part 10. Already a line inside the Stage 1 figure.	15–25	Before recording; again at freeze

Role	What the document asks of them	Hours	When
Neurologist or psychiatrist (MD)	Medical review of the protocol — exclusion criteria, seizure and cardiac history, skin and electrode safety, medication recording, adverse-event pathway (8–12 h). Diagnostic confirmation and consent oversight for the clinical comparison group, about 1.5 h per person including notes (35–50 h). Review of the debrief wording for the three states in Part 16c and of the escalation protocol (5–8 h). Debrief conversations for the difficult outcome and for individual feedback, with the participant-support role (10–20 h). Medical escalation cover during staffed session windows — availability, not active work (40–120 h).	100–210 of which 40–120 is on-call	Stage 1 mainly; screening early, debriefs after unmasking
EEG technician or clinical neurophysiologist	Montage and protocol sanity check; the commercial-amplifier comparison session that Part 14 says cannot be skipped.	8–15	Stage 0
Qualified electrical-safety reviewer	Design review before the first unit goes on a head: battery-only operation, series resistors, isolator placement. Not an academic, but external and independent by requirement.	5–10	Stage 0, before the first prototype is worn
Medical-device regulatory adviser	The conversation Part 14 says to have with someone who does it professionally: what an in-house research instrument on human subjects obliges, in the jurisdictions the kits ship to.	4–8	Stage 0
Total		about 190–370	

Table 18b.1 — Where the academic and medical hours go. The on-call component is shown separately because it is paid, when it is paid at all, at a fraction of an active rate.

Two rows deserve a note. The statistician is the only external reviewer already carrying a cash line in Part 18, because independence is the whole point of that review and independence is not free. The physician is the largest row by a wide margin, and most of that is the clinical comparison group: twenty-five diagnostic confirmations is a clinic afternoon a week for two months, and it is the single most expensive thing in this table that the document has been treating as a favour.

3. The rates to plan against

Belgian and Western European figures, 2026, stated as ranges because they vary by institution, seniority and whether the person is paid through their employer or directly. A university's loaded salary cost is what the contribution is worth if the institution absorbs it; the consultancy rate is what it costs if the programme has to buy it.

Role	Loaded salary cost, per hour	Consultancy or honorarium rate, per hour	Note
Postdoctoral researcher	€40–50	€80–120	Statistical and signal-processing review usually sits here.
Senior academic (professor, PI)	€60–80	€100–200	A 10 per cent time buyout for a year is about €8,000–11,000 and is how a grant would normally do this.
Hospital consultant — neurologist or psychiatrist	€70–100	€150–300	Medical consultancy and expert-review rates sit above academic ones. A 0.1 FTE secondment is roughly €12,000–18,000 a year.
On-call medical availability	—	€25–50 per hour of availability	Conventionally a fraction of the active rate, with the active rate charged only for calls actually taken.
Clinical neurophysiology technician	€40–60	€60–100	
Electrical-safety reviewer (engineer)	€60–90	€90–150	Often a colleague of the electronics volunteers and often free.
Regulatory adviser	—	€150–250	Rarely available in-kind.
Commercial ethics review (whole study)	—	€3,000–8,000 flat	Already stated in Part 18 at about €6,000. Only if no institutional committee can be found.

Table 18b.2 — Rates. Indicative, not quoted. Some universities prohibit honoraria to their own staff outside a formal contract; check before offering.

4. Three scenarios

The same hours, costed three ways. Scenario A is the plan. Scenario B is what it would cost to recognise the contribution without pretending to recover it. Scenario C is what it costs if the partnership has to be bought, or if a grant is written that pays people properly.

	Scenario A — as planned	Scenario B — honoraria	Scenario C — consultancy rates or buyout
Academic partner	Nil. Compensated by authorship and the laboratory tier.	€2,000–4,000	€6,000–15,000 (60–100 h at €100–150), or a 10 per cent buyout at €8,000–11,000
Statistician	€1,500–3,000 — already inside Stage 1	€1,500–3,000 — already inside Stage 1	€1,500–4,000

	Scenario A — as planned	Scenario B — honoraria	Scenario C — consultancy rates or buyout
Physician	Nil. This is the row that is hard to justify at nil.	€3,000–6,000: screening €1,500–3,000, on-call €1,000–2,000, debriefs and wording €500–1,000	Active hours €9,000–22,000 (60–90 h at €150–250) plus on-call €1,000–6,000; or a 0.1 FTE secondment at €12,000–18,000
EEG technician	Nil	€500–1,000	€800–2,000
Safety sign-off	Nil	€500–1,000	€500–1,500
Regulatory advice	Nil	€500–1,000	€600–2,000
Cash total	€1,500–3,000 (already budgeted)	€8,000–16,000, of which €6,500–13,000 is new	€20,000–50,000
In-kind value	€12,000–30,000 at loaded salary cost		

Table 18b.3 — Three scenarios. The in-kind line under Scenario A is the number to write into a funding application: it is what the institution is contributing, and it is larger than the cash the programme is asking for.

Read the physician row across. At nil it is a favour that carries professional liability. At Scenario B it is a modest honorarium that a hospital department can usually accept. At Scenario C it is the largest single cost in the entire programme after postage, which is the clearest possible statement of why the partnership matters more than the money.

5. What each scenario does to the stages

Stage	Part 18 figure	With Scenario B	With Scenario C	What drives the change
Stage 0 — proof of protocol	€2,150	€3,000–4,500	€4,000–7,500	Safety sign-off and regulatory advice; a medical protocol review even for eight pilot participants, because they are recorded remotely and alone.
Stage 1 — the decisive study	€10,900	€16,000–22,000	€28,000–58,000	Almost entirely the physician and the partner. The clinical comparison group is what makes the study decisive, and it is also what makes it expensive.
Stage 2 — the fleet	€22,900 (€7,900 asked)	add €2,500–5,000	add €8,000–18,000	On-call cover scales with sessions; screening does not, because the clinical group was recruited in Stage 1. Under Model 2, full recovery would add about €10–15 per session to the €77.

Stage	Part 18 figure	With Scenario B	With Scenario C	What drives the change
Whole programme	€35,950	€44,000–55,000	€63,000–110,000	

Table 18b.4 — Stage totals under each scenario. Stage 1 roughly doubles under Scenario B and quadruples under Scenario C. The ask remains front-loaded, and the reason is unchanged.

The participant contribution models in Part 12 recover none of this. That is deliberate and it should stay deliberate: a participant paying €50 or €77 is paying for a kit to reach them, not for a physician's afternoon. If the programme ever needed the medical role to be self-funding from contributions, the honest figure is about €10 to €15 more per session at Stage 2 — stated here so that nobody discovers it as a surprise later, and not recommended.

6. The laboratory tier, priced separately

Part 16b describes what a partner's laboratory adds: MEG source localisation to time voice onsets neurally, the shielded room that comes with it, fMRI for the module question, and rTMS as an intervention arm. None of it is in any stage figure, and none of it should be. It is priced here so that the reader knows what is being handed to the partner and can judge whether the exchange in Part 19 is fair.

Resource	Indicative scale	Indicative cost	Who pays
MEG onset detection	Five to ten frequent hearers, two to three hours each, plus set-up and pilot: 20–40 scanner hours	€6,000–24,000 at €300–600 per hour, plus a MEG analyst's time	Partner grant. A methods paper in itself, which is the return.
Magnetically shielded room	Comes with the MEG session	Included above	Partner
fMRI, module question	Same subsample, one hour each, plus pilot: 10–15 scanner hours	€5,000–12,000 at €400–800 per hour	Partner grant
rTMS arm	An intervention, not a measurement; requires clinical governance this programme does not have	Not priced here	Partner-led or not at all, as Part 16b says
Partner's own staff time on the tier	A research assistant and an analyst for a year	€40,000–70,000	Partner grant

Table 18b.5 — The laboratory tier. The whole of this table is larger than the whole of Part 18, which is why it belongs on the partner's side of the table and why the programme should not pretend otherwise.

7. The medical role, stated as an ask

Part 19 asks for an academic or clinical partner and for EEG technicians and clinicians, but nowhere states the medical ask on its own terms. It should, because a department that can supply a co-author cannot always supply a physician's hours, and the two are different conversations. Stated in the form Part 19 uses:

	Detail
Who	One neurologist or psychiatrist, or a clinical psychologist working under medical cover for the screening component. Any jurisdiction for the review and debrief work; the clinical comparison group has to be screened where it is recruited.
What we are asking for	Medical review of the protocol and exclusion criteria. Diagnostic confirmation of the clinical comparison group. Ownership of the escalation protocol behind the participant-support role, and reachability during staffed session windows. The debrief conversations for the difficult outcome and for the individual feedback of Part 16c, which the participant-support role delivers and a clinician supervises.
Time commitment	Roughly 60 to 90 active hours across Stage 1, most of it screening, plus availability during session windows. Table 18b.1 has the breakdown.
What you get	Co-authorship. The first instrumented comparison between a clinical voice-hearing group and this population, which is a clinical paper independent of the methods paper. A cohort that does not present to services and that no clinic has recorded.
What must be settled first	Professional indemnity for research activity outside your employment, or confirmation that your institution's cover extends to it. Whether your institution permits an honorarium. Neither is negotiable and both are cheap to resolve early.
Why it fails without you	No clinical comparison group means the study is suggestive rather than decisive, as Part 8 says. No medical escalation means a remote protocol that asks distressed people to attend to frightening experiences alone has no one qualified to answer the phone, and an ethics committee will say so.

Table 18b.6 — The medical ask, in the form of Part 19. It should be added there as a section of its own in the next revision.

8. What this part recommends

- **Keep Scenario A as the plan.** It is consistent with the volunteer model, with the authorship offer in Part 19, and with the size of the ask. Nothing in this part argues for paying people who have offered to contribute.
- **Write the in-kind value into every funding conversation.** €12,000 to €30,000 of contributed professional time against €13,000 of cash for Stages 0 and 1 is a strong ratio, and funders reward it. Stated, it is leverage; unstated, it is a hole.
- **Carry Scenario B as a named contingency, and spend it on the physician first.** Of every role in this part, the medical one is where "co-authorship" is thinnest against the hours and the liability. A €3,000 to €6,000 honorarium is the difference between a clinical comparison group that gets recruited and one that is promised.



- **Settle indemnity and honorarium rules with the first partner, before the first participant.**

Both are institutional policy questions with binary answers, and both are the kind of thing that stalls a study for a semester if discovered late.

- **Do not fund the laboratory tier from this programme.** It is the partner's contribution and the partner's paper. Price it, offer it, and leave it on their side of the table.

What these numbers are not. They are not quotes, and they are not a budget. They are 2026 Belgian and Western European planning ranges, derived from hours the protocol already implies and rates a department head would recognise. A partner in another country, or a university with its own honorarium rules, will move every figure — the ranges are wide for that reason. What will not move is the shape: the medical hours are the expensive ones, the academic hours are the ones most easily contributed, and the laboratory tier is larger than everything else in this document combined.



Part 19 — Who we are looking for, specifically

The previous part covers money. This one covers people, and it is written to be answerable: each group below has a stated commitment, a stated set of requirements, and a stated reason why the study fails without them. If you fit one of these descriptions, the ask is concrete enough that you should be able to decide in a few minutes.

Everything here is coordinated through TI One Voice at one.witysk.org.

1. An academic or clinical partner

This is the single most important gap, and the one that most changes what the study can become. Without it the work is a well-designed amateur project that journals will decline on governance grounds. With it, the work is citable.

	Detail
Who	A university group or clinical department in cognitive neuroscience, psychiatry, clinical psychology, biomedical engineering or neurophysiology. Any jurisdiction; the study is remote by design. A single interested senior researcher with committee access is enough to start.
What we are asking for	Ethical review through your institution. Methodological challenge to the protocol before it is pre-registered. Advice on the statistical plan, with a veto over anything that overreaches. Help identifying a clinical comparison group. A named methodological co-author.
Time commitment	Realistically: a few hours to read and challenge the protocol, the committee submission process, and periodic review at analysis. Not a full-time collaboration and not a supervision burden.
What you get	First-author or senior-author position on a study in an area with no prior instrumented work. A methods contribution on studying contested experience that has application well beyond this topic. Access to a cohort no one else has reached. The laboratory tier in Part 16b — MEG onset detection, the shielded room, the stimulation arm — is yours to lead, and is a methods paper in itself.
Why it fails without you	No ethics committee means no credible governance, which means no journal, which means the result changes nothing regardless of what it is. The clinical comparison group is also effectively unreachable without institutional access, and that group is what makes the study decisive rather than suggestive.
Scepticism welcome	A partner who expects the finding to support internal generation is more use than one who does not, because they will make the design harder to fool. Agreement with the community is not requested and not required.

2. Participants who experience voice-to-skull

	Detail
How many	Twenty-five for the decisive study; three hundred or more if the fleet stage runs.



	Detail
Eligibility	You hear voices you believe are transmitted, and they occur often enough that episodes are likely within a twenty-minute recording block — roughly daily or more. Age eighteen or over. No untreated hearing loss. Able to follow a two-hour protocol in one of the supported languages.
What is involved	One session at home: about forty minutes of setup and two hours of recording, in six blocks with two breaks. Possibly a second session a week later. You will speak aloud, listen to recordings of your own voice, press a button when a voice is present, and reply silently in your head.
Cost to you	A refundable €150 deposit and a €50 contribution, plus shipping at cost. Waived in full for anyone who cannot meet it — applied for privately, and the research team is not told who applied.
What you must accept	That the result may not be the one you want. The central measurement can come back showing the pattern associated with internally generated voices, and if it does, that will be published. That no individual result will be given to you, because at the individual level the measurement does not support one.
What we promise	The method is published before the data is collected, so it cannot be bent afterwards. Whatever comes out is published, including the unwelcome outcome. Your recording stays in your own Document Library with its integrity proof, and you can withdraw it at any point.
Why it fails without you	Obviously. But specifically: without people who experience episodes frequently, a twenty-minute block yields too few events and the second test is unrunnable.

3. Participants who do not hear voices

This is the recruitment need that gets forgotten and the one most likely to sink the study. It is also the easiest to fill, because the requirements are almost nothing.

	Detail
How many	Twenty-five, matched to the study group on age band, sex and handedness.
Eligibility	You do not hear voices without an external source. Age eighteen or over, no untreated hearing loss. That is the entire list.
A note for the community	Many people who identify as targeted individuals report harassment, surveillance or bodily sensations without any auditory phenomenon at all. If that describes you, you are eligible for this group and you would be genuinely valuable in it — it removes the objection that any difference found is really a difference in life circumstances, stress or sleep rather than in voice-hearing.
What is involved	Exactly the same session as everyone else, on the same hardware. Two hours. The same deposit and contribution terms, with the same waiver.
Why it fails without you	Every number in this study is a comparison. With no reference for what a normal measurement looks like on this equipment in this kind of room, the study group's scores float free and mean nothing at all. This group is not a formality; it is half the experiment.



4. Electronics and firmware

	Detail
How many	Two or three people. One with real analogue and biosignal experience is worth more than five generalists.
Skills	Schematic capture and PCB layout for mixed-signal boards. Experience with the ADS1299 or a comparable biopotential front end. Embedded C on ESP32 or similar. USB descriptor work, including composite devices and the Windows binding descriptors. Someone qualified to sign off the electrical safety review is separately essential and may be a different person.
First tasks	Prototype board with two converters cascaded. The audio envelope conditioning circuit. The timing self-test that the whole protocol depends on — build this first, because if it cannot be made to work the study does not happen.
Time commitment	Heavy for six to eight weeks during the prototype phase, light afterwards. Assembly of the fleet is a separate and much easier task that more people can help with.
Why it fails without you	The commercial route costs four times as much per unit and puts the fleet out of reach. Everything about the economics in Part 12 depends on the instrument being built rather than bought.

5. Software

Role	What is needed
Frontend — the session runner	A progressive web app that acts as the operator: guided electrode placement with live impedance, block sequencing, audio playback and capture, offline buffering, and the device connection flow. TypeScript and modern browser APIs. The harder half of this job is the interface decisions where a well-meaning improvement could mislead a vulnerable person — the constraints in Part 13 and Part 15 are not negotiable.
Backend — streaming ingest	Websocket ingest as a separate service, append-only binary write path, gap detection and retransmit, session state machine. Python and PostgreSQL against an existing self-hosted stack.
Signal processing and statistics	The pipeline in Parts 9 and 10, and the analysis. Someone who has done real event-related potential work and will argue about time windows is worth considerably more here than a general data scientist. Familiarity with MNE-Python, scikit-learn, and ideally Riemannian methods for small-sample EEG.
Data protection and legal	The impact assessment, the consent architecture, the deposit and loan terms across jurisdictions, and the separation between the research use and any evidentiary use. European practice specifically.
Translation	Consent materials and session instructions into the languages participants actually use. A protocol someone half-understands is a protocol they cannot consent to, and this is a consent requirement rather than a convenience.



6. Operations and logistics

The least glamorous role and the one that determines whether the fleet works at all. A kit that sits in a cupboard for three weeks between participants halves the throughput of the whole programme.

	Detail
How many	Two or three at Stage 1. More at Stage 2, ideally distributed — see the note below.
What is involved	Packing and dispatch, tracking, chasing returns, refurbishing units between participants (clean and disinfect electrodes, replace consumables, re-run the timing self-test), managing the deposit ledger, and handling the participant queue.
Regional volunteers are especially valuable	At three hundred sessions, postage is the single largest line in the budget — larger than the hardware. Volunteers in different countries each holding two or three kits and shipping domestically would cut that line roughly in half, which is a bigger financial contribution than most donations.
Skills	Reliability and record-keeping. No technical background required for the shipping side; the refurbishment checklist is written down and takes about twenty minutes per unit.
Why it fails without you	Twenty-five kits with nobody circulating them is a cupboard full of electronics. This role converts capital into data.

7. Participant support

Remote sessions ask people to attend closely to experiences they find frightening, with nobody in the room. Someone has to be reachable during and after, and it should not be the person analysing the data.

- Two or three people able to staff scheduled session windows, respond to a stop control being used, and follow a written escalation protocol.
- Experience in peer support, crisis lines or mental health services is valuable. Willingness to follow a protocol written by someone else is essential.
- This role is structurally separated from the research team, and deliberately so: a participant should be able to say they are struggling without feeling it affects their standing in the study.

If you are unsure which group you fall into, say so. The most common message we expect is from someone who is not certain they qualify as a participant, or who can offer three hours a week and does not know whether that is useful. Both are worth writing about. The single thing that would move this proposal from a document to an experiment fastest is not money — it is one academic willing to put it through an ethics committee.



Closing

The design in this article is not exotic. Every component of it is a standard technique with an established literature: an evoked-response measurement from the 1960s, a self-monitoring paradigm from the late 1990s, a readiness potential from 1965, a muscle-activity recording used in silent-speech research, and a permutation test any competent analyst can implement in an afternoon. Built rather than bought, the instrument costs less than a mid-range phone.

What is missing is not capability. What is missing is the willingness to treat the question as a question. The measurement exists. It has never been pointed here under conditions rigorous enough that either answer would be believed.

There are two halves to it, and they are not alternatives. A small, precise study establishes whether there is a signature and how large it is. A circulating fleet of kits — the shape Mykhailo Sokolov's white paper describes, adapted to a shared pool and delivered through infrastructure that largely already exists — then tests whether that specific signature survives contact with hundreds of ordinary homes. Run in that order, each makes the other worth more. Run in reverse, the second produces volume the first can no longer interpret.

Point it here properly — pre-registered, blinded, with a clinical comparison group and adequate numbers — and you get an answer that survives contact with critics. It might not be the answer anyone wants. It would still be the first one worth having.

None of which has happened yet. Today this is a document. It becomes an experiment when someone finds two thousand euros of parts, an academic willing to put a study through their ethics committee, twenty-five people willing to be measured, twenty-five willing to sit as controls, and a handful prepared to do the unglamorous work of soldering boards and shipping boxes. Two thousand euros. That is what currently stands between a design and a measurement. If you can supply any part of it, Parts 18 and 19 say how, in specific terms.

A note on the figures: every chart in this article is a schematic drawing of a predicted pattern, produced to illustrate the design. None of them are measured data. No study described here has been conducted. If it is conducted, the pre-registration should be published before any of these figures are redrawn with real numbers in them.